

Mathematical Logic

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※ Introduction

Intuitively two Sentences "say the same thing" **iff** they're true in exactly the same circumstances. Starting on logic: from the sentences

Example 1.0.1. This is an example of **syllogism**:

- Every man is mortal
 - Socrates is a man
-
- Therefore, Socrates is mortal

Consequences from some assumptions, which are considered true. The concept is that **if the assumptions are true, then the consequences must also be true.**

Logic doesn't really think about each part of the sentence in a literal sense (we don't care what they "mean" in the real world), but **each sentence has to be formalized in a way that makes the "shape" of the information the most important part.**

Example 1.0.2. Another example:

- All cats have seven legs
 - Bob is a cat
-
- Therefore Bob has seven legs

We can say this because the "shape" of the argument is the same as before, even if rationally it doesn't make sense, it's not "right", but **in a world in which these informations are true then the consequences must be true.**

We don't care about the content, **we only care about the shape and what our clauses imply.**

Example 1.0.3. Yet Another example:

- Every tik is tok
 - Tak is tik
-
- Therefore tak is tok

It doesn't mean shit, but this is true whatever being tik, tak or tok means. We'll eventually want to model pieces of the real world, but logic doesn't necessarily care about that.

this introduction should be redone

Formalizing the shape of these examples:

- First sentence

$$\forall x \quad P(x) \rightarrow Q(x)$$

For every individual in the universe, possessing the property $P()$ implies possessing $Q()$.

- Second sentence

$$P(s)$$

- Therefore

$$Q(s)$$

We **formalized** the sentences, but we **lost some information in the process**, we lose the facts that we're talking about Socrates, or cats, we don't know that we're talking about.

Example 1.0.4. Lets Consider another example:

- Every flower is perfumed
 - The rose is perfumed
-
- Therefore the rose is a flower?

Is this syllogism correct? Let's formalize it to find out.

$$\forall x \quad P(x) \rightarrow Q(x)$$

$$Q(s)$$

$$\text{Therefore } P(s)?$$

This obviously **doesn't hold**, our first predicate doesn't automatically works bidirectionally, we're **reversing our implication**. With our assumptions there's no "proof" that $P(s)$ is true.

Example 1.0.5. Final example:

- If it rains I take my umbrella

Which one is the **equivalent**?

1. If it doesn't rain I don't take my umbrella
2. If I don't take my umbrella then it doesn't rain
3. If I take the umbrella then it rains
4. Either it doesn't rain or I take the umbrella
5. It rains only if I take the umbrella
6. It rains if I take the umbrella
7. It rains if and only if I take the umbrella
8. None of the above

To answer we need to formalize what we mean by equivalent: for logicians it means "given a world in which its true then the other is true and given a world in which it's false then the other is false".

With this definition then the **second** one is **true**:

$$P(x) \rightarrow Q(x) \text{ then } \neg Q(x) \rightarrow \neg P(x)$$

The first one is a simple negation, it can't be true as well.

The third one is the opposite of the original sentence.

There's at least two types of *or* in a natural language, usually it's exclusive. The **fourth sentence is correct if the *or* is inclusive**, if it's a *xor* it can't be.

Five is also a **correct** translation, once we agree on the meaning of words like “if”, “only if”, ...; It basically means that the first part (rain) can or can not happen based on the “umbrella” fact, which is the same as the original.

We're speaking in both cases about a necessary and sufficient condition for rain. Taking the umbrella is a necessary condition for rain, it doesn't cause the rain. There's no causation between clauses.

Six and seven are false, since five is true.

Equivalence: What does “saying the same thing” means in logic terms?

Two sentences **say the same thing if and only if (iff) they are true in exactly the same circumstances**, a.k.a. “possible world”, a.k.a. “truth evaluation”.

If *it rains* then *I take my umbrella*: the cursive ones are stand alone sentences, each one is an **atomic sentence** (in logic terms), they cannot be further disassembled.

So, **formalizing** the sentence

- it rains = P
- I take my umbrella = Q
- if/then $\Rightarrow$

$$P \rightarrow Q$$

There can be many (infinite) world in which “it rains”, i.e. P , is true and another set of worlds in which “I take my umbrella”, i.e. Q , is true. These sets can (will) intersect.

Imagining something along the lines of a Venn diagram (I don't care enough to actually draw it), there will be the following regions:

- P : worlds in which it only rains
- Q : worlds in which I only take my umbrella
- $P \cap Q$: worlds in which it rains and I take my umbrella
- $\neg(P \vee Q)$: worlds in which it doesn't rain and I don't take my umbrella (outside all)

For each region we have

- it rains but I don't take my umbrella (inside the P region only) $\Rightarrow$ for the worlds in here the preposition is false
- it doesn't rain but I take my umbrella (Q region) $\Rightarrow$ for the worlds in here the preposition is true

- inside both regions, it rains and I take my umbrella $\implies$ for the worlds here the preposition is true
- outside both regions, doesn't rain, don't take my umbrella $\implies$ for the worlds here the preposition is true

Now we have a sort of “image” on which to evaluate if other sentences are equivalent. **If another preposition gives us the same image**, i.e. the same truth values in the same regions of the diagram, **it's equivalent to our original**.

Considering our earlier example and taking sentence #4:

$$(\neg P) \vee Q$$

- $\neg P$ is true when P is not (duh)
- $P \vee Q$ is true inside the P , Q and intersection region between them

So, combining these two gives us that, $(\neg P) \vee Q$ is true

- inside Q region
- inside intersection between P and Q
- outside everything

So it's the same as our initial example $P \rightarrow Q$, they give the same “picture”, so they say the same thing, they are equivalent. **They are true in the same circumstances**. Each world is false (or true), i.e. has the same value, in both cases.

Another case, sentence #2:

$$\neg Q \rightarrow \neg P$$

gives the same picture, while sentence #1

$$\neg P \rightarrow \neg Q$$

does not (obviously, they're opposite to one another, they can't be both true).

Considering sentences #5

$$Q \leftarrow P \implies P \rightarrow Q$$

and #6

$$Q \rightarrow P$$

#6 can be rephrased in #6' as: If I take my umbrella it rains. Doesn't sound too correct, does it?

While in sentence #5 “only if” goes the other way in respect to “if”.

Basically:

- **if** goes **one way**
- **only if** goes the **other way**
- **if and only if (iff)** goes **both ways**

Inside

$$P \rightarrow Q : \text{true}$$

It means that:

- P is **sufficient** for Q to be true (if there's P there must be Q)
- Q is **necessary** for P to be true (there can't be P without Q)

There can be no case in which P (it rains) is true and Q isn't (I take my umbrella).

When is our preposition true? " $P \rightarrow Q$ is true" can be read as:

"When P is true then Q is true"

or

"Each single time P is true then Q is true (in every possible world)"

※ Introduction to Formulas

A “Sentence” (or **proposition**) in natural language is a **grammatically correct sequence of words such that it makes sense to ask whether** (in the given circumstance) **it is true or false**.

Example 2.0.1. Let’s make some examples of propositions

- It rains 
- If it rains, I stay at home 
- What’s the time?  Questions are not sentences.
- $2 + 2 = 4$ 
- $2 + 2 = 4$  It’s wrong, but still a sentence.

2.1 Meaning: Denotation vs Connotation

Consider:

- 4
- 2^2
- $6 - 2$
- $5 + 1$

They’re **not sentences** because they **can’t be true or false**, they just are. But all these expressions **mean the same thing**, they all **denote** the natural number 4 (apart from the last one). This is the denotation of these expressions. They are all expressions that describe distinct ways to obtain 4. The **denotation** is the number 4, **all the other informations given by the expressions are connotations**.

For each given an expression E we have the denotation of E and the connotation(s) of E .

A **denotation** is just a **name** (a pointer), **referring to a uniquely determined object** of discourse.

The **connotations** are the **names with all the actual information they contain**. A name can be a complex thing and as such can contain informations.

We’ll primarily focus on denotations, as connotations are more challenging to handle, and denotations possess the following useful property.

Property 2.1.1 (Invariance under substitutions). Two different ways to connote the same denoted object can be substituted as they mean the same thing. For example we can exchange 4 and $2 + 2$ because they’re different strings but evaluate to the same object.

Ok but what is the denotation of sentences in standard arithmetic?

- $4 = \text{pred}(5)$ *true*
- $4 = \text{succ}(5)$ *false*
- $4 = 3 + 1$ *true*
- $4 = 4$ *true*
- $2 = 2$ *true*
- $4 = 6$ *false*

The **denotation of a sentence**, in a given possible world, is just **one of two truth values**, *true* or *false*. **Each sentence is evaluated to a precise truth value.**

2.2 Formalization

What is propositional logic about? Propositions (a.k.a. sentences) and their semantics (denotation, their truth values) in every possible circumstance.

To formalize sentences we distinguish among **two types of sentences**:

- **Atomic sentences**: Simple sentences (e.g., “It rains,” “Paul runs,” “I stay at home”) that cannot be simplified further.
- **Complex sentences**: Formed by combining atomic sentences with connectives (It rains and Paul runs, Paul runs or I stay at home, ...). These can be arbitrarily complex.

The **formalization** of these types of sentences is:

- **Atomic sentences** $\rightarrow$ **Atomic formulas**, they **become symbols** ($p, q, r, p_1, \dots$).
We fix a (countably) infinite set of symbols $\mathcal{L}$ as our set of atomic formulas, $\mathcal{L}$ is called “propositional language” and its elements are called “atomic formulas” or “propositional letters/variables”. In every possible world, every propositional letter can either be true or false.

- **Complex sentences**: Fix a propositional language $\mathcal{L}$ and the **set of propositional formulas on this language**.

$\mathcal{F}_{\mathcal{L}}$ is defined as follows (3 ways):

1. The smallest set such that:
 - $\forall p \in \mathcal{L}$ then $p \in \mathcal{F}_{\mathcal{L}}$ $(\mathcal{L} \subseteq \mathcal{F}_{\mathcal{L}})$
 - $\forall A, B \in \mathcal{F}_{\mathcal{L}}$ then $(A \cap B), (A \cup B), (A \rightarrow B), (\neg A) \in \mathcal{F}_{\mathcal{L}}$
2. $\mathcal{F}_{\mathcal{L}}$ is the **intersection** of all sets $\mathcal{X}$ such that:
 - $\mathcal{L} \subseteq \mathcal{X}$
 - $\forall A, B \in \mathcal{X}$ then $(A \cap B), (A \cup B), (A \rightarrow B), (\neg A) \in \mathcal{X}$
3. Inductive definition: $\mathcal{F}_{\mathcal{L}}$ is the set such that the following properties hold:
 - (Base) $\mathcal{L} \subseteq \mathcal{F}_{\mathcal{L}}$
 - (Inductive Step) If $A, B \in \mathcal{F}_{\mathcal{L}}$ then $(A \cap B), (A \cup B), (A \rightarrow B), (\neg A) \in \mathcal{F}_{\mathcal{L}}$
 - (3) Nothing else belongs to $\mathcal{F}_{\mathcal{L}}$

The difference between first and last one is that in the former we specify “smallest set”, while in the last it’s a property.

Example 2.2.1. given $p, q, r \in \mathcal{L}$, then

- $((p \wedge q) \rightarrow r) \in \mathcal{F}_{\mathcal{L}}$ this is a valid formula and as such is in $\mathcal{F}_{\mathcal{L}}$
- $p \vee \wedge(q \wedge r) \notin \mathcal{F}_{\mathcal{L}}$ this is a string but it’s not a formula

We must be able to identify formulas from meaningless strings. To achieve this, we require a certificate, some type of proof, that confirms whether a given string is a formula.

It is important to note that the set of connectives used in any definition is chosen arbitrarily; there can be many, few, or even infinitely many connectives. The formalization of complex sentences depends on the specific connectives under consideration.

2.3 Certifying formulas

We need to **produce a certificate that a string is a formula**, a kind of proof that something is a formula. We don't need to prove that something is NOT a formula since we indirectly get that from the fact that there's no certificate for such string.

2.3.1 $\mathcal{L}$ -construction

Given a word $w \in (\{\wedge, \vee, \neg, \rightarrow, \}, (\} \cup \mathcal{L})^*$, an $\mathcal{L}$ -construction is a way to **show that, w is actually a formula**, i.e. $w \in \mathcal{F}_{\mathcal{L}}$.

more formally an $\mathcal{L}$ -construction of w is a sequence finitely many strings $w_1, w_2, \dots, w_u$ such that:

- $w_u = w$
- $\forall i = 1, 2, \dots, u$, either.
 - $w_i \in L$
 - $\exists j < i$ such that $w_i = (\neg w_j)$
 - $\exists j, k < i$ such that
 - $w_i = (w_j \wedge w_k)$
 - $w_i = (w_j \vee w_k)$
 - $w_i = (w_j \rightarrow w_k)$

The string must be the negation of something else, the conjunction of something else or the disjunction of something else, implication, ecc.; it must be made of something else which is simpler.

Example 2.3.1. Let's show more than one $\mathcal{L}$ -construction that proves $(p \wedge q) \rightarrow r \in \mathcal{F}_{\mathcal{L}}$

- $p, \quad q, \quad r, \quad (p \wedge q), \quad ((p \wedge q) \rightarrow r)$
- $r, \quad q, \quad p, \quad p, \quad (p \rightarrow p), \quad (p \wedge q), \quad ((p \wedge q) \rightarrow r)$

These aren't the only $\mathcal{L}$ -construction possible, there are infinitely many possible ones (we can add useless stuff and even repetitions).

But we can have a **shortest** $\mathcal{L}$ -construction, its easy to prove that such $\mathcal{L}$ -construction will be as long as the number of letters + the number of connectives used in the formula we need to certify.

2.3.2 Unique readability

Directly from the definition of $\mathcal{L}$ -constructions comes another property: **unique readability**. From the definition of $\mathcal{L}$ -construction, it can be observed that if $w \in \mathcal{F}_{\mathcal{L}}$ then **exactly one** of the following cases **holds**

- $w \in \mathcal{L}$
- or there is $v \in \mathcal{F}_{\mathcal{L}}$ such that $w = (\neg v)$
- or there are $v_1, v_2 \in \mathcal{F}_{\mathcal{L}}$ such that $w = (v_1 \wedge v_2)$

- or there are $v_1, v_2 \in \mathcal{F}_{\mathcal{L}}$ such that $w = (v_1 \vee v_2)$
- or there are $v_1, v_2 \in \mathcal{F}_{\mathcal{L}}$ such that $w = (v_1 \rightarrow v_2)$

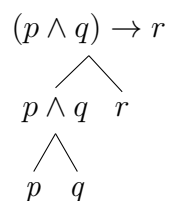
Essentially, either w it's not decomposable, or it's decomposable in a unique way.

Not every formal language possesses the psproperty of unique readability. A simple example of a formal language lacking this property is the case where $w \in \Sigma^*$ and $w = v_1 \cdot v_2$ (the concatenation of strings). For instance, consider $w = \text{hello}$. This string can be decomposed in 6 different ways of the form $w = v_1 \cdot v_2$:

- ϵ, hello
- h, ello
- he, llo
- hel, lo
- $hell, o$
- $hello, \epsilon$

The unique readability of formulas grants us additional forms of certification for “Formulicity” Thanks to this property, we can construct a **unique parsing tree** for any given formula.

Example 2.3.2. Lets draw the parsing tree of $(p \wedge q) \rightarrow r$:



For a given string $w \in (\{\wedge, \vee, \neg, \rightarrow, \}, \{\} \cup \mathcal{L})^*$ if we have a single parsing tree, it's for sure a formula.

2.3.3 Structural induction principle

Since we can **define** $\mathcal{F}_{\mathcal{L}}$ **as an inductive set** (third definition), it comes with its own **Induction principle**.

Definition 2.3.3. Let $\mathcal{P}$ be some property that can be asked of formulas (note that $\mathcal{P}$ may hold or not for any given $\mathcal{F} \in \mathcal{F}_{\mathcal{L}}$).

This property $\mathcal{P}$ holds for all formulas $\mathcal{F} \in \mathcal{F}_{\mathcal{L}}$ iff:

1. **base:** $\mathcal{P}$ holds for all propositional letters $p \in \mathcal{L}$
2. **step(s):** if $\mathcal{P}$ holds for generic formulas $A, B \in \mathcal{F}_{\mathcal{L}}$ then $\mathcal{P}$ holds for all the ways to build new formulas starting from A and B : $(\neg A), (A \wedge B), (A \vee B), (A \rightarrow B)$

Proof.

- Let $\mathcal{P}$ be a property satisfying **base** and **step(s)**.
- Let $\mathcal{I} = \{\mathcal{F} \in \mathcal{F}_{\mathcal{L}} : \mathcal{P} \text{ holds for } \mathcal{F}\}$ (the set of formulas for which $\mathcal{P}$ holds).

We now need to show that $\mathcal{F}_{\mathcal{L}} = \mathcal{I}$ (the set of formulas in which the property holds coincides with the complete set of formulas). To do that we need to show that:

1. $\mathcal{F}_{\mathcal{L}} \subseteq \mathcal{I}$
2. $\mathcal{I} \subseteq \mathcal{F}_{\mathcal{L}}$

2. is trivial since $\mathcal{I}$ is defined as a subset of $\mathcal{F}_{\mathcal{L}}$. We only need to prove that all the formulas are in $\mathcal{I}$.

Since $\mathcal{P}$ satisfies the **base** case, it follows that $\mathcal{P}$ holds for all $p \in \mathcal{L}$. Therefore, $\mathcal{L} \subseteq \mathcal{I}$.

If $A, B \in \mathcal{I}$, then $\mathcal{P}$ holds for A and B . By **step(s)**, $\mathcal{P}$ also holds for $(\neg A)$, $(A \wedge B)$, $(A \vee B)$, and $(A \rightarrow B)$. Consequently, the same constructions we defined for $\mathcal{F}_{\mathcal{L}}$ belong to $\mathcal{I}$.

Thus, each formula $\mathcal{F}$ belongs to $\mathcal{I}$, which implies $\mathcal{F}_{\mathcal{L}} \subseteq \mathcal{I}$. $\square$

2.4 Semantics of propositional logic

Classical propositional logic is based on a couple of **underlying design ideas**:

- **Bivalence principle**: a sentence (formula) in every possible universe **is either true or false**.
- **Truth-functionality principle**: (also called compositionality, extensionality) the **truth value of a sentence** (a formula) in a given possible universe **only** depends on the truth values of the sentences composing it and the fixed meaning of connectives.

An easy example of a non truth-functional realm is probability we can prove this with a simple example

Example 2.4.1. given $A :=$ Tomorrow it will rain and $B :=$ France is going to win the next world cup and given the probabilities of those two events being $v(A) = v(B) = 0.5$ then by truth functionality $v(A \rightarrow B) = v(\frac{1}{2} \rightarrow \frac{1}{2}) = v(A \rightarrow A)$ but this makes no sense since the probability of $v(A \rightarrow A)$ is obviously 1.

For instance, take conjunction

$$A, B \in \mathcal{F}_{\mathcal{L}} \quad A \wedge B$$

We want to evaluate the value in a given possible world. By bivalence we know that this is either true or false. By truth-functionality this value depends only on:

- whether A is true or false
- whether B is true or false

The notion of “possible world”: it’s a **function** from the set of propositional letters to the set of truth values

$$v : \mathcal{L} \mapsto \{0, 1\}$$

It assigns a truth value to each of the propositional letters. The set of all possible worlds is the set of all possible truth evaluations.

Now Let’s formalize these notions, by truth functionality lets fix the semantics of our connectives

$$\tilde{v}(A \wedge B) := I_{\wedge}(\tilde{v}(A), \tilde{v}(B))$$

$$I_{\wedge} : \{0, 1\}^2 \mapsto \{0, 1\} = a \cdot b = \min\{a, b\}$$

a	b	$a \wedge b$
0	0	0
0	1	0
1	0	0
1	1	1

$$I_{\vee} : \{0, 1\}^2 \mapsto \{0, 1\} = a + b = \max\{a, b\}$$

a	b	$a \vee b$
0	0	0
0	1	1
1	0	1
1	1	1

$$I_{\neg} : \{0, 1\} \mapsto \{0, 1\} = 1 - a$$

a	$\neg a$
0	1
1	0

$$I_{\rightarrow} : \{0, 1\}^2 \mapsto \{0, 1\} = \max\{1 - a, b\}$$

a	b	$a \rightarrow b$
0	0	1
0	1	1
1	0	0
1	1	1

$$I_{\leftrightarrow} : \{0, 1\}^2 \mapsto \{0, 1\} = ((a + b) \bmod 2) + 1$$

a	b	$a \leftrightarrow b$
0	0	1
0	1	0
1	0	0
1	1	1

$$I_{\oplus} : \{0, 1\}^2 \mapsto \{0, 1\} = ((a + b) \bmod 2)$$

a	b	$a \oplus b$
0	0	0
0	1	1
1	0	1
1	1	0

Definition 2.4.2. A truth evaluation is an arbitrarily chosen map

$$v : \mathcal{L} \mapsto \{0, 1\}$$

Definition 2.4.3. the canonical extension $\tilde{v} : \mathcal{F}_{\mathcal{L}} \mapsto \{0, 1\}$ of $v : \mathcal{L} \mapsto \{0, 1\}$ is defined as follows:

- if $F \in \mathcal{L}$ then $\tilde{v}(F) := v(F)$
- if $A, B \in \mathcal{F}_{\mathcal{L}}$ then
 - $\tilde{v}(A \wedge B) := I_{\wedge}(\tilde{v}(A), \tilde{v}(B))$
 - $\tilde{v}(A \vee B) := I_{\vee}(\tilde{v}(A), \tilde{v}(B))$
 - $\tilde{v}(A \rightarrow B) := I_{\rightarrow}(\tilde{v}(A), \tilde{v}(B))$
 - $\tilde{v}(\neg A) := I_{\neg}(\tilde{v}(A))$

2.4.1 Tautologies and other definitions

Definition 2.4.4 (Tautology). a formula $F \in \mathcal{F}_{\mathcal{L}}$ is a **Tautology** iff $\forall v : \mathcal{L} \mapsto \{0, 1\}, \tilde{v}(F) = 1$, i.e. it's always true for all assignment.

Definition 2.4.5 (Satisfiable Formula). a formula $F \in \mathcal{F}_{\mathcal{L}}$ is **Satisfiable** iff $\exists v : \mathcal{L} \mapsto \{0, 1\}, \tilde{v}(F) = 1$, i.e. there's at least one assignment that makes it true.

Definition 2.4.6. a formula $F \in \mathcal{F}_{\mathcal{L}}$ is (**contradiction, unsatisfiable, refutable**): iff $\forall v : \mathcal{L} \mapsto \{0, 1\}, \tilde{v}(F) = 0$, i.e. it's never true for all assignment.

Fact 2.4.1. Let $F \in \mathcal{F}_{\mathcal{L}}$ then F is a tautology iff

$$\neg F \text{ is a contradiction}$$

Proof. F is a tautology iff (by definition of tautology) $\forall v : \mathcal{L} \mapsto \{0, 1\}, \tilde{v}(F) = 1$. But this by definition of (truth table) $I_{\neg}$ means that $\tilde{v}(\neg F) = 0$ for all $v : \mathcal{L} \mapsto \{0, 1\}$, and thus by definition of contradiction $\neg F$ is unsatisfiable. $\square$

To **prove** that the **implication is correct** the way it is:

2.4.2 Digression on combinatorics

$$|\{F : \{0, 1\}^n \mapsto \{0, 1\}\}| = 2^{2^n}$$

Which is a specific version of a more general formula: with A and B as finite sets

$$|\{F : A \mapsto B\}| = |B|^{|A|}$$

Intuitively this formula can be explained by the fact that there are $|B|$ options for each element of A . Let $\mathcal{S}$ be a set of n many elements. There are 2^n possible subsets of $\mathcal{S}$, we can use the aforementioned formula to prove this

$$2^n = |\mathcal{P}(\mathcal{S})| = 2^{|\mathcal{S}|} = |\{F : \mathcal{S} \mapsto \{0, 1\}\}|$$

Note that we are counting all characteristic functions on $\mathcal{S}$, where a characteristic function is:

$$X_A : \mathcal{S} \mapsto \{0, 1\} \quad \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Remember that characteristic functions are essentially subsets; there is an isomorphism between characteristic functions and subsets.

Example 2.4.7. Let's make an example to justify the interpretation of implications.

“If a natural number is divisible by 4 then it is divisible by 2”

$$n = 4k \implies n = (2 \cdot 2)k \implies 2 \cdot (2k)$$

Here the only assumption is that n is divisible by 4, so we are skipping over the case where $n \neq 4k$. However, let us now examine all cases, even when this assumption does not hold, keeping in mind that the proposition is obviously true over all numbers.

“If 4 is divisible by 4, then 4 is divisible by 2”
 $true \rightarrow true = true.$

“If 3 is divisible by 4, then 3 is divisible by 2”
 $false \rightarrow false = true.$

“If 6 is divisible by 4 then 6 is divisible by 2”.
 $false \rightarrow true = true.$

2.4.3 Assignment Encoding

We want to demonstrate how it's possible to **encode** the **infinite information** given by a general truth value assignment on all letters of $\mathcal{L}$ in **finite time**.

Lemma 2.4.8. Let $F \in \mathcal{F}_{\mathcal{L}}$ and let $v, v' : \mathcal{L} \mapsto \{0, 1\}$ be two (possibly) distinct truth value assignments. Then: if $v(p) = v'(p)$ for all $p \in \mathcal{L}$ letters **actually occurring** in F (so occurs in every $\mathcal{L}$ -construction of F) then $\tilde{v}(F) = \tilde{v}'(F)$. The only relevant part of an assignment is the **letters actually occurring**, we don't care about all the infinite non-occurring letters.

Proof. Structural induction of F

Base: If $F = p$, $p \in \mathcal{L}$ then p occurs in F , trivially then by hypothesis $v(p) = v'(p)$, then by definition of $\tilde{v}$, $\tilde{v}(p) = \tilde{v}'(p)$.

Step(s):

- If $F = (\neg A)$ for some $A \in \mathcal{F}_{\mathcal{L}}$, then by induction hypothesis (I.H.) $\tilde{v}(A) = \tilde{v}'(A)$. Since each $p \in \mathcal{L}$ occurring in F occurs in A too, by definition of $I_{\neg}$:

$$v(F) = \tilde{v}(\neg A) \stackrel{\text{def}}{=} \neg \tilde{v}(A) \stackrel{\text{I.H.}}{=} \neg \tilde{v}'(A) \stackrel{\text{def}}{=} \tilde{v}'(\neg A) = v'(F)$$

- If $F = (A \wedge B)$ for some $A, B \in \mathcal{F}_{\mathcal{L}}$, then each $p \in \mathcal{L}$ occurring in A occurs in F too and each $p \in \mathcal{L}$ occurring in B occurs in F too. So we can apply the I.H. $\tilde{v}(A) = \tilde{v}'(A)$ and $\tilde{v}(B) = \tilde{v}'(B)$. By definition of $I_{\wedge}$:

$$v(F) = \tilde{v}(A \wedge B) \stackrel{\text{def}}{=} I_{\wedge}(\tilde{v}(A), \tilde{v}(B)) \stackrel{\text{I.H.}}{=} I_{\wedge}(\tilde{v}'(A), \tilde{v}'(B)) \stackrel{\text{def}}{=} \tilde{v}'(A \wedge B) = v'(F)$$

- If $F = (A \vee B)$ for some $A, B \in \mathcal{F}_{\mathcal{L}}$, then each $p \in \mathcal{L}$ occurring in A occurs in F too and each $p \in \mathcal{L}$ occurring in B occurs in F too. So we can apply the I.H. $\tilde{v}(A) = \tilde{v}'(A)$ and $\tilde{v}(B) = \tilde{v}'(B)$. By definition of $I_{\vee}$:

$$v(F) = \tilde{v}(A \vee B) \stackrel{\text{def}}{=} I_{\vee}(\tilde{v}(A), \tilde{v}(B)) \stackrel{\text{I.H.}}{=} I_{\vee}(\tilde{v}'(A), \tilde{v}'(B)) \stackrel{\text{def}}{=} \tilde{v}'(A \vee B) = v'(F)$$

- If $F = (A \rightarrow B)$ for some $A, B \in \mathcal{F}_{\mathcal{L}}$, then each $p \in \mathcal{L}$ occurring in A occurs in F too and each $p \in \mathcal{L}$ occurring in B occurs in F too. So we can apply the I.H. $\tilde{v}(A) = \tilde{v}'(A)$ and $\tilde{v}(B) = \tilde{v}'(B)$. By definition of $I_{\rightarrow}$:

$$v(F) = \tilde{v}(A \rightarrow B) \stackrel{\text{def}}{=} I_{\rightarrow}(\tilde{v}(A), \tilde{v}(B)) \stackrel{\text{I.H.}}{=} I_{\rightarrow}(\tilde{v}'(A), \tilde{v}'(B)) \stackrel{\text{def}}{=} \tilde{v}'(A \rightarrow B) = v'(F)$$

□

Remarks on the lemma: the lemma allow us **evaluate the behavior of any** $F \in \mathcal{F}_{\mathcal{L}}$ under all assignments (of which there are infinitely many) **just by computing the truth table of** F (finite object, albeit it can be large), whose columns are indexed by the propositional letters actually occurring in F .

If the formula F contains exactly the propositional letters $p_1, p_2, \dots, p_n$, how many rows will the truth table for F have? Based on the principles introduced in 2.4.1, there are 2^n rows. This growth is exponential with respect to the number of propositional letters in F . Each row corresponds to a possible truth value assignment (a function) for $p_1, p_2, \dots, p_n$ mapping to $\{0, 1\}$.

If the formula F contains exactly the propositional letters $p_1, p_2, \dots, p_n$, how many rows will the truth table for F have? Using what we introduced in 2.4.1, there are 2^n rows. This growth is exponential in the length of F . Each row is a possible assignment of $p_1, \dots, p_n \mapsto \{0, 1\}$.

shouldn't these two I.H. have different assignments? TOASK

Example 2.4.9. using Lemma 2.4.3 we can prove using truth table that, let $A, B \in \mathcal{F}_{\mathcal{L}}$:

1. $A \rightarrow \neg A$: satisfiable, but not a tautology
2. $A \wedge \neg A$: unsatisfiable
3. $A \rightarrow (A \vee B)$: tautology

You can check with truth tables (under each column the result of each operand):

A	$A \rightarrow \neg A$	$A \wedge \neg A$	A	B	$A \rightarrow (A \vee B)$
0	1	0	0	0	1
0	1	0	0	1	1
1	0	1	1	0	1
1	0	1	1	1	1

Below is a series of important tautologies. They are straightforward to prove using truth tables.

- $A \rightarrow (B \rightarrow A)$ (called “weakening”)
- $(\neg A \rightarrow A) \rightarrow A$ (called “consequentia mirabilis”)
- $(A \rightarrow B) \vee (B \rightarrow A)$ (called “prelinearity”)
- $(A \wedge B) \rightarrow A$
- $A \rightarrow (A \vee B)$
- $A \vee \neg A$ (called “tertium non datur” or “excluded middle”)
- $\neg \neg A \rightarrow A$ and $A \rightarrow \neg \neg A$ (called “double negation laws” or “involutiveness of negation”)

Let $A \in \mathcal{F}_{\mathcal{L}}$, $A \rightarrow \neg A$, the truth table:

A	$A \rightarrow \neg A$
0	1
1	0

This **does not take into consideration differences between A and simple letters**, but if we instantiate A with a specific formula, for example $A := p \wedge \neg p$ (a contradiction) then: $(p \wedge \neg p) \rightarrow \neg(p \wedge \neg p)$ is a tautology. But if $A := p \vee \neg p$ (tauto) then: $(p \vee \neg p) \rightarrow \neg(p \vee \neg p)$ is a contradiction.

Fact 2.4.2. A Tautology/Contradiction remains a Tautology/Contradiction if you replace letters with arbitrary formulas (the contrary doesn’t necessarily hold).

Two very important Tautologies are:

- **Prelinearity:** $(A \rightarrow B) \vee (B \rightarrow A) \quad \forall(A, B \in \mathcal{F}_{\mathcal{L}})$. This kinda means that “in every possible world $A \rightarrow B$ is true or $B \rightarrow A$ is true”.
- **Excluded middle:** $A \vee \neg A$. This tautology asserts that for any proposition A , it must be either true or false, there is no middle ground. In logical terms, this means that in every possible world, A is either true or $\neg A$ is true. The rejection of the principle of the excluded middle forms the basis for other logical systems, such as constructive or intuitionistic logic, which are particularly significant in computer science. Unlike classical logic, these systems do not assume $A \vee \neg A$ as universally valid, focusing instead on provability and constructibility.

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※ Semantics notions for sets of formulas

Let $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$ (Γ is a subset of the set of all formulas), then Γ is called a **theory**.

Definition 3.0.1. A theory $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$ is **satisfiable** iff $\exists v : \mathcal{L} \rightarrow \{0, 1\}$ such that $\forall \gamma \in \Gamma$ then $\tilde{v}(\gamma) = 1$ (in symbols $v \models \Gamma$).

Definition 3.0.2. A theory $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$ is **unsatisfiable** iff $\forall v : \mathcal{L} \rightarrow \{0, 1\}$ there exists at least $\gamma \in \Gamma$ such that $\tilde{v}(\gamma) = 0$. (in symbols $v \not\models \Gamma$).

$\leftarrow (v \models \gamma$ we mean v satisfies γ as a formula, $\tilde{v}(\gamma) = 1$)

$\leftarrow (v \not\models \gamma$ we mean v does not satisfy γ as a formula, $\tilde{v}(\gamma) = 0$)

3.1 Logical consequence

Of a formula $F \in \mathcal{F}_{\mathcal{L}}$ from a **theory** $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$.

Definition 3.1.1. Let $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$, $F \in \mathcal{F}_{\mathcal{L}}$, then F is a **logical consequence** of Γ , $\Gamma \models F$, iff for every assignment $v : \mathcal{L} \rightarrow \{0, 1\}$ satisfying Γ it holds that v satisfies F too:

$$v \models \Gamma \implies v \models F$$

To express the fact that a given formula $F \in \mathcal{F}_{\mathcal{L}}$ is a **tautology** by means of the notion of logical consequence there are several ways, for example:

$$T \models F$$

with T being an always true constant. Another solution is to let Ω be a set of tautologies:

$$\Omega \models F$$

Or even:

$$\emptyset \models F$$

since you can't falsify a formula in the $\emptyset$, there's no formula. For a **tautology** the standard notation most commonly used is:

$$\models F$$

3.1.1 Fundamental semantic properties of theories with respect to logical consequence

Lemma 3.1.2. Let $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$ be a theory and $A \in \mathcal{F}_{\mathcal{L}}$, then $\Gamma \models A$ iff $\Gamma \cup \{\neg A\}$ is unsatisfiable.

Proof.

$\Gamma \models A$ iff $\forall v : \mathcal{L} \rightarrow \{0, 1\}, \tilde{v}(\gamma) = 1 \forall \gamma \in \Gamma \rightarrow \tilde{v}(A) = 1$ by definition of $\models$
iff $\forall v : \mathcal{L} \rightarrow \{0, 1\}, \neg(\tilde{v}(\gamma) = 1 \forall \gamma \in \Gamma) \vee \tilde{v}(A) = 1$ by definition of $I_{\rightarrow}$
iff $\forall v : \mathcal{L} \rightarrow \{0, 1\}, \neg(\tilde{v}(\gamma) = 1 \forall \gamma \in \Gamma) \vee \tilde{v}(\neg A) = 0$ by definition of $I_{\neg}$
iff $\forall v : \mathcal{L} \rightarrow \{0, 1\}, \exists \gamma \in \Gamma \tilde{v}(\gamma) = 0 \vee \tilde{v}(\neg A) = 0$
iff $\Gamma \cup \{\neg A\}$ is UNSAT by definition of unsatisfiable

□

Lemma 3.1.3 (Semantical Deduction Lemma). Let $P, Q \in \mathcal{F}_{\mathcal{L}}$, $\{P\} \models Q$ **iff** $P \rightarrow Q$ is TAUTO. The logical consequence relation can be internalized using the logical implication connective.

Proof.

$$\begin{aligned} \{P\} \models Q &\text{ iff } \forall v : \mathcal{L} \rightarrow \{0, 1\}, \tilde{v}(P) = 1 \rightarrow \tilde{v}(Q) = 1 && \text{by definition of } \models \\ &\text{ iff } \forall v : \mathcal{L} \rightarrow \{0, 1\}, \tilde{v}(P \rightarrow Q) = 1 && \text{by definition of } I_{\rightarrow} \\ &\text{ iff } P \rightarrow Q \text{ is TAUTO} && \text{by definition of TAUTO} \end{aligned}$$

□

Lemma 3.1.4 (Semantical deduction). Let $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$ and $P, Q \in \mathcal{F}_{\mathcal{L}}$, then $\Gamma \cup \{P\} \models Q$ (which will be also written as $\Gamma, P \models Q$) **iff** $\Gamma \models P \rightarrow Q$.

Proof.

$$\begin{aligned} \Gamma, P \models Q &\text{ iff } \forall v : \mathcal{L} \rightarrow \{0, 1\}, (v \models \Gamma \wedge \tilde{v}(P) = 1) \rightarrow \tilde{v}(Q) = 1 && \text{by definition of } \models \\ &\text{ iff } \forall v : \mathcal{L} \rightarrow \{0, 1\}, v \models \Gamma \rightarrow \tilde{v}(P \rightarrow Q) = 1 && \text{by definition of } I_{\rightarrow} \\ &\text{ iff } \Gamma \models P \rightarrow Q && \text{by definition of } \models \end{aligned}$$

□

3.2 Compactness Theorem

Theorem 3.2.1 (Compactness Theorem).

Let $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$, Γ is SAT **iff** $\forall \Gamma' \subseteq_{\omega} \Gamma$, Γ' is SAT.

“IMPORTANT”

$\leftarrow$ (with $\subseteq_{\omega}$ meaning finite subset.)

Do we *really* need to consider the case Γ is infinite (contains infinitely many formulas)? Yeah, it will be shown that in first order logic a single formula may contain the amount of information of infinitely many propositional formulas.

Compactness theorem allows us to **reduce the analysis of satisfiability** of a possibly infinite theory **to the analysis of its finite sub-theories**, but observe that **there are infinitely many finite sub-theories of an infinite theory**.

Proof.

- ($\Rightarrow$) Proving the statement from **left to right** is trivial, if $v \models \Gamma$ then, clearly, by the definition of satisfiability and the notion of subset, $v \models \Gamma'$, for each $\Gamma' \subseteq \Gamma$ too (not even $\subseteq_{\omega}$, just any subset, even infinite).
- ($\Leftarrow$) Proving the statement from **right to left** is not so trivial.

□

Consider the statement **equivalent** to the theorem **obtained via contraposition** ($A \rightarrow B$ says the same thing as $\neg B \rightarrow \neg A$):

- ($\Rightarrow$) If there is $\Gamma' \subseteq_{\omega} \Gamma$ UNSAT then clearly Γ is UNSAT.
- ($\Leftarrow$) If Γ is UNSAT then $\exists \Gamma' \subseteq_{\omega} \Gamma$ such that Γ' is UNSAT

Some terminology before the proof: we say that a given theory $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$ is **finitely satisfiable** (FINSAT) iff $\forall \Gamma' \subseteq_{\omega} \Gamma$, Γ' is **satisfiable**.

We shall proceed with the proof of $\Gamma \text{ FINSAT} \implies \Gamma \text{ SAT}$

Proof.

Preparation

We fix any **arbitrarily chosen listing** $F_1, F_2, \dots, F_k, \dots$ (enumerably infinite, without repetitions) of all formulas in $\mathcal{F}_{\mathcal{L}}$. To concretely implement such an enumeration we have two issues to face:

- 1st issue: the **alphabet of $\mathcal{F}_{\mathcal{L}}$** : $\{\wedge, \vee, \neg, \rightarrow\} \cup \{\}, \{\} \cup \mathcal{L}$. This contains an infinite set and thus is an **infinite alphabet**. To **switch to a finite alphabet**:

$$\{\wedge, \vee, \neg, \rightarrow\} \cup \{\}, \{\} \cup \{p, l\}$$

with $p_k \in \mathcal{L}$ written as $p_k = plll \dots l$, with k many l s (just a unary encoding, we just need some kind of finite encoding, let it be a unary encoding, binary, base 10, hex, ...).

- 2nd issue: take the alphabet $\{a, b, \dots\}$: list all the words on this alphabet, but they are infinite and we must define an order. If we use a lexicographic ordering we get:

ϵ (empty word)
a
aa
$a \dots aa$
$\dots$
$b \leftarrow$ never reached

So the letter b appears after an infinite amount of words. Using a short-lex ordering might be fairer:

ϵ (empty word)
a
$\dots$
z
aa
ab
$\dots$
zz

A ordering is **fair** if each word appears at a finite stage.

Construction

We define an **infinite succession of theories** (sets of formulas) as follows:

$$D_0, D_1, \dots, D_k, \dots$$

Inductively defined as follows:

- **Base:**

$$D_0 := \Gamma$$

- **Step:**

$$D_{n+1} := \begin{cases} D_n \cup \{F_{n+1}\} & \text{if } D_n \cup \{F_{n+1}\} \text{ is FINSAT} \\ D_n \cup \{\neg F_{n+1}\} & \text{otherwise} \end{cases}$$

N.B. this definition is well given and in the second case does not directly imply that $D_n \cup \{\neg F_{n+1}\}$ is FINSAT, but just that $D_n \cup \{F_{n+1}\}$ is not FINSAT.

We also define:

$$D := \bigcup_{i \in \mathbb{N}} D_i$$

Now we shall prove some facts about the sequence

$$\Gamma = D_0 \subseteq D_1 \subseteq D_2 \subseteq \dots \subseteq D_k \subseteq \dots \subseteq D$$

To prove $\Gamma \text{ FINSAT} \implies \Gamma \text{ SAT}$, we first need to prove some facts about the sequence above.

Fact 1

Every member of the sequence D_n (for all $n \in \mathbb{N}$) is FINSAT.

Proof. By induction on n

- **Base:** $n = 0$, D_0 is FINSAT by definition, since $D_0 = \Gamma$ and Γ is FINSAT by assumption.
- **Step:** Assume that Fact 1 is true for $D_0, D_1, \dots, D_n$ and prove it for D_{n+1} . To prove this we shall proceed by contradiction. So we assume D_{n+1} is not FINSAT with the aim of finding a contradiction. Absurdum Hypothesis: D_{n+1} is not FINSAT. Then by the definition of the sequence, $D_n \cup \{F_{n+1}\}$ and $D_n \cup \{\neg F_{n+1}\}$ are both NOT FINSAT.

Then by definition of FINSAT

- there exists $D' \subseteq_{\omega} D_n \cup \{F_{n+1}\}$ such that D' is UNSAT.
- there exists $D'' \subseteq_{\omega} D_n \cup \{\neg F_{n+1}\}$ such that D'' is UNSAT.

Lazy version

We can assume, without loss of generality that $F_{n+1} \in D'$, $\neg F_{n+1} \in D''$. We can argue that they are already in there, otherwise we can add them without major problems (if it's unsatisfiable adding something keeps it that way).

Actual version

If $F_{n+1} \notin D'$ then $D' \subseteq D_n$, but D_n is FINSAT, so D' is FINSAT, which is a contradiction.

That is we can display D' and D'' as follows:

$$D' = E' \cup \{F_{n+1}\}, \quad D'' = E'' \cup \{\neg F_{n+1}\}$$

with $E' = D' \setminus \{F_{n+1}\}$ and $E'' = D'' \setminus \{\neg F_{n+1}\}$.

We observe that by construction that $E', E'' \subseteq_{\omega} D_n$.

Note: $E' \cup E'' \cup \{F_{n+1}\}$ is UNSAT, since it contains D' . For the same reason $E' \cup E'' \cup \{\neg F_{n+1}\}$ is UNSAT since it contains D'' .

But $E' \cup E'' \subseteq_{\omega} D_n$, so E', E'' **must be** FINSAT since we know that D_n is FINSAT by I.H, so we get that $E' \cup E''$ is SAT.

This **satisfiable** set **differs** from the **unsatisfiable** ones by **one formula**, notably the same formula.

So, we have:

- $E' \cup E''$ **satisfiable**
- $E' \cup E'' \cup \{F_{n+1}\}$ **unsatisfiable** ($\dagger$)
- $E' \cup E'' \cup \{\neg F_{n+1}\}$ **unsatisfiable** ($\dagger\dagger$)

By definition of satisfiability $\exists v : \mathcal{L} \rightarrow \{0, 1\}$ such that $v \models E' \cup E''$ ($\tilde{v}(F) = 1$ for all $F \in E' \cup E''$).

What about $\tilde{v}(F_{n+1})$?

- If it's 0 it will satisfy $\tilde{v}(\neg F_{n+1})$ and then $v \models E' \cup E'' \cup \{\neg F_{n+1}\}$, **contradicting** ($\dagger\dagger$)
- If it's 1 it will satisfy $\tilde{v}(F_{n+1})$ and then $v \models E' \cup E'' \cup \{F_{n+1}\}$, **contradicting** ($\dagger$)

And, by **bivalence**, there are **no other cases to consider**. We have reached in any case the contradiction that proves the induction step (D_{n+1} is actually FINSAT).

□

Fact 2

$$D = \bigcup_{i \in \mathbb{N}} D_i \text{ is FINSAT}$$

Do we *really* need to prove this? Yes, there are some properties of some sets, such as finiteness, that don't live into their union (with a collection of finite sets, their union might not be a finite set). **Properties of the members don't automatically carry over to their union.**

Proof. Let D' be a generic finite subset of D ($D' \subseteq_{\omega} D$). It is sufficient to prove that D' is SAT.

We can display the finite set D' by listing its formulas:

$$D' = \{F_{i1}, F_{i2}, \dots, F_{im}\} \text{ for some } m \in \mathbb{N}$$

We give these indexes since we don't know exactly what formulas are inside D' , but there must be a maximum, whatever that number is and whatever the numbers in the set are.

Let k be the maximum index

$$k = \max_{j=1}^m (ij)$$

Then:

$$D' \subseteq D$$

$$D' \subseteq \{F_1, \neg F_1, F_2, \neg F_2 \dots, F_k, \neg F_k\}$$

$$D' \subseteq \{F_1, \neg F_1, F_2, \neg F_2 \dots, F_k, \neg F_k\} \cap D \subseteq D_k$$

← (CHANGED from slides)

So $D' \subseteq_\omega D_k$ (as D_k is defined in our sequence), but by Fact 1 D_k is FINSAT. So D' is satisfiable since it's one of its sub-theories. $\square$

Fact 3

For each formula $F_t \in \mathcal{F}_{\mathcal{L}}$, **exactly one among** F_t and $\neg F_t$ **belongs to** D .

Proof. By construction of the sequence, $F_t \in D_t$ or $\neg F_t \in D_t$. Since $D_t \subseteq D$: $F_t \in D$ or $\neg F_t \in D$. This proves that at least one is in the set.

We now have to exclude that both of them are in the set: $(\{F_t, \neg F_t\} \subseteq D)$. But D is FINSAT, and clearly $\{F_t, \neg F_t\}$ would contradict Fact 2 since it can't be satisfiable, by the definition of $I_{\neg}$. So $F_t \in D$ or $\neg F_t \in D$ and $\{F_t, \neg F_t\} \not\subseteq D$. Exactly one is $\in D$. $\square$

Construction

We define the following **assignment** $v_D : \mathcal{L} \rightarrow \{0, 1\}$ such that $\forall p \in \mathcal{L}$, $v_D(p) = 1$ **iff** $p \in D$.

← (Notice that v_D is **well defined** (either $p \in D$ or $p \notin D$ for each possible $p \in \mathcal{L}$ also notice that $p \in \mathcal{L}$ its a letter so $p \in D$ means p is inside one of the formulas of D)

Fact 4

$$v_D \models D \quad (\forall F \in D, \tilde{v}_D(F) = 1)$$

If we can prove this then we have essentially proved that $v_D \models \Gamma$ **since** $\Gamma \subseteq D$ (by construction of D). So Γ is SAT.

Proof. By structural induction, we prove $\forall F \in \mathcal{F}_{\mathcal{L}} : \tilde{v}_D(F) = 1$ **iff** $F \in D$.

Base: $F = p$, with $p \in \mathcal{L}$: so the statement we need to prove is the very definition of $v_D(p)$:

$$\tilde{v}_D(p) = v_D(p) = 1 \quad \text{iff} \quad p \in D$$

← (Notice that would suffice to prove just one side of the iff, namely the side that says if $F \in D$ then $\tilde{v}_D(F) = 1$ (but the stronger proof is easier). We ask for the biconditional (iff) for better using the I.H.)

Step(s): Proof by structural induction on the definition of formula and prove that it's $\in D$:

- $F = (\neg G)$, F is the negation of some formula G .

$$\begin{aligned} \tilde{v}_D(F) = 1 & \text{ iff } \tilde{v}_D(G) = 0 && \text{by definition of } I_{\neg} \\ & \text{ iff } G \notin D && \text{by I.H.} \\ & \text{ iff } \neg G \in D && \text{by Fact 3} \\ & \text{ iff } F \in D \end{aligned}$$

- $F = (G \wedge H)$.

Lemma. $G \in D \wedge H \in D$ **iff** $G \wedge H \in D$

Proof.

- ($\Rightarrow$) Lets suppose by absurdum $G \wedge H \notin D$

$$G \wedge H \notin D \text{ iff } \neg(G \wedge H) \in D \quad \text{by Fact 3}$$

but then $\{G, H, \neg(G \wedge H)\} \subseteq_{\omega} D$ contraddicting Fact 2 since this set is obviously UNSAT ⚡

- ($\Leftarrow$) Lets suppose by absurdum $\neg(G \in D \wedge H \in D)$ that is $\neg G \in D \vee \neg H \in D$ lets assume $\neg G \in D$ (its the same otherwise) but then $\{G \wedge H, \neg G\} \subseteq_{\omega} D$ contraddicting Fact 2 since this set is obviously UNSAT ⚡

$\leftarrow$ (destruct the assumption like in coq)

□

Now lets continue with the proof.

$$\begin{aligned} \tilde{v}_D(F) = 1 & \text{ iff } \tilde{v}_D(G) = 1 \wedge \tilde{v}_D(H) = 1 && \text{by definition of } I_{\wedge} \\ & \text{ iff } G \in D \wedge H \in D && \text{by I.H.} \\ & \text{ iff } G \wedge H \in D && \text{by Lemma} \\ & \text{ iff } F \in D \end{aligned}$$

- $F = (G \vee H)$.

Lemma. $G \in D \vee H \in D \text{ iff } G \vee H \in D$

Proof.

- ($\Rightarrow$) Lets suppose by absurdum $G \vee H \notin D$

$$G \vee H \notin D \text{ iff } \neg(G \vee H) \in D \quad \text{by Fact 3}$$

but then assuming that $G \in D$ (its the same otherwise) $\{G, \neg(G \vee H)\} \subseteq_{\omega} D$ contraddicting Fact 2 since this set is obviously UNSAT ⚡

$\leftarrow$ (destruct the assumption like in coq)

- ($\Leftarrow$) Lets suppose by absurdum $\neg(G \in D \vee H \in D)$ that is $\neg G \in D \wedge \neg H \in D$ but then $\{\neg G, \neg H, G \vee H\} \subseteq_{\omega} D$ contraddicting Fact 2 since this set is obviously UNSAT ⚡

□

Now lets continue with the proof.

$$\begin{aligned} \tilde{v}_D(F) = 1 & \text{ iff } \tilde{v}_D(G) = 1 \vee \tilde{v}_D(H) = 1 && \text{by definition of } I_{\vee} \\ & \text{ iff } G \in D \vee H \in D && \text{by I.H.} \\ & \text{ iff } G \vee H \in D && \text{by Lemma} \\ & \text{ iff } F \in D \end{aligned}$$

- Prove implication for fun (remember that $G \rightarrow H$ has the same truth table as $(\neg G) \vee H$)

□

$\leftarrow$ (End of proof of Fact 4)

Since $v_D \models D$ then a fortiori (for stronger reasons) $v_D \models \Gamma$ (the assignment satisfies D , which is larger than Γ , so v_D must satisfy Γ).

So we have proved that:

$$\Gamma \text{ FINSAT} \Leftrightarrow \Gamma \text{ SAT}$$

□

$\leftarrow$ (End of compactness proof)

By **contraposition**:

$$\Gamma \text{ UNSAT} \implies \exists \Gamma' \subseteq_{\omega} \Gamma, \Gamma' \text{ UNSAT}$$

3.2.1 Remarks on Compactness Theorem:

D is “Large” and satisfiable. We **cannot expand D further without losing** the property of D **being satisfiable**.

Proof. First of all, Since D contains no contradiction it's obviously not $\mathcal{F}_{\mathcal{L}}$, so $D \neq \mathcal{F}_{\mathcal{L}}$.

Let $F_t \notin D$ then by Fact 3 $\neg F_t \in D$ and thus given $\overline{D} = D \cup \{F_t\}$ we have that $\{F_t, \neg F_t\} \subseteq_{\omega} \overline{D}$ so $\overline{D}$ is UNSAT.

If I add anything not already in D , the negation of that formula must already be in D so the set cannot remain satisfiable if I add anything else.

□

D is called a **maximal expansion**, with respect to **satisfiability**, of Γ . We call a set E of formulas **maximally satisfiable** if it has the same property of D that is E is a set and for all $F \in \mathcal{F}_{\mathcal{L}}, F \notin E$ the set $E \cup \{F\}$ is UNSAT (if I add anything the set cannot remain satisfiable). D is a maximal expansion of Γ , but there can be **many**, in our proof D is fixed, after fixing the listing of formulas.

If we reason about the **information contained** inside a **maximally satisfiable theory** D we notice that it carries the **same amount of information** as a **truth value assignment** $v_D : \mathcal{L} \rightarrow \{0, 1\}$ defined as $v_D(p) = 1$ **iff** $p \in D$ for all $p \in \mathcal{L}$. We can go the other way around, **starting from any assignment** $v : \mathcal{L} \rightarrow \{0, 1\}$ and **construct the set of formulas** D_v :

$$D_v := \{F \in \mathcal{F}_{\mathcal{L}} : \tilde{v}(F) = 1\}$$

This way

- D_v is a MAXSAT theory
- $v_{D_v} = v$, with v truth assignment $v : \mathcal{L} \rightarrow \{0, 1\}$
- Same thing with: $D_{v_E} = E$ with E a MAXSAT theory

Associated with every MAXSAT theory there is an assignment satisfying the theory ($v_D \models D$), and similarly for every assignment there can be a max sat theory satisfied by that assignment.

How can we tell if:

$$(\Gamma \subseteq \mathcal{F}_{\mathcal{L}}, A \in \mathcal{F}_{\mathcal{L}}) \quad \Gamma \models A?$$

We have already proven in Lemma 3.1.2 that we can **reduce this problem to** the unsatisfiability problem: $\Gamma \cup \{\neg A\}$ **is unsatisfiable**.

Assume Γ is a finite theory (Γ contains finitely many formulas)

$$\Gamma = \{B_1, B_2, \dots, B_u\}$$

$$\Gamma \models A \text{ iff}$$

$$\{B_1, B_2, \dots, B_u\} \models A \text{ iff}$$

$$\{B_1, B_2, \dots, B_u, \neg A\} \text{ is UNSAT (as a theory) iff}$$

$$B_1 \wedge B_2 \wedge \dots \wedge B_u \wedge \neg A \text{ is UNSAT (as a formula) iff} \quad \text{by definition of } I_{\wedge}$$

What if Γ is infinite? The conjunction of infinitely many things it's not a formula and its satisfiability it's not decidable in finite time. But fortunately by compactness theorem, if $\Gamma \cup \{\neg A\}$ is UNSAT then there exists $\Gamma' \subseteq_{\omega} \Gamma \cup \{\neg A\}$ is UNSAT

Good news: we can have a finite theory which is unsatisfiable. Bad news: we don't know how to find it since there are infinitely many but this means that the procedure is not decidable but semidecidable.

3.3 Semantic Equivalence (logical equivalence)

Are these the same?

$$A \wedge B \quad B \wedge A$$

These are syntactically different, they are different strings of symbols but semantically, **for all assignments** $v : \mathcal{L} \rightarrow \{0, 1\}$ it holds that $\tilde{v}(A \wedge B) = \tilde{v}(B \wedge A)$, by the definition of $I_\wedge$. They always **get the same resulting truth values**.

3.3.1 Equivalence relation

What is a relation? An n -ary relation R on a set S is just a subset of the n -th Cartesian power S^n of S :

$$R \subseteq S^n$$

A relation, mathematically, is just this.

Definition 3.3.1 (Equivalence relation). An equivalence relation R on a set S , R is a **binary relation** ($R \subseteq S^2$) such that it **satisfies 3 conditions**:

- R is **reflexive**: $\forall s \in S, (s, s) \in R$
- R is **symmetric**: $\forall s_1, s_2 \in S$ if $(s_1, s_2) \in R$ then also $(s_2, s_1) \in R$
- R is **transitive**: $\forall s_1, s_2, s_3 \in S$ if $(s_1, s_2) \in R$ and $(s_2, s_3) \in R$ then $(s_1, s_3) \in R$

Definition 3.3.2. Let $R \subseteq S^2$ be an equivalence relation, then for each $s \in S$ the **equivalence class** of s , called $[s]_R$ or s/R is **defined as**

$$[s]_R := \{t \in S : (s, t) \in R\}$$

Observe: $(s, t) \in R$ **iff** $[s]_R = [t]_R$.

Definition 3.3.3 (Partitions of a set). A partition of S is a set $\{B_1, \dots, B_k, \dots\} = \{B_i : i \in I\}$ of subsets of S (not necessarily finite, $B_i \subseteq S$) such that:

- $B_i \cap B_j = \emptyset \quad \forall i \neq j \in I$
- $\bigcup_{i \in I} B_i = S$

In simple words a partition is a way of **cutting S into blocks**.

Every **equivalence relation** $R \subseteq S^2$ **determines uniquely a partition** P_R of S

$$P_R = \{[s]_R : s \in S\}$$

We can go the other way around, every **partition** P of S **determines uniquely an equivalence relation** R_P on S ($R_P \subseteq S^2$)

$$R_P : (s, t) \in R_P \text{ **iff** } s, t \text{ belong to the unique same block } B_i \in P$$

3.3.2 Definition of Semantic equivalence

Let us verify the the following **is an equivalence relation on $\mathcal{F}_{\mathcal{L}}$**

$$\equiv \subseteq \mathcal{F}_{\mathcal{L}}^2$$

For each $(A, B) \in \mathcal{F}_{\mathcal{L}}^2$ (or simpler, $A, B \in \mathcal{F}_{\mathcal{L}}$), then $(A, B) \in \equiv$, also written as $A \equiv B$ (A is **logically equivalent to B** iff, $\forall v : \mathcal{L} \rightarrow \{0, 1\}$, $\tilde{v}(A) = \tilde{v}(B)$). Is it an equivalence relation?

- $\equiv$ is **reflexive**: for all $A \in \mathcal{F}_{\mathcal{L}}$, $A \equiv A$, for each $v : \mathcal{L} \rightarrow \{0, 1\}$ $\tilde{v}(A) = \tilde{v}(A)$
- $\equiv$ is **symmetric**: for all $A, B \in \mathcal{F}_{\mathcal{L}}$, $A \equiv B$ implies $B \equiv A$, for all $v : \mathcal{L} \rightarrow \{0, 1\}$: $\tilde{v}(A) = \tilde{v}(B)$ implies $\tilde{v}(B) = \tilde{v}(A)$
- $\equiv$ is **transitive**: for all $A, B, C \in \mathcal{F}_{\mathcal{L}}$, $A \equiv B$ and $B \equiv C$ implies $A \equiv C$, for all $v : \mathcal{L} \rightarrow \{0, 1\}$: $\tilde{v}(A) = \tilde{v}(B)$ and $\tilde{v}(B) = \tilde{v}(C)$ implies $\tilde{v}(A) = \tilde{v}(C)$

$\equiv$ **determines a partition of $\mathcal{F}_{\mathcal{L}}$** . We subdivided the world in such a way that formulas that say the same thing are in the same block

$$\mathcal{F}_{\mathcal{L}}/\equiv = P_{\equiv} = \{[A]_{\equiv} : A \in \mathcal{F}_{\mathcal{L}}\}$$

$[A \wedge B]_{\equiv} = [B \wedge A]_{\equiv}$ says that the formulas are equivalent, which is the same as saying $A \wedge B \equiv B \wedge A$.

Two formulas are **logically equivalent** if they have the **same truth value under any assignment**. In general if $R \subseteq S^2$ is an equivalence relation, we write $S/R = \{[S]_R : s \in S\}$. This is called “ S modulo R ” or “the quotient of S over R ”.

We eventually will enrich the set of logical equivalence classes $\mathcal{F}_{\mathcal{L}}/\equiv$ with operations which shall confer to the enriched set the status of an algebraic structure (namely: Boolean algebra).

3.4 Substitution (letter with a formula)

Substitution of a propositional letter with a formula (inside a formula). We want to get new formula from another one, while remaining semantically equivalent.

For each $A \in \mathcal{F}_{\mathcal{L}}$, for each $p \in \mathcal{L}$, for each $B \in \mathcal{F}_{\mathcal{L}}$, let us define the formula obtained by **replacing in A each occurrence of p with B** , $A[B/p] \in \mathcal{F}_{\mathcal{L}}$. Define it inductively on the definition of formula as follows:

- **Base**: $A = q$, $q \in \mathcal{L}$, define

$$q[B/p] := \begin{cases} B & q = p \\ q & q \neq p \end{cases}$$

- **Step**: cases, if we have to substitute:

$$(\neg A)[B/p] := \neg(A[B/p])$$

$$(A_1 \wedge A_2)[B/p] := A_1[B/p] \wedge A_2[B/p]$$

$$(A_1 \vee A_2)[B/p] := A_1[B/p] \vee A_2[B/p]$$

$$(A_1 \rightarrow A_2)[B/p] := A_1[B/p] \rightarrow A_2[B/p]$$

Lemma 3.4.1 (Substitution). Let $v : \mathcal{L} \rightarrow \{0, 1\}$ be an assignment and $S, T \in \mathcal{F}_{\mathcal{L}}$. If $\tilde{v}(S) = \tilde{v}(T)$, then $\forall A \in \mathcal{F}_{\mathcal{L}}$, and $\forall p \in \mathcal{L}$

$$\tilde{v}(A[S/p]) = \tilde{v}(A[T/p])$$

Proof. By induction on A :

- **Base:** $A = q$, $q \in \mathcal{L}$, two cases:

- $q \neq p$, then $q[S/p] = q = q[T/p]$, then

$$\tilde{v}(q[S/p]) = \tilde{v}(q) = \tilde{v}(q[T/p])$$

- $q = p$, then $q[S/p] = S$ and $q[T/p] = T$, so

$$\tilde{v}(q[S/p]) = \tilde{v}(S) = \tilde{v}(T) = \tilde{v}(q[T/p])$$

- **Step:** cases are:

- $A = \neg B$, by I.H. $\tilde{v}(B[S/p]) = \tilde{v}(B[T/p])$ we have that:

$$\tilde{v}((\neg B)[S/p]) = I_{\neg}(\tilde{v}(B[S/p])) = I_{\neg}(\tilde{v}(B[T/p])) = \tilde{v}((\neg B)[T/p])$$

- $A = (B \wedge C)$, by I.H. $\tilde{v}(B[S/p]) = \tilde{v}(B[T/p])$ and $\tilde{v}(C[S/p]) = \tilde{v}(C[T/p])$ we have that:

$$\tilde{v}((B \wedge C)[S/p]) = I_{\wedge}(\tilde{v}(B[S/p]), \tilde{v}(C[S/p])) = I_{\wedge}(\tilde{v}(B[T/p]), \tilde{v}(C[T/p])) = \tilde{v}((B \wedge C)[T/p])$$

- $A = (B \vee C)$
- $A = (B \rightarrow C)$

The other cases are left as an exercise for the reader.

□

Theorem 3.4.2 (Substitution). Let $S, T \in \mathcal{F}_{\mathcal{L}}$, if $S \equiv T$, then $\forall A \in \mathcal{F}_{\mathcal{L}}$ and $\forall p \in \mathcal{L}$:

$$A[S/p] \equiv A[T/p]$$

Proof.

For all $v : \mathcal{L} \rightarrow \{0, 1\}$ it holds that $\tilde{v}(S) = \tilde{v}(T)$, since $S \equiv T$. So by the Lemma 3.4.1 (substitution), we have $\tilde{v}(A[S/p]) = \tilde{v}(A[T/p])$, and then, by definition of $\equiv$

$$A[S/p] \equiv A[T/p]$$

□

Theorem 3.4.3. Given $F, G, H \in \mathcal{F}_{\mathcal{L}}$ the following are logical equivalences:

- **Idempotence:**

- $F \wedge F \equiv F$
- $F \vee F \equiv F$

- **Commutativity:**

- $F \vee G \equiv G \vee F$
- $F \wedge G \equiv G \wedge F$

- **Associativity:**

- $F \wedge (G \wedge H) \equiv (F \wedge G) \wedge H$
- $F \vee (G \vee H) \equiv (F \vee G) \vee H$

- **Distributivity:**

- $F \wedge (G \vee H) \equiv (F \wedge G) \vee (F \wedge H)$
- $F \vee (G \wedge H) \equiv (F \vee G) \wedge (F \vee H)$

- **Absorption:**

- $F \vee (F \wedge G) \equiv F$
- $F \wedge (F \vee G) \equiv F$

Exercise: prove that idempotence $F \vee F \equiv F$, $F \wedge F \equiv F$ follows only by absorption (using substitution theorem).

- **Double negation** (involutiveness of negation): $\neg\neg F \equiv F$

- **De Morgan:**

- $\neg(F \vee G) \equiv \neg F \wedge \neg G$
- $\neg(F \wedge G) \equiv \neg F \vee \neg G$

- **Interdefinability:**

- $F \rightarrow G \equiv (\neg F) \vee G$
- $F \rightarrow G \equiv \neg(F \wedge \neg G)$
- $F \vee G \equiv (\neg F) \rightarrow G$
- $F \wedge G \equiv \neg(F \rightarrow \neg G)$
- $F \vee G \equiv \neg(\neg F \wedge \neg G)$
- $F \wedge G \equiv \neg(\neg F \vee \neg G)$

Proof. Just one case (the remaining ones are left as an exercise for the reader). We want to show that

$$\neg(F \vee G) \equiv \neg F \wedge \neg G$$

For all $v : \mathcal{L} \rightarrow \{0, 1\}$ $\tilde{v}(\neg(F \vee G)) = 1$

iff by the semantics of $\neg$: $I_{\neg}$: $\tilde{v}(F \vee G) = 0$

iff by the semantics of $\wedge$: $\tilde{v}(F) = 0$ and $\tilde{v}(G) = 0$

iff by the semantics of $\neg$: $\tilde{v}(\neg F) = 1$ and $\tilde{v}(\neg G) = 1$

iff by the semantics of $\wedge$: $\tilde{v}(\neg F \wedge \neg G) = 1$

By bivalence $\neg(F \vee G) \equiv \neg F \wedge \neg G$ (we've proved that it holds for 1, the only other possible value is 0 so we don't need to prove that). $\square$

3.5 Derived connectives

We define these operators as **shortened versions of longer formulas**.

We define $F \leftrightarrow G$, as $(F \rightarrow G) \wedge (G \rightarrow F)$. Specify

$$I_{\leftrightarrow} : \{0, 1\}^2 \rightarrow \{0, 1\}$$

a	b	$a \leftrightarrow b$
0	0	1
0	1	0
1	0	0
1	1	1

Observation: $\models F \leftrightarrow G$ (is a tautology) **iff** $F \equiv G$ **iff** $\models \neg F \leftrightarrow \neg G$ **iff** $\models G \leftrightarrow F$ **iff** $\models \neg G \leftrightarrow \neg F$.

Observation: $F \models G$ and $G \models F$ **iff** $F \equiv G$.

We define other derived connectives $\perp, \top$. Technically they are zero-ary connectives (and as such they are formulas in themselves), they're **logical constants**, don't need arguments to be complete to a formula.

$$\begin{aligned} \perp, \top &\in \mathcal{F}_{\mathcal{L}} \\ I_{\perp} : \{0, 1\}^0 &\rightarrow \{0, 1\} \quad \text{identify with 0} \\ I_{\top} : \{0, 1\}^0 &\rightarrow \{0, 1\} \quad \text{identify with 1} \end{aligned}$$

The $\{0, 1\}^0$ represents the singleton set, so we can **identify the function with the result**. So

$$I_{\perp} := 0 \quad I_{\top} := 1$$

The $\top$ is in the **equivalence class of tautologies**:

$$\top \equiv A \text{ iff } A \text{ is an arbitrary tautology } (\models A), \text{ e.g. } \top \equiv p \rightarrow p$$

And $\perp$ is in the **equivalence class of contradictions**:

$$\perp \equiv A \text{ iff } A \text{ is an arbitrary contradiction } (\models \neg A), \text{ e.g. } \perp \equiv p \wedge \neg p$$

Some things **about these constants**:

$$\neg \top \equiv \perp, \quad \neg \perp \equiv \top$$

If $\models A$

$$\begin{aligned} \neg A &\equiv A \rightarrow \perp \\ A &\equiv \top \rightarrow A \end{aligned}$$

Some other equivalences:

- **Complementation**

$$\begin{aligned} A \wedge \neg A &\equiv \perp \\ A \vee \neg A &\equiv \top \end{aligned}$$

- **Unit and neutral elements**

$$\begin{aligned} A \wedge \perp &\equiv \perp \\ A \vee \perp &\equiv A \\ A \wedge \top &\equiv A \\ A \vee \top &\equiv \top \end{aligned}$$

3.6 $\equiv$ as a congruence

$\equiv$ is **more than** simply being an **equivalence relation**. $\equiv$ is a **congruence** with respect to $\vee, \wedge, \rightarrow, \neg$ **thought as operations** as such:

$$\begin{aligned} \wedge : (\mathcal{F}_{\mathcal{L}}/\equiv)^2 &\rightarrow \mathcal{F}_{\mathcal{L}}/\equiv \\ ([A]_{\equiv}, [B]_{\equiv}) &\rightarrow [A \wedge B]_{\equiv} \end{aligned}$$

If $A \equiv B$ and $C \equiv D$ then $A \wedge B \equiv C \wedge D$.

Exercise: verify that $\equiv$ is a congruence with respect to all the connectives $\wedge, \vee, \rightarrow, \neg$ (also $\top, \perp, \leftrightarrow$ even if they're derived so there would technically be no need).

Proof. We demonstrate just the case of $\equiv$ being a congruence with respect to $\neg$.

$A \equiv B$ **iff** $\neg A \equiv \neg B$. $\forall v : \mathcal{L} \rightarrow \{0, 1\}$, $\tilde{v}(A) = \tilde{v}(B)$ **iff** $I_{\neg}(\tilde{v}(A)) = I_{\neg}(\tilde{v}(B))$ **iff** $\tilde{v}(\neg A) = \tilde{v}(\neg B)$. $\square$

3.7 Partial order relation

Let R be a **binary relation** on S , $R \subseteq S^2$ such that:

- R is reflexive $(s, s) \in R \quad \forall s \in S$
- R is **antisymmetric** $\forall (s, t) \in R$, if $(s, t) \in R$ and $(t, s) \in R$ then $s = t$
- R is transitive if $(s, t) \in R$ and $(t, r) \in R$ then $(s, r) \in R$

← (don't confuse this property with the asymmetric property)

Any relation with these properties is a partial order relation (poset).

← (Symbols: $\leq, \preceq, \sqsubseteq$.)

Some examples of partial orderings for the naturals:

$$(\mathbb{N}, \leq)$$

$a \leq b$ if it exists $c \in \mathbb{N}$ such that $a + b = c$. This relation has a minimum element, which is 0.

$$(\mathbb{N}, \geq)$$

$a \geq b$ if it exists $c \in \mathbb{N}$ such that $a - b = c$. This relation has a maximum element, which is 0.

Both of the previous posets are also **total orders**, since this property holds:

- R is **total** $\forall a, b \in S \quad (a, b) \in R$ or $(b, a) \in R$

Another unituitive poset is this one:

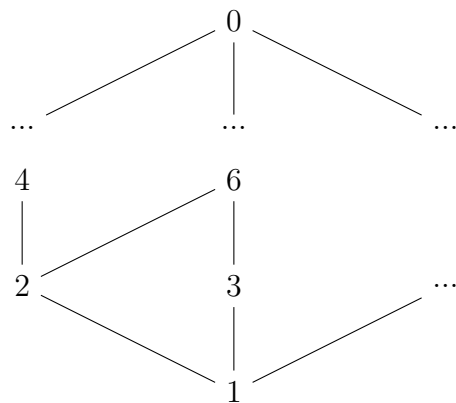
$$(\mathbb{N}, =)$$

In this relation every element is on the same level.

Divisibility order:

$$(\mathbb{N}, |)$$

$a|b$ (a divides b) if there is $c \in \mathbb{N}$ such that $a \cdot c = b$.



This order has 1 as a minimum, all the prime numbers in the first level (atoms), then all numbers created just by primes and so on until the maximal element, which is zero since there's always a number such that $n \cdot 0 = 0$.

A **set** P , equipped with a partial order relation $\leq \subseteq P^2$ is called a **partially ordered set**, for short **poset** $(P, \leq)$.

3.8 Lattice

a **Lattice** (Reticolo) is a poset such that each pair of element in the poset has a unique **greatest lower bound** and a unique **lowest upper bound**. where considering $S \subseteq P$, P poset, $(P, \leq)$.

We can define the **infimum** (greatest lower bound): $\inf S$ is the element $t \in P$ such that $\forall s \in S$ then $t \leq s$ and for all $z \in P$, if $z \leq s$ for all $s \in S$ then $z \leq t$ (t is the greatest lower bound).

← (the infimum for a given set can also not exists)

We can define the **supremum** (lowest upper bound) $\sup S$: is the element $t \in P$ such that $\forall s \in S$ then $s \leq t$ and for all $z \in P$, if $s \leq z$ for all $s \in S$ then $t \leq z$ (t is the lowest upper bound).

← (the supremum for a given set can also not exists)

We can also define:

$$\min S = t \text{ if } t = \inf S \text{ and } t \in S$$

$$\max S = t \text{ if } t = \sup S \text{ and } t \in S$$

Example. of a set lacking sup (and then lacking max)

$$S \subseteq \mathbb{Q} \quad S := \{x : x^2 \leq 2\}$$

This has a sup, but not in the rationals. It would be $\sqrt{2} \in \mathbb{R} \setminus \mathbb{Q}$.

While

$$S' \subseteq \mathbb{Q} \quad S' := \{x : x \leq 1\}$$

Has a sup but not a max, $\sup S' = 1$ but $1 \notin S'$, so $\max S'$ does not exist.

$$S'' \subseteq \mathbb{Q} \quad S'' := \{x : x \leq 1\}$$

$\sup S'' = 1$, $1 \in S''$ so $\max S'' = 1$

Exercise: is our poset a lattice?

$$(\mathbb{N}, \leq)$$

it's a lattice because every pair has an greatest lower bound (the smaller number) and a lowest upper bound (the largest number).

$$(\mathbb{N}, |)$$

Divisibility is a lattice, the inf is the greatest common divisor, while the sup is the lowest common multiple.

$$(\mathbb{N}, =)$$

Is NOT a lattice, every element is on the same level, there is no inf or sup.

3.9 Algebraic Structure

(Very sketchy) Definition of Algebraic Structure: Let S be a set, and $*_1, *_2, \dots, *_n$ a number of operations (can be infinite) on S (that is functions $*_i : S^{\nu(i)} \rightarrow S$, where $\nu(i)$ gives the arity of the operation).

Then $(S, *_1, *_2, \dots, *_n)$ is an **algebraic structure**, or is an algebra.

Example 3.9.1.

$$\begin{array}{ll}
 (\mathbb{Z}, +, -, 0) & \\
 + : \mathbb{Z}^2 \rightarrow \mathbb{Z} & (a, b) \rightarrow a + b \\
 - : \mathbb{Z}^1 \rightarrow \mathbb{Z} & a \rightarrow -a \\
 0 : \mathbb{Z}^0 \rightarrow \mathbb{Z} & 0 \text{ the constant/integer number } 0
 \end{array}$$

The arity of these operations is, respectively: $\nu(+) = 2$, $\nu(-) = 1$, $\nu(0) = 0$.

3.9.1 Isomorphism

Two **structures** are **isomorphic**

$$(S, *_1, *_2, \dots, *_n) \cong (R, \odot_1, \odot_2, \dots, \odot_m)$$

If they mathematically behave in the same manner, i.e. **iff**:

$$\begin{aligned}
 & m = n \\
 & \nu_S^{(i)} = \nu_R^{(i)} \quad \text{for each } i = 1, 2, \dots, n \\
 & \exists f : S \rightarrow R \text{ such as:} \\
 & \left\{ \begin{array}{ll} f : \text{ injective:} & x \neq y \in S \implies f(x) \neq f(y) \\ f : \text{ surjective:} & \forall y \in R \text{ there is } x \in S \text{ s.th. } f(x) = y \end{array} \right. \\
 & \implies f \text{ is bijective (1-1 correspondance)} \\
 & f : \text{ is a homomorphism : } f(*_i(s_1, \dots, s_k)) = \odot_i(f(s_1), f(s_2), \dots, f(s_k)) \\
 & \nu_S^{(i)} = k, \text{ for all } (s_1, \dots, s_k) \in S^k
 \end{aligned}$$

← (guarda slide in piu)

3.9.2 Lattices as algebraic structures

A **lattice as an algebraic structure** is a **set with binary operations**

$$\begin{array}{ll}
 (R, \sqcap, \sqcup) & \sqcap : R^2 \rightarrow R \\
 & \sqcup : R^2 \rightarrow R
 \end{array}$$

such that **both** these **operations** are **associative and commutative**. $\sqcap, \sqcup$ are related by absorption:

$$\forall x, y \in R \quad \begin{cases} x \sqcap (x \sqcup y) = x \\ x \sqcup (x \sqcap y) = x \end{cases}$$

A **lattice as a poset** can be **redefined as/associated** to a **lattice as an algebraic structure**

$$(R, \leq) \overset{\text{unique}}{\longleftrightarrow} (R, \sqcap, \sqcup)$$

each lattice meant as a poset can be thought of uniquely as an algebra and vice versa. from left to right:

$$(R, \leq) \rightsquigarrow (R, \sqcap, \sqcup)$$

we just have to define the operations. For all $a, b \in R$ we need to define:

$$a \sqcap b := \inf\{a, b\}$$

$$a \sqcup b := \sup\{a, b\}$$

While from right to left

$$(R, \sqcap, \sqcup) \rightsquigarrow (R, \leq)$$

For all $a, b \in R$, $a \leq b$ **iff** $a = a \sqcap b$ (or, equivalently, $b = a \sqcup b$).

Check: $a \leq b$ defined as above is such that $\leq$ is:

- reflexive: for all $a \in R$, $a \leq a$ iff $a = a \sqcap a$ (by idempotence)
- antisymmetric: ...
- transitive: ...

We now proved that $(R, \leq)$ is a poset, but the lattice is a special kind of poset. So now we need to check that the poset $(R, \leq)$ is indeed a lattice. We need to prove that **for each** $a, b \in R$ **there is** $\inf\{a, b\}$ **and** $\sup\{a, b\}$.

Letting $c \leq a$ and $c \leq b$ we have to show that $c \leq a \sqcap b$, since it means we've proven that $a \sqcap b = \inf\{a, b\}$, so it exists and is defined, $\inf\{a, b\} = a \sqcap b$.

$$c \underbrace{=}_{\text{def of } c \leq a} = a \sqcap c \underbrace{=}_{\text{def of } c \leq b} a \sqcap (b \sqcap c) \underbrace{=}_{\text{assoc}} (a \sqcap b) \sqcap c \underbrace{\text{iff}}_{\text{def of } \leq} c \leq a \sqcap b$$

The proof that $\sup\{a, b\} = a \sqcup b$ is analogous. We have proved that $(R, \leq)$ is a lattice (as a poset).

3.10 Boolean Algebras

A Boolean Algebra is a particular type of algebra:

$$(S, \sqcap, \sqcup, ()^c, 0, 1)$$

The arity of the operations is:

- $\sqcap, \sqcup : S^2 \rightarrow S$
- $()^c : S \rightarrow S$
- $0, 1 : S^0 \rightarrow S$

← (The $()^c$ is the complement)

Additional properties: $(S, \sqcap, \sqcup)$ is a lattice (has an order), which is:

- **Bounded:** for all $a \in S$ there's a minimum $0 \leq a$ and a maximum $a \leq 1$ such that

$$\begin{cases} a \sqcap 0 = 0 & a \sqcup 1 = 1 \\ a \sqcup 0 = a & a \sqcap 1 = a \end{cases}$$

- **Complemented:** for all $a \in S$, such that $a \sqcap a^c = 0$ and $a \sqcup a^c = 1$.
- **Distributive:** for all $a, b, c \in S$, $a \sqcap (b \sqcup c)$

$$\begin{aligned} a \sqcap (b \sqcup c) &= (a \sqcap b) \sqcup (a \sqcap c) \\ a \sqcup (b \sqcap c) &= (a \sqcup b) \sqcap (a \sqcup c) \end{aligned}$$

So, boolean algebra is a shorthand name for bounded, complemented and distributive lattices.

Example 3.10.1. Let A be a set and let $\mathcal{P}(A)$ be the powerset of A (set of all subsets of A):

$$(\mathcal{P}(A), \cap, \cup, \overline{}, \emptyset, A)$$

Check that this is a boolean algebra. It must be commutative, absorption, bounds, ...

If A is finite, say $|A| = n$, then $|\mathcal{P}(A)| = 2^n$. 2^A is a commonly used notation for the powerset.

Every **finite** Boolean algebra has 2^k elements for some $k \in \mathbb{N}$.

Every **finite** Boolean algebra can be conceived/is **isomorphic with** the **powerset of some finite set**.

$(\mathcal{P}(A), \subseteq)$, forall S, T , $S \subseteq T$ **iff** $S = S \cap T$ (or $T = S \cup T$). This is the lattice isomorphic to the boolean algebra defined above.

Example 3.10.2.

$$A = \{a, b\}, \quad |\mathcal{P}(A)| = |2^A| = 2^2 = 4$$

$$2^A = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

So

$$(2^A, \subseteq)$$

is a lattice.

3.10.1 Boolean algebra of truth-values:

$$(\{0, 1\}, I_\wedge = \min, I_\vee = \max, I_\neg = 1 - x, 0 = I_\perp, 1 = I_\top)$$

This is the smallest significant boolean algebra and in the ordering we have that 0 is below 1.

It's isomorphic to:

$$\cong (\mathcal{P}(\{*\}), \cap, \cup, \overline{}, \emptyset, \{*\})$$

$\leftarrow (\{*\}$ is the singleton set)

Boolean algebras of n -tuples of truth-values:

$$(\{0, 1\}^n, \overline{}, \min, \max, \bar{1} - x, (0, 0, \dots, 0), (1, 1, \dots, 1))$$

Operations are defined pointwise (componentwise):

$$\overline{\min}((b_1, \dots, b_n), (c_1, \dots, c_n)) := (\min\{b_1, c_1\}, \dots, \min\{b_n, c_n\})$$

$$\overline{\max}((b_1, \dots, b_n), (c_1, \dots, c_n)) := (\max\{b_1, c_1\}, \dots, \max\{b_n, c_n\})$$

$$\bar{1} - (b_1, \dots, b_n) = (1 - b_1, \dots, 1 - b_n)$$

3.10.2 Lindembaum Algebras of (equivalence classes of) formulas

It is a boolean algebra.

$$(\mathcal{F}_L / \equiv, \wedge, \vee, \neg, [\perp]_\equiv, [\top]_\equiv)$$

Since $\equiv$ is a congruence with respect to $\wedge, \vee, \neg$ the following **definition is well given** (does not depend on the representatives of the classes). For all $A, B \in \mathcal{F}_L$:

$$[A]_\equiv \wedge [B]_\equiv := [A \wedge B]_\equiv$$

$$[A]_\equiv \vee [B]_\equiv := [A \vee B]_\equiv$$

$$\neg[A]_\equiv := [\neg A]_\equiv$$

It has an **order relation** since it's also a **lattice**:

$$(\mathcal{F}_{\mathcal{L}}/\equiv, \leq) \quad [A]_{\equiv} \leq [B]_{\equiv} \text{ iff } [A]_{\equiv} = [A \wedge B]_{\equiv} \text{ or } [B]_{\equiv} = [A \vee B]_{\equiv}$$

Some combinatorics about $\mathcal{F}_{\mathcal{L}}/\equiv$

$$|\mathcal{F}_{\mathcal{L}}| = |\mathcal{L}| = |\mathbb{N}| = \aleph_0$$

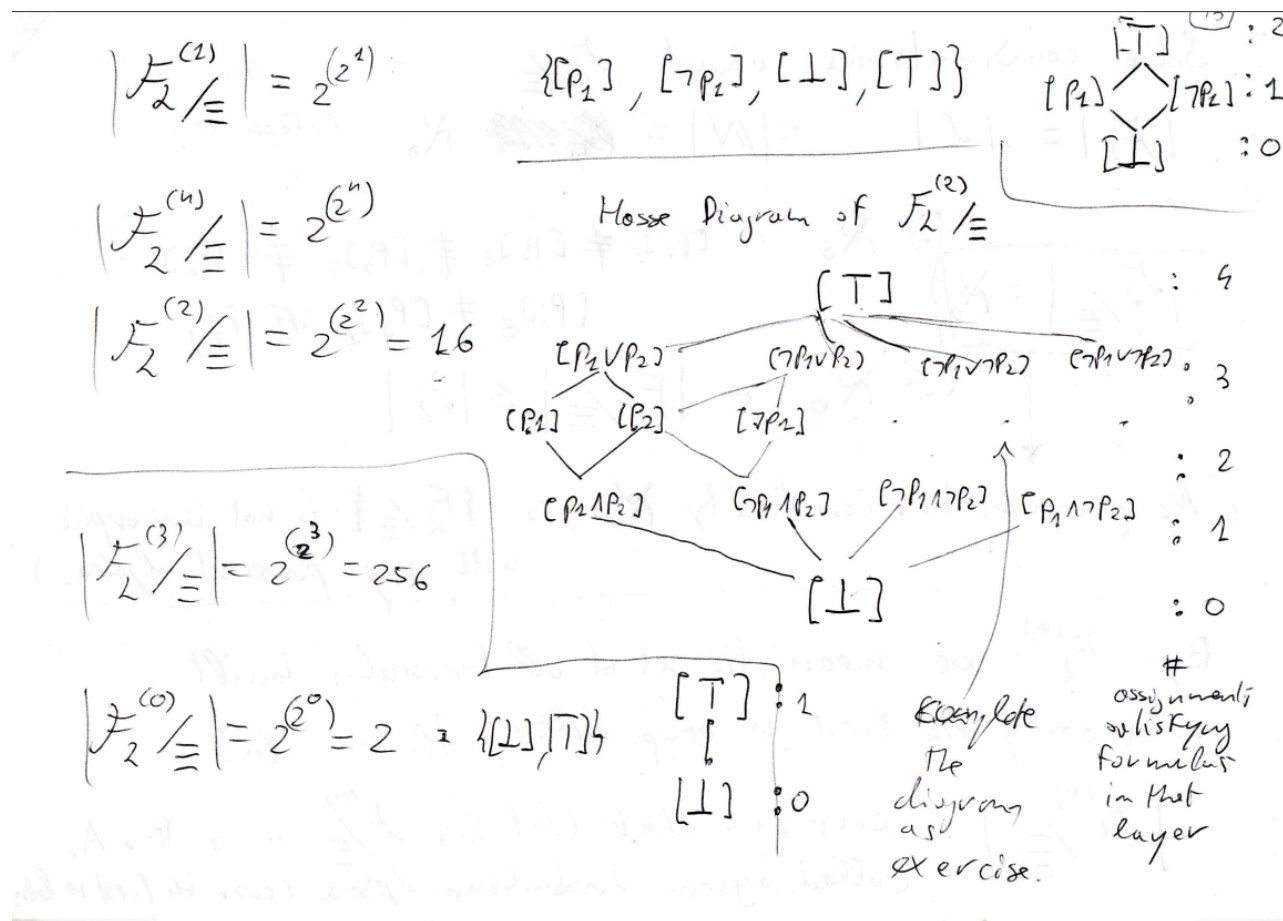
$$|\mathcal{F}_{\mathcal{L}}/\equiv| = \aleph_0$$

No powerset has cardinality $\aleph_0$, so $|\mathcal{F}_{\mathcal{L}}/\equiv|$ is **not isomorphic** with any powerset algebra.

By $\mathcal{F}_{\mathcal{L}}^{(n)}$ we mean the set of all formulas built using only the first n propositional letters $\{p_1, p_2, \dots, p_n\}$. Check that

$$|\mathcal{F}_{\mathcal{L}}^{(n)}/\equiv|$$

is a boolean algebra (called Lindenbaum algebra over the first n letters).



First exercise: taking

$$(\mathcal{F}_{\mathcal{L}}/\equiv, \leq)$$

thought as a poset, we know that

$$[A]_{\equiv} \leq [B]_{\equiv}$$

$$\text{iff } [A]_{\equiv} = [A \wedge B]_{\equiv}$$

$$\text{iff } A \equiv A \wedge B$$

$$\text{iff } \models A \leftrightarrow (A \wedge B)$$

$$\text{iff } \models A \rightarrow B$$

Looking at the truth table, we can see that it's structured in such a way that no assignment can make $A \rightarrow B$ a tautology.

Exercise: $F^{(n)}/\equiv$ is isomorphic with

$$2^{2^n} \cong (\mathcal{P}(\mathcal{P}(\dots)), \cap, \cup, ()^c, \emptyset, \dots)$$

Some other notable logically equivalent formulas:

- **De Morgan Laws on n many variables:**

$$\neg(F_1 \vee F_2 \vee \dots \vee F_n) \equiv \neg F_1 \wedge \neg F_2 \wedge \dots \wedge \neg F_n$$

$$\neg(F_1 \wedge F_2 \wedge \dots \wedge F_n) \equiv \neg F_1 \vee \neg F_2 \vee \dots \vee \neg F_n$$

- **Generalized distributivity:**

$$(F_1 \vee \dots \vee F_u) \wedge (G_1 \vee \dots \vee G_v) \equiv \bigvee_{i=1}^u \bigvee_{j=1}^v (F_i \wedge G_j)$$

$$(F_1 \wedge \dots \wedge F_u) \vee (G_1 \wedge \dots \wedge G_v) \equiv \bigwedge_{i=1}^u \bigwedge_{j=1}^v (F_i \vee G_j)$$

3.11 Term Function

Motivations: $\models F$ **iff** (by definition) for all $v : \mathcal{L} \rightarrow \{0, 1\}$ it holds that $\tilde{v}(F) = 1$.

We Desire that:

- the **formula** “becomes” the **function**
- the **assignments** “become” the **arguments**

We fix a **formula** $A \in \mathcal{F}_{\mathcal{L}}$ and let $Var(A)$ be the set of **propositional letters actually occurring** in A (since assignments are infinite by definition). Now fix a **suitable natural number** $n \in \mathbb{N}$ such that $Var(A) \subseteq \{p_1, p_2, \dots, p_n\}$ (just take n “large enough” for A).

Desideratum (what we want): we want to **define a function** $\hat{A} : \{0, 1\}^n \rightarrow \{0, 1\}$ **such that** for all assignments $v : \mathcal{L} \rightarrow \{0, 1\}$:

$$\hat{A}(v(p_1), v(p_2), \dots, v(p_n)) = \tilde{v}(A)$$

The **formula** has **become a function**, the **assignment** has become the **argument**. Sometimes it's more convenient.

Definition: of **term function** $\hat{A}$ associated with the formula $A \in \mathcal{F}_{\mathcal{L}}$ with respect to $n \in \mathbb{N}$ ($\text{Var}(A) \subseteq \{p_1, \dots, p_n\}$).

We define this by **induction on the structure of A** :

- **Base:** $A = p_i, p_i \in \{p_1, \dots, p_n\}$

$$\hat{A} : \{0, 1\}^n \rightarrow \{0, 1\}$$

$$\hat{p}_i : \{0, 1\}^n \rightarrow \{0, 1\}$$

for all $(t_1, \dots, t_n) \in \{0, 1\}^n$

$$\hat{p}_i(t_1, \dots, t_n) := t_i$$

so it satisfies the desideratum:

$$\hat{p}_i(v(p_1), \dots, v(p_i), \dots, v(p_n)) = v(p_i) = \tilde{v}(p_i)$$

- **Steps:** we have to define the cases:

$$- A = \neg B$$

$$\hat{A}(t_1, \dots, t_n) = (\neg \hat{B})(t_1, \dots, t_n) := I_{\neg}(\hat{B}(t_1, \dots, t_n)) = 1 - (\hat{B}(t_1, \dots, t_n))$$

$$- A = B \wedge C$$

$$\hat{A}(t_1, \dots, t_n) = (B \hat{\wedge} C)(t_1, \dots, t_n) := I_{\wedge}(\hat{B}(t_1, \dots, t_n), \hat{C}(t_1, \dots, t_n))$$

$$- A = B \vee C$$

$$\hat{A}(t_1, \dots, t_n) = (B \hat{\vee} C)(t_1, \dots, t_n) := I_{\vee}(\hat{B}(t_1, \dots, t_n), \hat{C}(t_1, \dots, t_n))$$

$$- A = B \rightarrow C$$

$$\hat{A}(t_1, \dots, t_n) = (B \hat{\rightarrow} C)(t_1, \dots, t_n) := I_{\rightarrow}(\hat{B}(t_1, \dots, t_n), \hat{C}(t_1, \dots, t_n))$$

$$- \top$$

$$\hat{\top}(t_1, \dots, t_n) = 1$$

$$- \perp$$

$$\hat{\perp}(t_1, \dots, t_n) = 0$$

3.11.1 Boolean algebra of term functions

$$\hat{\mathcal{F}}_{\mathcal{L}}^{(n)} := \left(\left\{ \hat{F} : \{0, 1\}^n \rightarrow \{0, 1\} : F \in \mathcal{F}_{\mathcal{L}}^{(n)} \right\}, \hat{\wedge}, \hat{\vee}, \hat{\neg}, \hat{\top}, \hat{\perp} \right)$$

where

$$(\hat{F} \hat{\wedge} \hat{G})(t_1, \dots, t_n) = I_{\wedge}\{\hat{F}(t_1, \dots, t_n), \hat{G}(t_1, \dots, t_n)\}$$

$$(\hat{F} \hat{\vee} \hat{G})(t_1, \dots, t_n) = I_{\vee}\{\hat{F}(t_1, \dots, t_n), \hat{G}(t_1, \dots, t_n)\}$$

$$\hat{\neg}\hat{F}(t_1, \dots, t_n) = I_{\neg}(\hat{F}(t_1, \dots, t_n))$$

We can prove that $\hat{\mathcal{F}}_{\mathcal{L}}^{(n)}$ is **isomorphic** with $\mathcal{F}_{\mathcal{L}}^{(n)} / \equiv$ ($\hat{\mathcal{F}}_{\mathcal{L}}^{(n)} \cong \mathcal{F}_{\mathcal{L}}^{(n)} / \equiv$) via: $\hat{F} \mapsto [F]_{\equiv}$ for each $F \in \mathcal{F}_{\mathcal{L}}$.

$$\hat{\mathcal{F}}_{\mathcal{L}}^{(n)} = \left(\{ \hat{F} : \{0, 1\}^n \rightarrow \{0, 1\} : F \in \mathcal{F}_{\mathcal{L}} \}, \hat{\wedge}, \hat{\vee}, \hat{\neg}, \hat{\top}, \hat{\perp} \right)$$

Compare with

$$Func^{(n)} = (\{g : \{0, 1\}^n \rightarrow \{0, 1\}\}, \hat{\wedge}, \hat{\vee}, \hat{\neg}, \hat{\top}, \hat{\perp})$$

Check that $Func^{(n)}$ is a **boolean algebra**.

This

$$\mathcal{F}_{\mathcal{L}}^{(n)} \subseteq Func^{(n)}$$

is trivially true since each term function belongs to $Func^{(n)}$, while

$$\mathcal{F}_{\mathcal{L}}^{(n)} \supseteq Func^{(n)}$$

is also true, called **functional completeness of propositional logic**.

So

$$\mathcal{F}_{\mathcal{L}}^{(n)} = Func^{(n)}$$

They're not isomorphic, it's **literally the same algebra**, they have the same universe and same operations.

3.12 Functional Completeness Theorem

Theorem 3.12.1. We have to prove that **for each natural number** $n \in \mathbb{N}$, the set of function $Func^{(n)} \subseteq \hat{\mathcal{F}}_{\mathcal{L}}^{(n)}$. That is, for each function $g : \{0, 1\}^n \rightarrow \{0, 1\}$ there exists a formula $F \in \mathcal{F}_{\mathcal{L}}^{(n)}$ such that $g = \hat{F}$ (that is, for each n -tuple $(t_1, \dots, t_n) \in \{0, 1\}^n$, $g(t_1, \dots, t_n) = \hat{F}(t_1, \dots, t_n)$).

Proof. Let $g : \{0, 1\}^n \rightarrow \{0, 1\}$, let us build $F \in \mathcal{F}_{\mathcal{L}}^{(n)}$ such that $g = \hat{F}$. We can build this truth table:

t_1	t_2	$\dots$	t_n	$g(t_1, \dots, t_n)$
0	0	$\dots$	0	b_0
0	0	$\dots$	1	b_1
		$\vdots$		$\vdots$
a_{i1}	a_{i2}	$\dots$	a_{in}	b_i
		$\vdots$		$\vdots$
1	1	$\dots$	1	b_{2^n-1}

Osservation. If $g(t_1, \dots, t_n) = 0$ for all $(t_1, \dots, t_n) \in \{0, 1\}^n$ (g always 0), we can take $\perp = F$

$$\hat{F} = \hat{\perp} = 0 = g$$

Otherwise, at least a bit is 1, let $I \subseteq \{0, 1, \dots, 2^n - 1\}$ be defined by $i \in I$ **iff** $b_i = 1$, by our earlier observation $I \neq \emptyset$.

For all $i \in I$ let us consider the i -th row of the table:

$$a_{i1} \ a_{i2} \ \dots \ a_{in} \mid b_i = 1 \quad \leftarrow (b_i = 1 \text{ since } i \in I.)$$

For each $j \in \{1, \dots, n\}$ we set

$$\begin{cases} A_{ij} := p_j & \text{if } a_{ij} = 1 \\ A_{ij} := \neg p_j & \text{if } a_{ij} = 0 \end{cases}$$

We set

$$F := \bigvee_{i \in I} \bigwedge_{j=1}^n A_{ij} \quad \leftarrow (\text{note } F \in \mathcal{F}_{\mathcal{L}}^{(n)}.)$$

We now have to prove that $\hat{F} = g$.

It's enough to prove that for each $i \in I$:

$$\left. \begin{array}{c} \widehat{\left(\bigwedge_{j=1}^n A_{ij} \right)}(t_1, \dots, t_n) = 1 \\ \text{iff} \\ (t_1, \dots, t_n) = (a_{i1}, a_{i2}, \dots, a_{in}) \end{array} \right\} (\dagger)$$

Proof of $(\dagger)$: Clearly

$$(A_{i1} \wedge \widehat{A_{i2} \wedge \dots \wedge A_{in}})(a_{i1}, a_{i2}, \dots, a_{in}) = 1$$

For convenience:

$$(A_{i1} \wedge A_{i2} \wedge \dots \wedge A_{in}) = \bigwedge_{j=1}^n A_{ij}$$

Since, by definition of A_{ij}

$$\hat{A}_{ij}(a_{i1}, a_{i2}, \dots, a_{in}) = \begin{cases} a_{ij} & \text{if } a_{ij} = 1 \\ 1 - a_{ij} & \text{if } a_{ij} = 0 \end{cases}$$

we have that $\hat{A}_{ij}(a_{i1}, \dots, a_{in}) = 1$, So.

$$\widehat{\left(\bigwedge_{j=1}^n A_{ij} \right)}(a_{i1}, a_{i2}, \dots, a_{in}) = 1$$

By definition of $I_\wedge$.

Let now us take $(t_1, \dots, t_n) \in \{0, 1\}^n$, with $(t_1, \dots, t_n) \neq (a_{i1}, a_{i2}, \dots, a_{in})$ (so $(t_1, \dots, t_n)$ is not the i th row). Let $j \in \{1, \dots, n\}$ be such that $t_j \neq a_{ij}$ (there must be at least one position in which they are different, otherwise they would be the same row)

$$\begin{cases} t_j = 1 & \text{iff } a_{ij} = 0 \\ t_j = 0 & \text{iff } a_{ij} = 1 \end{cases}$$

then

$$\hat{A}_{ij}(t_1, \dots, t_n) = \begin{cases} t_j = 1 - a_{ij} = 0 & \text{iff } t_j = 0 \text{ iff } a_{ij} = 1 \\ 1 - t_j = a_{ij} = 0 & \text{iff } t_j = 1 \text{ iff } a_{ij} = 0 \end{cases}$$

So

$$\widehat{\left(\bigwedge_{j=1}^n A_{ij} \right)}(t_1, \dots, t_n) = 0$$

By definition of $I_\wedge$. So we proved $(\dagger)$.

So, by $(\dagger)$

$$\widehat{\left(\bigwedge_{j=1}^n A_{ij} \right)}(t_1, \dots, t_n) = 1 \text{ iff } t_1, \dots, t_n \text{ is the } i\text{th row}$$

So

$$\widehat{\left(\bigvee_{i \in I} \bigwedge_{j=1}^n A_{ij} \right)}(t_1, \dots, t_n) = 1 \text{ iff } (t_1, \dots, t_n) \text{ is the } i\text{th row for some } i \in I \text{ iff } g(t_1, \dots, t_n) = 1. \quad \leftarrow \text{ (by definition of } I)$$

So we have concluded that F , defined as:

$$F := \bigvee_{i \in I} \bigwedge_{j=1}^n A_{ij}$$

is such that

$$\hat{F}(t_1, \dots, t_n) = g(t_1, \dots, t_n)$$

for all $(t_1, \dots, t_n) \in \{0, 1\}^n$, so

$$\hat{F} = g$$

□

Exercise: prove that for any $f : \{0, 1\}^n \rightarrow \{0, 1\}$ there is a formula $G \in \mathcal{F}_{\mathcal{L}}^{(n)}$ such that $\hat{G} = f$ and

$$G = \bigwedge_{i \in J} \bigvee_{j=1}^n B_{ij}$$

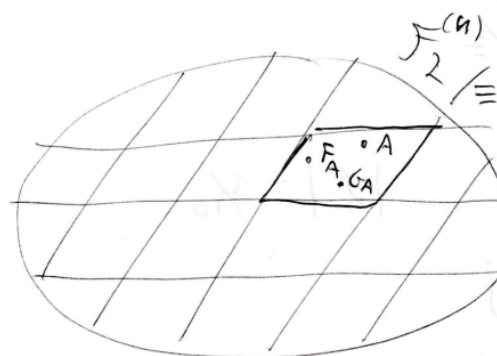
where $J \subseteq \{0, \dots, 2^n - 1\}$ such that $i \in J$ iff $b_j = 0$ and

$$\begin{cases} B_{ij} = \neg p_j & \text{if } a_{ij} = 1 \\ B_{ij} = p_j & \text{if } a_{ij} = 0 \end{cases}$$

Dual of before, just switch everything.

The functional completeness theorem
is a normal form theorem.

Given $f : \{0, 1\}^n \rightarrow \{0, 1\}$ $F = \bigvee_{i \in I} \bigwedge_{j=1}^n A_{ij}$ $G = \bigwedge_{i \in J} \bigvee_{j=1}^n B_{ij}$
are such that $\hat{F} = f = \hat{G}$.



$A \in \mathcal{F}_2^{(n)}$, then there are
 $F_A = \bigvee \bigwedge A_{ij}$ and
 $G_A = \bigwedge \bigvee B_{ij}$
such that

$$\hat{F}_A = \hat{A} = \hat{G}_A$$

$$\text{that is } [F_A]_{\equiv} = [A]_{\equiv} = [G_A]_{\equiv}$$

$$\text{that is } F_A \equiv A \equiv G_A$$

Normal Form: picking a "standard" element in every
one of the equivalence classes.

Observation: The functional completeness theorem **does not hold** for the **set of all formulas** $\mathcal{F}_{\mathcal{L}}$. It **works only when we restrict** to formulas and functions on **the first n propositional letters**, for any $n \in \mathbb{N}$

$$Func^{(n)} = \hat{\mathcal{F}}_{\mathcal{L}}^{(n)} \cong \mathcal{F}_{\mathcal{L}}^{(n)} / \equiv$$

← (Functional completeness theorem proves the first =.)

If we consider all propositional letters (remember $|\mathcal{L}| = \aleph_0$)

$$Func^{(\omega)} = \{f : \{0, 1\}^\omega \rightarrow \{0, 1\}\}$$

But $\{0, 1\}^\omega$ is the set of all infinitely long sequences of bits.

$$|Func^{(\omega)}| = |[0, 1] \cap \mathbb{R}| = |\mathbb{R}| = 2^{\aleph_0} = \aleph_1 > \aleph_0 = |\mathcal{F}_{\mathcal{L}} / \equiv|$$

The normal forms:

- **Conjunctive**

$$\bigwedge_{i \in J} \bigvee_{j=1}^n B_{ij}$$

- **Disjunctive**

$$\bigvee_{i \in I} \bigwedge_{j=1}^n A_{ij}$$

Are said **canonical** or **complete**, conjunctive/disjunctive normal forms, that is every one of the disjuncts/conjuncts **mentions all the letters** $p_1, p_2, \dots, p_n$.

BY HAND:		CANONICAL																																					
Example $Func^{(3)}$ p_1, p_2, p_3 $f: \{0, 1\}^3 \rightarrow \{0, 1\}$ For each (t_1, t_2, t_3) $f(t_1, t_2, t_3) = t_1$ ↑ projection on the 1st component	$\hat{F} = p_1$ observe $\hat{p}_1(t_1, t_2, t_3) = t_1$ so $\hat{p}_1 = f$	<table border="1"> <thead> <tr> <th>p_1</th><th>p_2</th><th>p_3</th><th>f</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>0</td><td>1</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td><td>1</td></tr> </tbody> </table>	p_1	p_2	p_3	f	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	1	1	0	1	1	1	1	0	1	1	1	1	1	$F :=$ $(p_1 \wedge \neg p_2 \wedge \neg p_3) \vee$ $(p_1 \wedge \neg p_2 \wedge p_3) \vee$ $(p_1 \wedge p_2 \wedge \neg p_3) \vee$ $(p_1 \wedge p_2 \wedge p_3)$
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3.13 Functionally complete set of connectives

A set $C \subseteq \{\wedge, \vee, \rightarrow, \neg, \leftrightarrow, \top, \perp, \dots\}$ (subset of connectives) is **functionally complete** iff for all $F \in \mathcal{F}_{\mathcal{L}}^{(n)}$ (for all $n \in \mathbb{N}$) there is some $G \in \mathcal{F}_{\mathcal{L}}^{(n)}$ written using only the connectives in C such that $\hat{F} = \hat{G}$ (or $F \equiv G$, or $[F]_{\equiv} = [G]_{\equiv}$, or $\models F \leftrightarrow G$, they're all equivalent).

Some examples of functional complete sets are:

- $\{\wedge, \vee, \neg\}$: is **functionally complete**, by the **functional completeness theorem**.
- $\{\neg, \wedge\}$: same as above, use De Morgan laws to obtain this set
- $\{\neg, \vee\}$: also De Morgan
- $\{\neg, \rightarrow\}$: can be proven remembering that $A \wedge B \equiv (\neg A) \rightarrow B$
- $\{\rightarrow, \perp\}$: since $\neg A \equiv A \rightarrow \perp$

NAND: Operator defined as negation of AND, it's usually defined as:

$$A \text{ NAND } B := \neg(A \wedge B)$$

$$I_{\text{NAND}} = \{0, 1\}^2 \rightarrow \{0, 1\}$$

$$I_{\text{NAND}}(x, y) = 1 - \min\{x, y\}$$

A	B	$\neg$	$(A \wedge B)$	NAND
0	0	1	0	1
0	1	1	0	1
1	0	1	0	1
1	1	0	1	0

Recalling that $\{\neg, \wedge\}$ is functionally complete, the set only composed by **NAND** is **functionally complete**:

$$\begin{cases} \neg A \equiv A \text{ NAND } A \\ A \wedge B \equiv \neg(A \text{ NAND } B) \equiv (A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B) \end{cases}$$

3.14 Normal Forms

EXERCISES

We have seen canonical/complete normal forms (CNF, DNF) and they are **generally too long**. We want to produce CNFs and DNFs without their being canonical, in order to obtain shorter formulas.

3.14.1 Implication Free Normal Form IFNF:

We're getting rid of implication.

Definition 3.14.1. A Formula $F \in \mathcal{F}_{\mathcal{L}}$ is in IFNF ($F \in IFNF$) iff F does not contain any occurrence of $\rightarrow$.

3.14.2 Negation Normal Form NNF

Definition 3.14.2. A formula $F \in IFNF$ is in NNF ($F \in NNF$) iff the **negation connective** is applied **only to propositional letters** (if $\neg B$ is a sub-formula of F , then $B = p_i$ for some $p_i \in \mathcal{L}$).

Example 3.14.3. Some Examples:

- Is $p_1 \wedge \neg(p_2 \vee p_3) \in \text{NNF}$? Obviously no, but its equivalent $p_1 \wedge (\neg p_2 \wedge \neg p_3)$
- Is $p_1 \rightarrow \neg p_2 \in \text{NNF}$? No, since it's not $\in IFNF$ (the negation is only on letters though)

Some interesting relations are:

$$\begin{array}{l} CNF \\ DNF \end{array} \subseteq NNF \subseteq IFNF$$

Notation: $G \preceq F$, G is a **Subformula** of F (for $G, F \in \mathcal{F}_{\mathcal{L}}$) if G occurs in every L -construction of F (you need G to build F , get the minimal L -construction and check if G is in there).

Transformations: We shall use the following equivalences to **replace subformulas** with **equivalent ones**:

1. $C \rightarrow D \equiv (\neg C) \vee D$
2. $\neg\neg C \equiv C$
3. $\neg(C \vee D) \equiv \neg C \wedge \neg D$ (De Morgan)
4. $\neg(C \wedge D) \equiv \neg C \vee \neg D$ (De Morgan)

We shall use these logical equivalences to transform subformulas of a given formula $F \in \mathcal{F}_{\mathcal{L}}$ using these from left to right, we're **using them as transformation rules** (we could do the opposite, but goes against our normal forms).

Lemma 3.14.4. For any $F \in \mathcal{F}_{\mathcal{L}}$ we can build another formula $G \in \mathcal{F}_{\mathcal{L}}$ (in finite number of steps) such that $F \equiv G$ and $G \in IFNF$. Any formula can be “transformed” into an equivalent one in IFNF.

Proof. For any $E \in \mathcal{F}_{\mathcal{L}}$ let $i(E)$ be the number of occurrences of $\rightarrow$ in E . We can observe that $E \in IFNF$ iff $i(E) = 0$. If $E \notin IFNF$ then $i(E) > 0$.

1. Take $C \rightarrow D \preceq E$ (which exists for any E with $i(E) > 0$).
2. Replace $C \rightarrow D$ with $(\neg C) \vee D$ and call E' the formula obtained.

$$E' = E [(\neg C) \vee D / C \rightarrow D]$$

(with a small abuse of notation, I think we defined the substitution only for letters but I think it's admissible).

Now $E \equiv E'$ for the transformation at step (2) thanks to the substitution theorem and now we have

$$i(E) > i(E')$$

since we removed at least one implication (possibly more than one).

So, given in input $F \in \mathcal{F}_{\mathcal{L}}$ with $F \notin IFNF$, we build a sequence

$$F = F_0 \equiv F_1 \equiv \dots \equiv F_u \in IFNF \quad (\dagger)$$

where

$$i(F_0) > i(F_1) > \dots > i(F_u) = 0$$

Since we're strictly descending natural numbers since each F_{i+1} is obtained from F_i by substituting some $C \rightarrow D \preceq F$ with $(\neg C) \vee D$. Trivially, the sequence $(\dagger)$ ends in a finite number of steps u with $u \leq i(F_0) = i(F)$, producing a formula F_u such that $F_u \equiv F$, $F_u \in IFNF$. $\square$

Lemma 3.14.5. for any $F \in IFNF$ we can build, in a finite number of steps, a **sequence**

$$F = F_0 \equiv F_1 \equiv \dots \equiv F_v = G$$

such that $G \equiv F$ and $G \in NNF$.

Proof. For any $A \in IFNF$ let $\ell(A) :=$ the **number of** occurrences of **connectives** $\{\wedge, \vee, \neg\}$ in A .

For any $E \in IFNF$ let

$$m(E) := \sum_{\neg A \preceq E} \ell(A)$$

Lets prove that $m(E) = 0$ iff $E \in NNF$.

- ($\Rightarrow$) If $m(E) = 0$ then by definition we have $\sum_{\neg A \preceq E} 0$ then $\ell(A) = 0$ for all $\neg A \preceq E$, then $A = p$ for some $p \in \mathcal{L}$, then $E \in NNF$ (by definition of NNF).
- ($\Leftarrow$) If $E \in NNF$ then $\neg A \preceq E$ iff $A = p$ for some $p \in \mathcal{L}$, so if A is a propositional letter, it contains no connectives $\ell(A) = 0$, so $m(E) = \sum_{\neg A \preceq E} 0 = 0$.

We now have to show that each application of our transformations (we list here) is such that one step from F_i to F_{i+1} has $m(F_i) > m(F_{i+1})$.

Cases:

- **Transformation:** $\neg\neg C \rightsquigarrow C$
Using $\neg\neg C \preceq F_i$ we need to prove

$$m(\neg\neg C) > m(C)$$

By definition:

$$m(C) = \sum_{\neg A \preceq C} \ell(A)$$

So:

$$\begin{aligned} m(\neg\neg C) &= \sum_{\neg A \preceq \neg\neg C} \ell(A) = \ell(\neg C) + \sum_{\neg A \preceq \neg C} \ell(A) \\ &= \ell(C) + 1 + \ell(C) + \sum_{\neg A \preceq C} \ell(A) \end{aligned}$$

So $m(\neg\neg C) > m(C)$ by at least 1.

- **Transformation:** $\neg(C \wedge D) \rightsquigarrow (\neg C \vee \neg D)$
Using $\neg(C \wedge D) \preceq F_i$ we need to prove:

$$m(\neg(C \wedge D)) > m(\neg C \vee \neg D)$$

By definition:

$$m(\neg C \vee \neg D) = \sum_{\neg A \preceq \neg C \vee \neg D} \ell(A) = \ell(C) + \ell(D) + \sum_{\neg A \preceq C} \ell(A) + \sum_{\neg A \preceq D} \ell(A)$$

So:

$$\begin{aligned} m(\neg(C \wedge D)) &= \sum_{\neg A \preceq \neg(C \wedge D)} \ell(A) = \ell(C \wedge D) && + \sum_{\neg A \preceq C} \ell(A) + \sum_{\neg A \preceq D} \ell(A) \\ &= \ell(C) + 1 + \ell(D) && + \sum_{\neg A \preceq C} \ell(A) + \sum_{\neg A \preceq D} \ell(A) \end{aligned}$$

then $m(\neg(C \wedge D)) > m(\neg C \vee \neg D)$.

- **Transformation:** $\neg(C \vee D) \rightsquigarrow (\neg C \wedge \neg D)$

Using $\neg(C \vee D) \preceq F_i$ we need to prove:

$$m(\neg(C \vee D)) > m(\neg C \wedge \neg D)$$

Same as before, just switch symbols.

□

Observe: if $G \preceq F$, going from G to H and subsequently from F to F' , $F' = F[H/G]$

$$\text{if } m(G) > m(H) \text{ then } m(F) > m(F')$$

As a matter of fact:

$$\begin{aligned} m(F) &= \sum_{\neg A \preceq F} \ell(A) = \sum_{\neg A \preceq G} \ell(A) + \sum_{\neg A \preceq F, \neg A \not\preceq G} \ell(A) = m(G) + \sum_{\neg A \preceq F, \neg A \not\preceq G} \ell(A) \\ m(F') &= \sum_{\neg A \preceq F'} \ell(A) = \sum_{\neg A \preceq H} \ell(A) + \sum_{\neg A \preceq F, \neg A \not\preceq H} \ell(A) = m(H) + \sum_{\neg A \preceq F, \neg A \not\preceq H} \ell(A) \\ &\implies m(F) > m(F') \end{aligned}$$

We have, by the proofs of the previous lemmas, efficient (linear time) algorithms to put a formula $F \in F_h$ into an equivalent one in IFNF and Len NNF.

$$F \xrightarrow{\text{efficiently}} F^{IF} \equiv F \xrightarrow{\text{efficiently}} F^N \equiv F \quad \left(\begin{array}{l} F^{IF} \in \text{IFNF} \\ F^N \in \text{NNF} \end{array} \right).$$

(at most $i(F) + m(F)$ steps)

Example

$$F_0 = F$$

$$, p, q, r \in \mathcal{L}$$

✓

$$F_0: \neg[(p \vee q) \rightarrow r] \wedge \neg(q \vee r)$$

$$i(F_0) = 1$$

$$F_1: \neg[(\neg(p \vee q) \vee r) \wedge \neg(q \vee r)]$$

$$i(F_1) = 0, m(F_1) = 6 + 1 + 1 = 8$$

$\in \text{IFNF}$

$$(4) F_2: (\neg(\neg(p \vee q)) \vee r) \vee \neg\neg(q \vee r)$$

$$m(F_2) = 3 + 1 + 2 + 1 = 7$$

$$(3) F_3: (\neg\neg(p \vee q) \wedge \neg r) \vee \neg\neg(q \vee r)$$

$$m(F_3) = 2 + 1 + 0 + 2 + 1 = 6$$

$$(2) F_4: ((p \vee q) \wedge \neg r) \vee \neg\neg(q \vee r)$$

$$m(F_4) = 0 + 2 + 1 = 3$$

$$(2) F_5: ((p \vee q) \wedge \neg r) \vee (q \vee r)$$

$$m(F_5) = 0$$

$\in \text{NNF}$

3.14.3 Conjunctive Normal Form CNF

Terminology:

- **Definition: Literal**, by literal we mean a formula in the form p or $\neg p$ for some $p \in \mathbb{L}$. A letter or a negation of one.

- **Definition: Opposite Literal**, if A is a literal, then we denote by $\overline{A}$, called opposite of A , the literal so defined:

$$\begin{cases} \text{if } A = p & \text{then } \overline{A} = \neg p \\ \text{if } A = \neg p & \text{then } \overline{A} = p \end{cases}$$

Equivalently

$$\overline{A} := (\neg A)^N$$

- **Definition: Clause**, a clause is a disjunction of a finite number of literals:

$$l_1 \vee l_2 \vee \dots \vee l_k$$

where each l_i is a literal.

- **Definition: CNF**, a CNF is a conjunction of a finite number of clauses:

$$C_1 \wedge C_2 \wedge \dots \wedge C_n$$

where each C_j is a clause.

- **Definition: Empty Clause**, an empty clause ($\square$) is, by definition, a disjunction of zero literals.

- **Definition: Empty CNF**, the empty CNF ($\emptyset$) is the conjunction of zero clauses.

Thinking about l_1 as a literal, and the disjunctions

$$\begin{array}{l} 0 \quad \square \\ 1 \quad l_1 \\ 2 \quad l_1 \vee l_2 \\ 3 \quad l_1 \vee l_2 \vee l_3 \\ \vdots \\ k \quad l_1 \vee l_2 \vee l_3 \vee \dots \vee l_k \\ \vdots \end{array}$$

If any of these is satisfiable, then all of the ones below are satisfiable, so it makes sense to put $\square$ on top of the first literal l_1 , so, by definition, the empty literal $\square$ **is unsatisfiable and a contradiction**. Same game with clauses and empty CNF $\emptyset$:

$$\begin{array}{l} 0 \quad \emptyset \\ 1 \quad C_1 \\ 2 \quad C_1 \wedge C_2 \\ \vdots \end{array}$$

But here the satisfiability flows upwards, if one is satisfied then all of the above are satisfied. The empty CNF $\emptyset$ is still above all, and as such is satisfied by all assignments and is a tautology

3.14.4 Disjunctive Normal Form DNF

A DNF is a disjunction of a finite number of formulas:

$$D_1 \vee D_2 \vee \dots \vee D_h$$

Where each D_i (dual of a clause, hasn't got a proper name, maybe co-clause) is a conjunction of a finite number of literals:

$$l_1 \wedge l_2 \wedge \dots \wedge l_m$$

Every definition stated before has a DNF dual but they don't have a proper name since *nobody cares*.

Lemma 3.14.6. Every formula $F \in \mathcal{F}_{\mathcal{L}}$ is logically equivalent to a formula $F^C \in CNF$ and to a formula $F^D \in DNF$

$$(CNF \ni F^C \equiv F \equiv F^D \in DNF)$$

Proof. By structural induction of F :

- **Base:** $F = p$, $p \in \mathcal{L}$ then $p^C \equiv p \equiv p^D$. A single literal is already in CNF and DNF, $p^C = p = p^D$.

- **Step:**

- Case $F = \neg G$, by I.H., there are $G^C \in CNF$, $G^D \in DNF$ with $G^C \equiv G \equiv G^D$

$$F = \neg G \equiv \neg(G^D) \in CNF$$

$$F = \neg G$$

$$\neg(G^D) = \neg(D_1 \vee D_2 \vee \dots \vee D_v)$$

with each D_i

$$D_i = l_{i1} \wedge l_{i2} \wedge \dots \wedge l_{iq}$$

but by De Morgan

$$\neg(G^D) \equiv \neg D_1 \wedge \neg D_2 \wedge \dots \wedge \neg D_v$$

And, still by De Morgan, each $\neg D_i$

$$D_i \equiv (\bar{l}_{i1} \vee \bar{l}_{i2} \vee \dots \vee \bar{l}_{iu_i})$$

So $\neg(G^D)$ is the conjunction $\neg D_1 \neg D_2 \wedge \dots \neg D_v$ of a finite number of formulas ($\neg D_i$), each one equivalent to a disjunction of a finite number of literals, that is $\neg(G^D) \in CNF$, and $F \equiv \neg G \equiv \neg(G^D)$.

- Case: $F = (G \wedge H)$, by I.H.

$$G^C, H^C \in CNF, \quad G^D, H^D \in DNF$$

$$\text{with } G^C \equiv G \equiv G^D, \quad H^C \equiv H \equiv H^D$$

We want to build **2 new formulas**:

- $F \equiv F^D$, defining F^D as $:= G^D \vee H^D \in DNF$, then by definition is $\in DNF$, and by construction $G^D \vee H^D \equiv F$. Disjunction of disjunctions of finitely many co-clauses, so still $\in DNF$.
- $F \equiv F^C$, then

$$G^C = G_1 \wedge G_2 \wedge \dots \wedge G_u$$

$$H^C = H_1 \wedge H_2 \wedge \dots \wedge H_v$$

So

$$G \vee H \equiv G^C \vee H^C \equiv (G_1 \wedge G_2 \wedge \dots \wedge G_u) \vee (H_1 \wedge H_2 \wedge \dots \wedge H_v) \equiv$$

Using generalized distributivity:

$$\equiv \bigwedge_{i=1}^u \bigwedge_{j=1}^v (G_i \wedge H_j)$$

Each G_i and each H_j are clauses, i.e., disjunctions of finitely many literals, so each term in the form $G_i \wedge H_j$ is again a clause, and then the whole formula is a finite conjunction of clauses and, by definition, a CNF, also $\equiv G \vee H = F$.

- Then we have the case $(G \wedge H)$, which is left as an exercise to the reader.
- Case (to complete our set of connectives): $(G \rightarrow H)$, which is also left as an exercise, because implication can be reduced to $G \rightarrow H \equiv (\neg G) \vee H$.

□

Exercise $n=3$ $p_1, p_2, p_3, q_1, q_2, q_3 \in \mathcal{L}$ $\xrightarrow{\text{Exercise}} 3 \text{ disjuncts of 2 literals each}$

$(p_1 \wedge q_1) \vee (p_2 \wedge q_2) \vee (p_3 \wedge q_3) \in DNF \rightsquigarrow CNF$

$\equiv \text{distributivity}$

$[p_1 \vee ((p_2 \wedge q_2) \vee (p_3 \wedge q_3))]$

$\wedge$

$[q_1 \vee ((p_2 \wedge q_2) \vee (p_3 \wedge q_3))]$

$\equiv \text{distributivity}$

$(p_1 \vee p_2 \vee (p_3 \wedge q_3))$

$\wedge$

$(p_1 \vee q_2 \vee (p_3 \wedge q_3))$

$\wedge$

$(q_1 \vee p_2 \vee (p_3 \wedge q_3))$

$\vee$

$(q_1 \vee q_2 \vee (p_3 \wedge q_3))$

$\equiv \text{distributivity}$

$(p_1 \vee p_2 \vee p_3)$

$\wedge$

$(p_2 \vee p_2 \vee q_3)$

$\wedge$

$(p_2 \vee q_2 \vee p_3)$

$\wedge$

$(p_2 \vee q_2 \vee q_3)$

$\wedge$

$(q_1 \vee p_2 \vee p_3)$

$\wedge$

$(q_1 \vee p_2 \vee q_3)$

$\wedge$

$(q_1 \vee q_2 \vee p_3)$

$\wedge$

$(q_1 \vee q_2 \vee q_3)$

$\in CNF$

8 conjuncts of 3 literals each

Exercise: generalize to each $n \in \mathbb{N}$.

$(p_1 \wedge q_1) \vee (p_2 \wedge q_2) \vee \dots \vee (p_n \wedge q_n)$

DNF n disjuncts of 2 literals each

CNF 2^n conjuncts of n literals

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✂ Notions of computational complexity

Definition 4.0.1. A (decision) **problem** (or “*language*”) is a subset $L \subseteq \Sigma^*$, i.e., a subset L of the set of all finite strings (words) over a finite alphabet (set) Σ .

For instance, formulas of $\mathcal{F}_{\mathcal{L}}$ can be written over the alphabet

$$\Sigma = \{\wedge, \vee, \neg, \rightarrow\} \cup \{(), () \cup \{p, |\}\} \quad p_i \in \mathcal{L}$$

So we can say that:

$$\mathcal{F}_{\mathcal{L}} \subseteq \Sigma^*$$

Given a $w \in \Sigma^*$, we know that it belongs to $\mathcal{F}_{\mathcal{L}}$ **iff** w is a formula, there is an L -construction for w . There are other decision problems we’ve seen, for example:

$$NNF \subseteq \Sigma^*, \quad CNF \subseteq \Sigma^*, \quad DNF \subseteq \Sigma^*$$

These happen to be **syntactical problem** which are (*usually*) **easy**.

But there are **semantics problems**, which are (*usually*) **not so easy**. For instance:

$$SAT \subseteq \Sigma^*$$

Input $w \in \Sigma^*$, definition: $w \in SAT$ iff $w \in \mathcal{F}_{\mathcal{L}}$ and w is satisfiable.

We can decide if a formula is satisfiable, but it’s long (think of a truth table, it’s exponential in length). Another semantical problem:

$$TAUTO \subseteq \Sigma^*$$

Same as before, but all entries must be 1, not at least one. Same with checking if a formula is unsatisfiable:

$$UNSAT \subseteq \Sigma^*$$

and complement of SAT, which is checking if it’s unsatisfiable but it doesn’t have to be a formula

$$SAT^C \subseteq \Sigma^*$$

We can solve all these problems, although inefficiently.

4.1 Computational Model

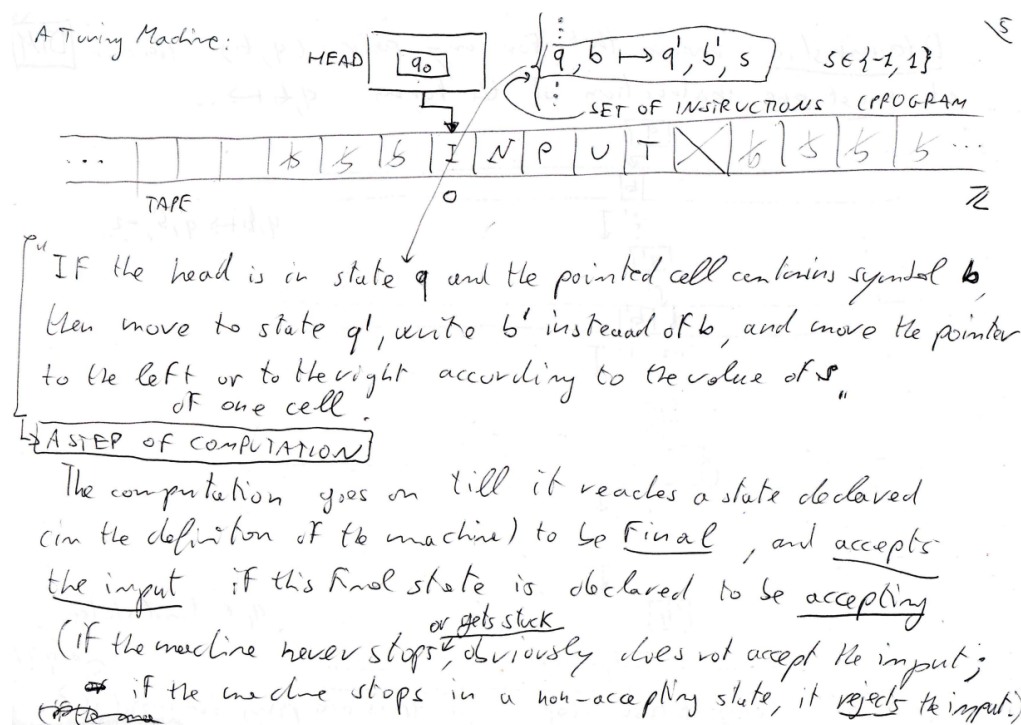
We want to **fix a computational model** which will provide a reasonable notion of **elementary step of computation**.

Church-Turing (extended) Thesis: Tells us that there are **many** different reasonable **computational models**, but the **set of computable functions** (what can actually be computed on these models) is **always the same**. Every “reasonable” computational model is “equivalent”, in the sense that every one of these models defines the same set of efficiently computable functions (decidable problems).

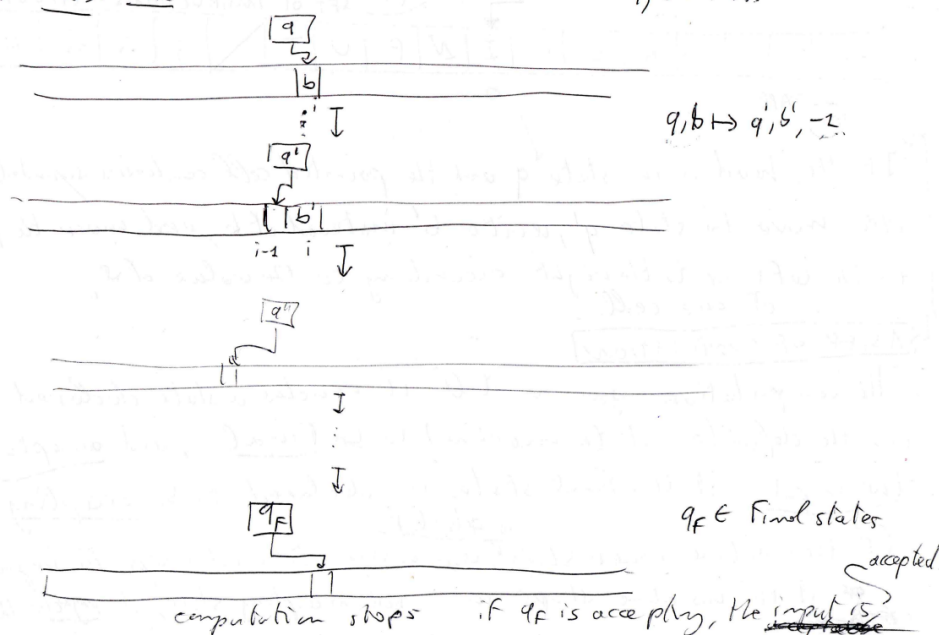
We choose the **model of Turing Machines**, which comes in several flavors:

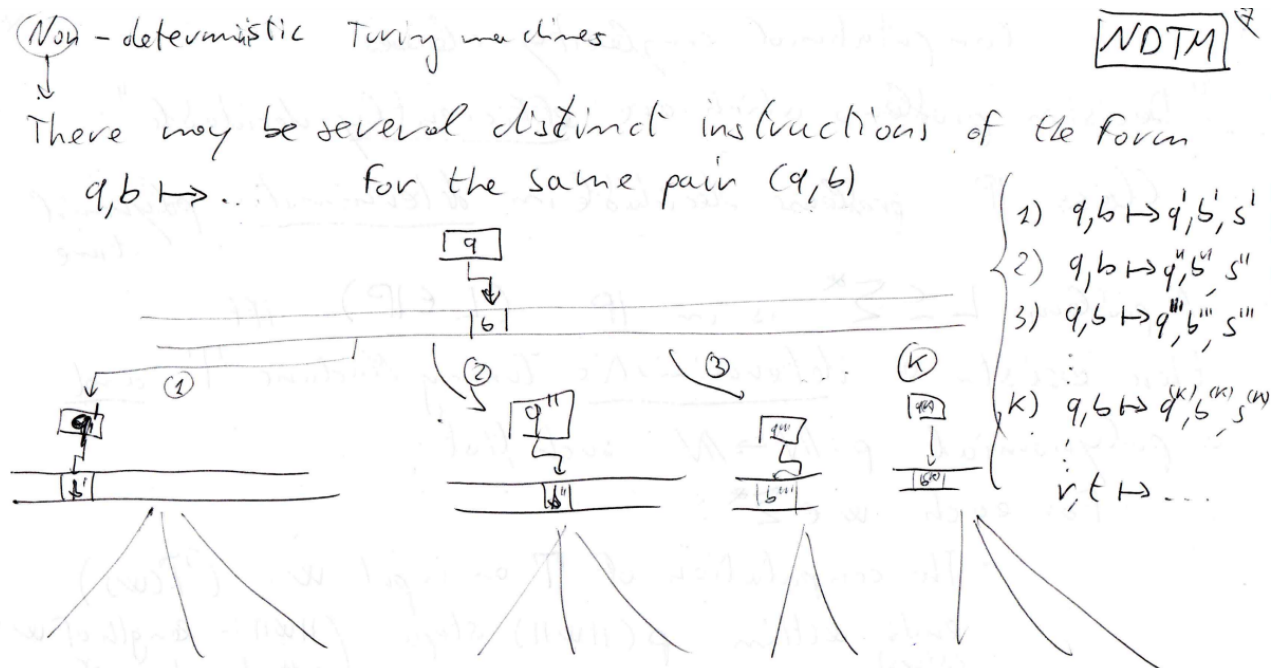
- Vanilla: **Deterministic Turing Machines DTM**
- Chocolate: **Non-Deterministic Turing machines NDTM**

Which will be useful to define complexity classes.



Deterministic: means that for any pair (q, b) , there is at most one instruction of the form $q, b \mapsto \dots$ DTM





An input is accepted if at least one branch of the computation tree ends in a final accepting state. (# of steps: length of the shortest accepting branch)

The computation goes on until it reaches a state declared (in the definition of the machine) to be final, and accepts the input if this final state is declared to be accepting.

The machine could also never stop, thus not accepting the input, and it could also explicitly reject the input. **Distinction** between **deterministic** (vanilla) and **non-deterministic** (chocolate) Turing machines:

- **Deterministic** (called DTM): means that for **any state-symbol pair** (q, b) there is **at most one instruction** of the form $q, b \rightarrow \dots$.
- **Non-Deterministic** (called NDTM): there may be **several distinct instruction** for **any state-symbol pair** of the form $q, b \rightarrow \dots$; several instruction for the same state and symbol. When such an instruction is reached, there's a universe for each possible instruction the machine can execute. A DTM is a sequence of steps, a NDTM is a tree of possibilities and an input is accepted when at least one of the leafs in the NDTM tree reaches an accepting state.

4.2 Computational Complexity Classes

Decision problems which are **efficiently** decidable.

Definition 4.2.1. Class $\mathcal{P}$: problems decidable in deterministic polynomial time. A problem $L \subseteq \Sigma^*$ is in $\mathcal{P}$ ($L \in \mathcal{P}$) **iff** there exists a Deterministic Turing Machine T and a polynomial $p: \mathbb{N} \rightarrow \mathbb{N}$ such that for each possible input $w \in \Sigma^*$

- the computation of T on input w ($T(w)$) ends within $p(\|w\|)$ steps (with $\|w\|$ meaning the length of w , the number of characters composing w).
- for each $w \in L$, $T(w)$ accepts. For each $w \notin L$, $T(w)$ does not accept.

Definition 4.2.2. Class $\mathcal{NP}$: problems decidable in non-deterministic polynomial time.

Notion of Efficiency

The **composition of polynomials is a polynomial**, so multiple polynomials together are still a polynomial and thus efficient (but with a **higher degree**, useful for combining algorithms). Composition of polynomials of a fixed degree (if greater than 1) is not a polynomial of that degree. **Multiplication of polynomials is, again, a polynomial**, but, again, the result is of a **different degree**. We want that our notion of “**efficient computation**” to be **robust under composition, multiplication and change of the computational model**. So we’re lead to **use polynomial** (with no fixed bound on the degree).

Previously seen syntactical problems are $\in \mathcal{P}$.

Definition 4.2.3. Class $\mathcal{NP}$: A problem $L \subseteq \Sigma^*$ is $\in \mathcal{NP}$ **iff** there exists a deterministic Turing Machine T and two polynomials $p, q : \mathbb{N} \rightarrow \mathbb{N}$ such that for all $w \in \Sigma^*$ and each $z \in \Gamma^*$ (Γ may be the same as Σ) we have:

- $T(w, z)$ ends within $p(|w|)$ steps
- For each $w \in L$ there exists additional information $z \in \Gamma^*$ such that $|z| \leq q(|w|)$ (is polynomial with respect to the length of w) and $T(w, z)$ accepts
- For each $w \notin L$ and for all $z \in \Gamma^*$ such that $|z| \leq q(|w|)$, $T(w, z)$ does not accept

Equivalently: A problem L is in $\mathcal{NP}$ iff there **exists a NDTM** T and a **polynomial** $p : \mathbb{N} \rightarrow \mathbb{N}$ such that $w \in L$ iff $T(w)$ accepts in a number of steps $\leq p(|w|)$.

The class $\mathcal{NP}$ can be equivalently thought as the class of **polynomially verifiable guess and check procedures**: if $w \in L$, guess an information $z \in \Gamma^*$ witnessing that w is actually in L then check in deterministic polynomial time that z witnesses that $w \in L$.

Example 4.2.4. SAT, taking $w \in \Sigma^*$ with the alphabet:

$$\Sigma = \{\wedge, \vee, \neg, \rightarrow\} \cup \{\}, \{\} \cup \{p, \mid\} \quad p_i \in \mathcal{L}$$

$w \in SAT$ **iff** $w \in \mathcal{F}_{\mathcal{L}}$ and w is **satisfiable**.

- Is $SAT \in \mathcal{P}$? Nobody knows (since nobody knows whether $\mathcal{P} = \mathcal{NP}$, we can guess no).
- Is $SAT \in \mathcal{NP}$? Looking at the definitions given for the class, can I guess and check? If a relevant truth assignment *descended from above*, could I **check that assignment in polynomial time**? Of course I can. We can guess a (restricted, obviously) assignment and check in deterministic polynomial time in $|w|$ that the assignment equals 1 (satisfies satisfiability).

Definition 4.2.5. The class $\text{co-}\mathcal{NP}$ is the “complement” of the class $\mathcal{NP}$ and can be thought as the class of **polynomially verifiable guess and check procedures** (BUT with an important caveat): if $w \notin L$, guess an information $z \in \Gamma^*$ witnessing that w is actually **NOT** in L then check in deterministic polynomial time that z witnesses that $w \notin L$.

What about **other problems**?

- $UNSAT \in \text{co-}\mathcal{NP}$, I can’t say that something is $UNSAT$ only from a random assignment, but i can check if its not $UNSAT$ namely if the assignment returns 1.
- $SAT^C \in \text{co-}\mathcal{NP}$ for the same reason (think about it)
- $TAUTO \in \text{co-}\mathcal{NP}$ for the same reason (think about it)

On $\mathcal{NP}$ **problems** we can do **guess and check if an answer is correct**, while on $\text{co-}\mathcal{NP}$ **problems** we can’t do that, equivalently we can **guess and check if the answer is NOT correct**.

Definition 4.2.6. $\mathcal{NP}$ -complete ($\mathcal{NP}_c \subseteq \mathcal{NP}$): we say that L is $\mathcal{NP}_c$ **iff**:

- $L \in \mathcal{NP}$
- Every $L' \in \mathcal{NP}$ is such that L' is **polynomially reducible to** L (we say that L is $\mathcal{NP}$ -hard)

4.3 Polynomial reducibility

Definition 4.3.1. A problem $L_1 \subseteq \Sigma^*$ is **polynomially reducible** to a problem $L_2 \subseteq \Gamma^*$ **iff** there exists a **DTM** T_{L_1, L_2} and a polynomial $p : \mathbb{N} \rightarrow \mathbb{N}$ such that for all $w \in \Sigma^*$, T_{L_1, L_2} transforms w into $w' \in \Gamma^*$ using a number of steps $\leq p(|w|)$ and such that $w \in L_1$ **iff** $w' \in L_2$. We write $L_1 \preceq_p L_2$.

This means that solving problem L_2 means solving L_1 , each input of one problem can be **efficiently** (in polynomial time) **transformed** into an instance of the other. If $L \in \mathcal{NP}_c$ and $L \in \mathcal{P}$ then $\mathcal{NP} = \mathcal{P}$, since every other problem $\in \mathcal{NP}$ could be reduced to a problem $\mathcal{NP}_c$, making $\mathcal{P}$ and $\mathcal{NP}$ collapse.

4.3.1 Cook-Levin Theorem

Basically, it says that there are $\mathcal{NP}_c$ problems. Cook proved this with *CNF-SAT*, so *CNF-SAT* $\in \mathcal{NP}_c$ and thus *SAT* $\in \mathcal{NP}_c$

Example 4.3.2. Examples of polynomial reduction:

- TAUTO and UNSAT

$$TAUTO \preceq_p UNSAT$$

$$w \mapsto \neg w$$

$$w \in TAUTO \text{ **iff** } \neg w \in UNSAT$$

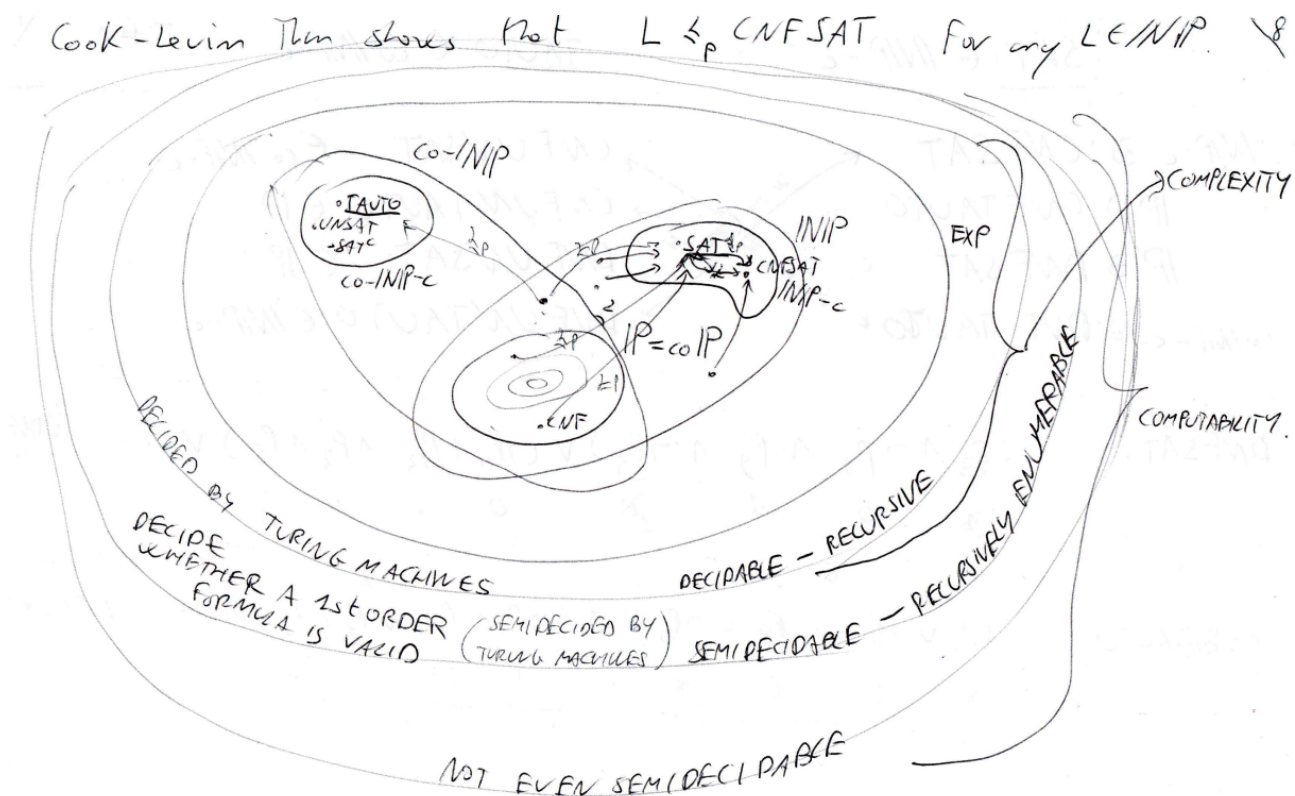
- CNF-SAT to SAT: $w \in \Sigma^*$, $w \in CNFSAT$ **iff** $w \in CNF$ and $w \in SAT$

$$CNF-SAT \preceq_p SAT$$

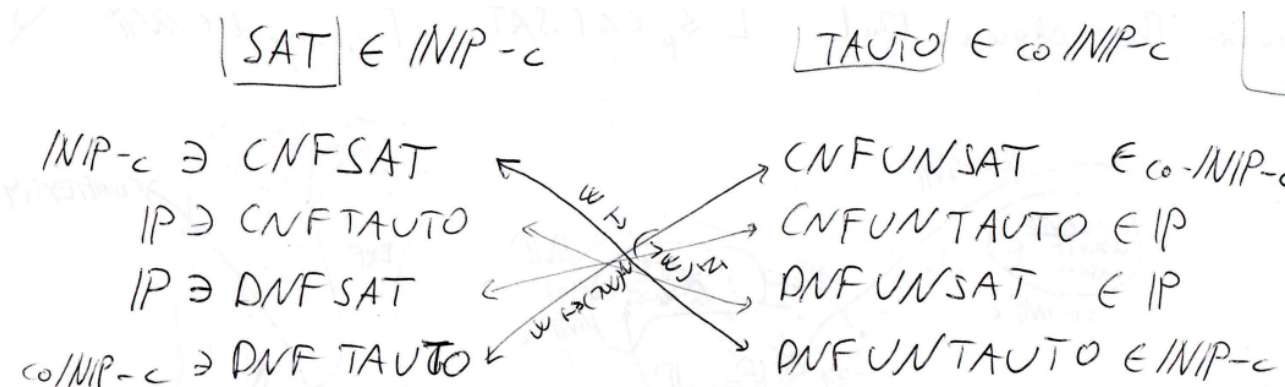
$$w \mapsto \begin{cases} w & \text{if } w \in CNF \\ \perp & \text{if } w \notin CNF \end{cases}$$

By Cook-Levin Theorem $CNFSAT \preceq_p SAT$ and $SAT \preceq_p CNFSAT$. The proof given by Cook of the Cook-Levin theorem is by “simulation”. Given a problem $L \subseteq \Sigma^*$, $L \in \mathcal{NP}$, we have to show that $L \preceq_p CNFSAT$. By def of $\mathcal{NP}$ there is a NDTM T_L deciding L in non-deterministic polynomial time. The idea of the proof is, for each $w \in \Sigma^*$ to build a CNF which mimics the computation of T_L on w . Finally, the produced CNF is satisfiable **iff** $T_L(w)$ accepts.

Since T_L works in polynomial time with respect to $|w|$, a careful analysis of the produced CNF shows that its length is polynomial with respect to $|w|$. Cook-Levin theorem **shows that** $L \preceq_p CNFSAT$ **for any** $L \in \mathcal{NP}$ (equivalent to saying that $CNFSAT \in \mathcal{NP}_c$).



$\leftarrow (\mathcal{P} = \text{co-}\mathcal{P} \text{ since } L \in \mathcal{P} \implies L^c \in \mathcal{P})$



Questions/Observations:

- Is it computationally efficient transforming a CNF into a logically equivalent DNF? It's *very unlikely* since it would mean reducing CNFSAT to DNFSAT efficiently thus making $\mathcal{P} = \text{NP}$. $\text{CNFSAT} \leq_p \text{DNFSAT} \implies \mathcal{P} = \text{NP}$.
- Is it computationally efficient transforming a DNF into a logically equivalent CNF? No, same as above but with $\text{NP}^c \ni \text{DNFUNTAUTO} \leq_p \text{CNFUNTAUTO} \in \mathcal{P}$

Nonetheless, we'll show a poly reduction from SAT to CNFSAT

$$\text{SAT} \leq_p \text{CNFSAT}$$

$\leftarrow (\text{if } \text{SAT} \leq_p \text{DNFSAT} \text{ then } \mathcal{P} = \text{NP})$

4.4 Equisatisfiability

Definition 4.4.1. Let $A, B \in \mathcal{F}_L$, then A is equisatisfiable with B (A and B are equisatisfiable) **iff** A is satisfiable **iff** B is satisfiable. That is: either A and B are both satisfiable or A and B are both unsatisfiable (might not be the same assignments, just satisfiability).

$\leftarrow (A \equiv B \implies A \text{ equisat } B \text{ the other way does not hold})$

4.4.1 Equisat as an equivalence relation

Is equisat an equivalence relation? We need to prove reflexivity, symmetry and transitivity.

- Reflexivity: $A \text{ equisat } A$ ✓
- Symmetry: $A \text{ equisat } B$ then $B \text{ equisat } A$ ✓
- Transitivity: $A \text{ equisat } B$ and $B \text{ equisat } C$ then $A \text{ equisat } C$ ✓

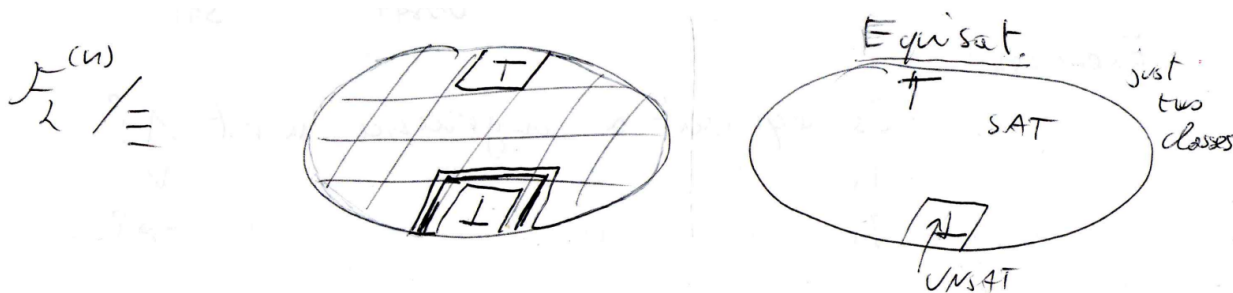
Is equisat a congruence with respect to the operations considered (connectives)?

Example 4.4.2. Consider for example $\neg$, let $A \in TAUTO$ and $B \in SAT, B \notin TAUTO$, we have:

- $A \text{ equisat } B$
- $\neg A \text{ equisat } \neg B$

We can find equisat formulas not always satisfied by all the same assignments (we can prove this for any of the connectives). So it's **not a congruence**.

$A \equiv B$ implies that $A \text{ equisat } B$. $\equiv$ is, as a relation, more refined than equisat (i.e., equisat is coarser than $\equiv$). The **equivalence classes of equisat** are **unions of equivalence classes** of $\equiv$, more specifically, equisat cares only between unsatisfiable and satisfiable (takes the $\perp$ class and the union of all the rest), while $\equiv$ has classes for each assignment.



4.4.2 SAT to CNFSAT

We shall **reduce** SAT polynomially $\preceq_p$ to $CNFSAT$ using **preservation of equisatisfiability** instead of preservation of logical equivalence.

Observation: Recall that $SAT \preceq_p NNF SAT$, preserving logical equivalence (since transforming a formula to NNF can be done in polynomial time while preserving logical equivalence).

Observation: If $A \in NNF$ and $A \notin CNF$ then A contains at least one subformula of the following form:

$$(\dagger) : C \vee (D_1 \wedge D_2) \quad \text{or} \quad (D_1 \wedge D_2) \vee C : (\dagger\dagger)$$

for some $C, D_1, D_2 \in NNF$.

Since $A \in NNF$, A is a $\wedge$ - $\vee$ combination of literals. If every occurrence of $\vee$ is in a subformula with the main connective $\wedge$ then $A \in CNF$. If $A \notin CNF$ there must be a subformula of the shape $(\dagger)$ or $(\dagger\dagger)$.

We call any occurrence of $(\dagger)$ or $(\dagger\dagger)$ a “**violation**”. Preliminary step: since $(\dagger) \equiv (\dagger\dagger)$, we use substitution theorem to **replace every occurrence of the kind $(\dagger\dagger)$ with $(\dagger)$** . This requires constant time. We can now assume that **every violation is in the form $(\dagger)$** .

Let $A \in NNF$ and let $B = (\dagger)$ with $B \preceq A$. Then B is a violation of A .

← (Keep in mind that a formula made only of $\vee$ is in CNF , so A must have both $\vee$ and $\wedge$)

The **original formula** may contain a **finite number n of violations**, so we **eliminate at least one at a time**, while **maintaining equisatisfiability** and doing this in **efficient time**, to obtain something equisat.

For **each violation B** we **introduce** a new (not occurring in A) propositional **letter $a_B \in \mathcal{L}$** . We **define**

$$B' := B [a_B / D_1 \wedge D_2] \wedge (\neg a_B \vee D_1) \wedge (\neg a_B \vee D_2)$$

$\leftarrow$ (We call the part after the first $\wedge$ “**tail**”)

Where

$$B'' := B [a_B / D_1 \wedge D_2]$$

is the formula obtained by replacing in B every occurrence of $D_1 \wedge D_2$ with the letter a_B .

So:

$$B' = (C' \vee a_B) \wedge (\neg a_B \vee D_1) \wedge (\neg a_B \vee D_2)$$

with

$$C' = C [a_B / D_1 \wedge D_2]$$

We shall prove that A **and** A' , obtained by **replacing B in A with B'** , are **equisat** (while, in general, not being $\equiv$).

Example 4.4.3. Examples $p, q, r \in \mathcal{L}$ and a fresh $a \in \mathcal{L}$ ($a \neq p, a \neq q, a \neq r$).

We can show how a simple violation can be removed:

$$\text{CNF} \not\vdash p \vee (q \wedge r) \mapsto (p \vee a) \wedge (\neg \vee q) \wedge (\neg a \vee r) \in \text{CNF}$$

The first is not a CNF, while the second is $\in \text{CNF}$. We now have to show that they're equisat.

With the assignment

$$v(q) = v(r) = v(a) = 1, \quad v(p) = 0$$

we can see that

$$\tilde{v}(p \vee (q \wedge r)) = 1$$

$$\tilde{v}((p \vee a) \wedge (\neg \vee q) \wedge (\neg a \vee r)) = 1$$

So they're **both satisfiable** $\implies$ they're **equisat**.

But *are they logical equivalent?* Considering

$$w(q) = w(r) = 1, \quad w(p) = w(a) = 0$$

Then

$$\tilde{w}(p \vee (q \wedge r)) = 1$$

$$\tilde{w}((p \vee a) \wedge (\neg \vee q) \wedge (\neg a \vee r)) = 0$$

So they're equisat, but **not** $\equiv$.

Recall: No shortest CNF $\equiv$ to the DNF for some n has 2^n clauses of n literals each

$$\begin{aligned}
 & \boxed{n=3} \quad (p_1 \wedge q_1) \vee (p_2 \wedge q_2) \vee (p_3 \wedge q_3) \mapsto \\
 & \quad \underbrace{\hspace{10em}}_C \quad \underbrace{\hspace{10em}}_{D_1 \wedge D_2} \\
 & ((p_1 \wedge q_1) \vee (p_2 \wedge q_2) \vee q_3) \wedge (\neg a_3 \vee p_3) \vee (\neg a_3 \vee q_3) \mapsto \\
 & \quad \underbrace{\hspace{10em}}_C \quad \underbrace{\hspace{10em}}_{D_1 \wedge D_2} \\
 & ((p_1 \wedge q_1) \vee q_2 \vee q_3) \wedge (\neg a_3 \vee p_3) \vee (\neg a_3 \vee q_3) \wedge (\neg a_2 \vee p_2) \wedge (\neg a_2 \vee q_2) \\
 & \mapsto \underbrace{\hspace{10em}}_{D_1 \wedge D_2} \quad \underbrace{\hspace{10em}}_C \\
 & (a_1 \vee a_2 \vee a_3) \wedge (\neg a_3 \vee p_3) \vee (\neg a_3 \vee q_3) \wedge (\neg a_2 \vee p_2) \wedge (\neg a_2 \vee q_2) \\
 & \quad \wedge (\neg a_1 \vee p_1) \wedge (\neg a_1 \vee q_1) \\
 & \hspace{15em} \in \text{CNF}
 \end{aligned}$$

For general $n \geq 0$: $2n+1$ clauses, one with n literals $(a_1 \vee a_2 \vee \dots \vee a_n)$ and the other ones with 2 literals each $(\neg a_i \vee p_i)$ or $(\neg a_i \vee q_i)$.

4.4.3 Algorithm

- Let $F \in \mathcal{F}_L$.
- Let $A \in \text{NNF}$ such that $A \equiv F$. A is built in polynomial time with respect to $\|F\|$.
- Build a sequence

$$F \equiv A := A_0 \rightsquigarrow A_1 \rightsquigarrow \dots \rightsquigarrow A_u \in \text{CNF}$$

where $u \leq$ the number of violations occurring in A ($u \leq \|A\|$, max number of steps) and A_{i+1} is obtained from A_i by removing at least one violation as follows:

- Remove a violation from A_i by substituting B with B' : the outcome is A_{i+1}
- Then we shall guarantee that for each $i = 0, 1, \dots, u-1$, A_i equisat A_{i+1}
- Then $F \equiv A$ is equisat with $A_u \in \text{CNF}$ by transitivity

Notice that in passing **from** A_i **to** A_{i+1} (from B to B') we have a **dilatation of the formulas** $\|A_{i+1}\| \geq \|A_i\|$. But by **how much**? Remembering that

$$B \rightsquigarrow B' = B[a_b/D_1 \wedge D_2] \wedge (\neg \vee D_1) \wedge (\neg a_b \vee D_2)$$

So $\|B'\| \leq \|B\| + k$ where k is a constant (12 counting all symbols). So $\|A_{i+1}\| \leq \|A_i\| + k$.

If the initial number of violations in $A = A_0$ is v , after at most $u \leq v$ steps we have produced $A_u \in \text{CNF}$, and $\|A_u\| \leq \|A\| + u \cdot k \leq (k+1)\|A\|$. It increases by a constant number of symbols k , for at most u times. $\leftarrow (u < \|A\|)$

So in the end, the **algorithm runs in deterministic polynomial time**.

4.4.4 Proving correctness

To prove the correctness of the algorithm it suffices to prove that for each index $i = 0, 1, \dots, u-1$, A_i **equisat** A_{i+1} , then by the transitivity of equisat $F \equiv A_0$ equisat $A_u \in CNF$

Let $B \preceq A_i$ be a violation for A_i . Then:

$$B = C \vee (D_1 \wedge D_2)$$

$$B' = B'' \wedge (\neg a_b \vee D_1) \wedge (\neg a_b \vee D_2) \quad \text{with} \quad B'' = [a_b/D_1 \wedge D_2]$$

Then we can say that

$$\begin{cases} T & := A_i [q/B] \\ A_i & := T [B/q] \\ A_{i+1} & := T [B'/q] \end{cases}$$

where q is a fresh propositional letter.

Observation: Take the tail $(\neg a_b \vee D_1) \wedge (\neg a_b \vee D_2)$, we can see that

$$(\neg a_b \vee D_1) \wedge (\neg a_b \vee D_2) \equiv a_b \rightarrow (D_1 \wedge D_2)$$

also

$$(\neg a_b \vee D_1) \wedge (\neg a_b \vee D_2) \wedge (\neg D_1 \vee \neg D_2 \vee a_b) \equiv a_b \leftrightarrow (D_1 \wedge D_2)$$

← (we added the other side of the implication this makes the tail longer tho)

Assume A_i is **satisfiable** and we'll prove that A_{i+1} is **satisfiable too**.

Definition 4.4.4. Let $v : \mathcal{L} \rightarrow \{0, 1\}$ be such that $\tilde{v}(A_i) = 1$, then we define v_{a_b}

$$\begin{cases} v_{a_b}(p) := v(p) & \text{for all } p \in \mathcal{L}, p \neq a_b \\ v_{a_b}(a_b) := \tilde{v}(D_1 \wedge D_2) & \text{otherwise} \end{cases}$$

← (it must exist because A_i is SAT)

It's well defined.

Notice that $\tilde{v}_{a_b}(a_b \rightarrow (D_1 \wedge D_2)) = \tilde{v}(\neg(D_1 \wedge D_2)) \vee \tilde{v}(D_1 \wedge D_2) = 1$, by the semantics of implication. Then $(\dagger) = \tilde{v}_{a_b}(B') = \tilde{v}_{a_b}(B'')$, by the semantics of $\wedge$, since $B' = B'' \wedge$ tail and tail=1. Now:

$$\begin{cases} B & = (B [a_b/D_1 \wedge D_2]) [D_1 \wedge D_2/a_b] \\ B'' & = (B [a_b/D_1 \wedge D_2]) [a_b/a_b] \end{cases}$$

← (We can do this since a_b does not occur in $D_1 \wedge D_2$)

We can now do

$$\tilde{v}_{a_b}(a_b) = v_{a_b}(a_b) \stackrel{\text{def}}{=} \tilde{v}(D_1 \wedge D_2) = \tilde{v}_{a_b}(D_1 \wedge D_2)$$

Apply to first and last the substitution Lemma:

$$\tilde{v}_{a_b}(B) = \tilde{v}_{a_b}(B'') \quad (\dagger\dagger)$$

← (last = it's true because we just introduced the new variable a_b , so a_b can't be in $(D_1 \wedge D_2)$)

Now putting everything together we have:

$$\begin{aligned} 1 &= \tilde{v}(A_i) \\ &= \tilde{v}_{a_b}(A_i) && a_b \text{ does not occur in } A_i \\ &= \tilde{v}_{a_b}(T [B/q]) && \text{By definition of } A_i \\ &= \tilde{v}_{a_b}(T [B''/q]) && \text{By } (\dagger\dagger) \\ &= \tilde{v}_{a_b}(T [B'/q]) && \text{By } (\dagger) \\ &= \tilde{v}_{a_b}(A_{i+1}) && \text{By definition of } A_{i+1} \end{aligned}$$

So $\tilde{v}_{a_b}(A_{i+1}) = 1$, and thus $A_{i+1} \in SAT$. □

Assume now, **for the other way around**, that $A_{i+1} \in SAT$ with the aim of proving that A_i is *SAT* too. We have two cases:

1. **Lucky case:** Assume further that A_{i+1} is satisfied by an assignment $w : \mathcal{L} \rightarrow \{0, 1\}$ of the form $\tilde{v}_{a_b}$, that is

$$w(a_b) = \tilde{w}(D_1 \wedge D_2)$$

at least one satisfying assignment has this property. Then we *kinda* reverse the earlier chain:

$$\begin{aligned} 1 &= \tilde{w}(A_{i+1}) \\ &= \tilde{w}([B'/q]) \\ &= \tilde{w}([B''/q]) \\ &= \tilde{w}([B/q]) \\ &= \tilde{w}(A_i) \end{aligned}$$

So $\tilde{w}(A_i) = 1$, that is $A_i \in SAT$.

2. **Second, unlucky, case:** A_{i+1} is satisfiable but all satisfying assignments $w : \mathcal{L} \rightarrow \{0, 1\}$ ($\tilde{w}(A_{i+1}) = 1$) are such that $\tilde{w}(a_b) \neq \tilde{w}(D_1 \wedge D_2)$.

The condition

$$\tilde{w}(a_b) \neq \tilde{w}(D_1 \wedge D_2)$$

implies that

$$w(a_b) = 0 \quad \text{and} \quad \tilde{w}(D_1 \wedge D_2) = 1$$

or

$$w(a_b) = 1 \quad \text{and} \quad \tilde{w}(D_1 \wedge D_2) = 0$$

By bivalence, but the second one can be ruled out since $\tilde{w}$ satisfies the tail, which means that $\tilde{w}(a_b \rightarrow (D_1 \wedge D_2)) = 1$.

TODO (FINISH THIS PROOF also fuck you, also there is an exercise/direction in the slides)

Remark 4.4.5. This technique to reduce $SAT \preceq_p CNFSAT$ is inspired to Karp's technique to prove that

$$SAT \preceq_p 3\text{-}CNFSAT$$

Where by $k\text{-}CNFSAT$ we mean the problem of deciding satisfiability of CNFs where each clause has exactly/at most k literals each. Karp proved that $3\text{-}CNFSAT$ is already $\in \mathcal{NPc}$. He also proved that $2\text{-}CNFSAT \in \mathcal{P}$. These are called **dichotomy results**, up to a certain degree the problem is “easy”, then it becomes difficult.

We want to deal with **problems of the kind** SAT, TAUTO or $\Gamma \stackrel{?}{\models} A$. About these problems we know that:

- $SAT \preceq_p CNFSAT \in \mathcal{NPc}$
- $TAUTO \preceq_p UNSAT \preceq_p CNFUNSAT \in co\text{-}\mathcal{NPc}$

Given the problem of **deciding** whether $\Gamma \models A$ (for Γ a finite theory)

$$\Gamma = \{\gamma_1, \dots, \gamma_n\} \quad \gamma_i \in \mathcal{F}_{\mathcal{L}}, \quad A \in \mathcal{F}_{\mathcal{L}}$$

We can **reduce it** to a problem of CNFUNSAT:

$$\begin{aligned} \Gamma \models A &\Leftrightarrow \Gamma \cup \{\neg A\} && UNSAT \quad (\text{as a theory}) \\ &\Leftrightarrow \gamma_1 \wedge \gamma_2 \wedge \dots \wedge \gamma_n \wedge \neg A && UNSAT \quad (\text{as a formula}) \\ &\Leftrightarrow S \in CNF, \quad S \in CNFUNSAT && \text{For some } S \text{ equisat } \gamma_1 \wedge \gamma_2 \wedge \dots \wedge \gamma_n \wedge \neg A. \end{aligned}$$

4.5 Notational Variant for CNFs

Take a **family of formulas**

$$F_i \in \mathcal{F}_{\mathcal{L}} \quad i = 1, \dots, k$$

Then we can “*forget*” to parenthesize these expressions due to **associativity**

$$\begin{cases} F_1 \wedge F_2 \wedge \dots \wedge F_k \\ F_1 \vee F_2 \vee \dots \vee F_k \end{cases}$$

We can also forget about order, due to **commutativity**

$$\begin{cases} F_1 \wedge F_2 \equiv F_2 \wedge F_1 \\ F_1 \vee F_2 \equiv F_2 \vee F_1 \end{cases}$$

And repetitions, for **idempotence**:

$$\begin{cases} F_1 \wedge F_1 \equiv F_1 \\ F_1 \vee F_1 \equiv F_1 \end{cases}$$

So from now on a **clause** $l_1 \vee l_2 \vee \dots \vee l_u$ will be denoted as the **set of its literals** $\{l_1, l_2, \dots, l_u\}$. A **CNF** $C_1 \vee C_2 \vee \dots \vee C_u$ will be denoted by the **set of its clauses** $\{C_1, C_2, \dots, C_u\}$. A CNF will be a set of sets of literals. A theory made of CNFs will be a set of sets of literals.

What? It says the same thing? Also on the lecture notes? TOASK

Example 4.5.1.

$$\begin{aligned} CNF : & (p \vee q \vee \neg r) \wedge (q \vee r \vee a) \wedge p \\ \downarrow \\ CNF : & \{ \{p, q, \neg r\}, \{q, r, a\}, \{p\} \} \end{aligned}$$

A theory of CNFs is just a set of clauses. An infinite theory is a set containing infinitely many clauses. Now the notation of empty CNF as $\emptyset$ *makes sense* (thinking about an empty set of clauses), while an empty clause is $\square$ (empty set of literals). A CNF containing an **empty clause** is **unsatisfiable**, the shortest CNF containing $\square$ is just $\{\square\}$. on the other hand the empty CNF $\emptyset$ is SAT, even tautological.

We want to deal with $\Gamma \stackrel{?}{\models} A$. We want to reduce it to the problem of deciding whether a **set of clauses** S (possibly infinite) is **unsatisfiable**. S is SAT **iff** there is an assignment $v : \mathcal{L} \rightarrow \{0, 1\}$ such that $v \models C$ for each $C \in S$ (satisfies each clause in the set). For all $C \in S$ there is at least a literal $L \in C$ such that $v \models L$ ($\tilde{v}(L) = 1$, it satisfies the literal).

If we want to prove that S is *UNSAT*, a good strategy is to **enlarge** S into a new set S' ($S \subseteq S'$) in such a way that $S \equiv S'$ (or at least S equisat S'). For instance, if enlarging S , iteratively

$$S \subseteq S' \subseteq S'' \dots \subseteq S^{(k)}$$

If we find that $\square \in S^{(k)}$, then we can conclude that $S^{(k)}$ is UNSAT, so S is UNSAT too.

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※ Refutational Methods

They are **deductive methods** whose aim is to **prove the unsatisfiability of a set of clauses** (of a formula or of a theory). Many of these methods are based on the inference rule called “*resolution principle*”.

They are calculi particularly **fit to be automated** (for proof search, we can just give theory and formula to a machine and make it decide). This is the basic behind Automated Deduction. There are **different possible branches**:

- SAT Solver: decide if a given set of clauses is satisfiable or not (DPLL).
- Theorem Prover: designed to automatically prove mathematical theorem (Coq).
- Logic Programming: asking a machine to refute a theorem $\in CNF$ (Prolog).

← (Technically Coq is a proof assistant not prover but you get the gist)

5.1 Resolution Principle

Given two clauses C_1 and C_2 we say that the clause D is the **resolvent** of C_1 and C_2 on the pivot l (where l is a literal) **iff**:

- $l \in C_1$
- $\bar{l} \in C_2$
- $D := (C_1 \setminus \{l\}) \cup (C_2 \setminus \{\bar{l}\})$

We shall write $D = \mathbb{R}(C_1, C_2; L, \bar{L})$.

Example 5.1.1.

$$S := \{\{x, y, \neg t\}, \{u, \neg y, t\}\}$$

How many resolvents can we have? 2, namely:

$$D_1 : \mathbb{R}(C_1, C_2; y, \neg y) = \{x, \neg t, u, t\}$$

$$D_2 : \mathbb{R}(C_1, C_2; \neg t, t) = \{x, y, u, \neg y\}$$

Lemma 5.1.2 (Correctness). (of the resolution principle, also called “Soundness”).

Let $D = \mathbb{R}(C_1, C_2; l, \bar{l})$ be the resolvent of some clauses, then

$$\{C_1, C_2\} \models D$$

Proof. We must show that for each assignment $v : \mathcal{L} \rightarrow \{0, 1\}$, if $v \models C_1$ and $v \models C_2$ then $v \models D$ too. Let $v : \mathcal{L} \rightarrow \{0, 1\}$ be such that $v \models C_1$ and $v \models C_2$, then $v \models M$ and $v \models N$ for some literals $M \in C_1$ and $N \in C_2$. If it were the case that $M = L$ and $N = \bar{L}$ then $\tilde{v}(L) = 1$ and $\tilde{v}(\bar{L}) = 1$ but this is clearly a contradiction so it can’t be the case.

Then at least one in M or N is such that $M \neq L$ or $N \neq \bar{L}$. We assume without loss of generality that $M \neq L$ then $M \in D$ (by definition of D), so $\tilde{v}(M) = 1$ implies that $\tilde{v}(D) = 1$.

□

← (small exercise on notes)

Corollary 5.1.3. If $\mathbb{R}(C_1, C_2; L, \bar{L}) = \square$ then $\{C_1, C_2\}$ is UNSAT.

Proof. $\{C_1, C_2\} \models \square$ by Correctness Lemma. □

Corollary 5.1.4. Let S be a set of clauses. Let $S = S_0, \dots, S_k$ be a sequence of sets of clauses such that:

1. For all indexes $i = 0, 1, \dots, k-1$, S_{i+1} is obtained from S_i by adding one or more resolvents of clauses in S_i
2. $\square \in S_k$

Then S is UNSAT.

Proof. $S = S_0 \models S_1 \models S_2 \models \dots \models S_k \ni \square$, the sequence is guaranteed by correctness Lemma, so $S \models \square$ that is S is UNSAT. □

Our aim is to show that the **refutational method** based on iterated applications of resolution principle is **correct** and **refutationally complete**.

In general, a logical calculus may have (or not) the two following **properties**:

- **Correctness (Soundness):** a calculus is correct (sound) if the “*certificates*” (proofs) outputs witness a true state of things (it does not produce “fake certificates”). In our case the sequence $S_0, S_1, \dots, S_k \ni \square$ is a correct “certificate”, or proof, that S is UNSAT.
- **Completeness:** a calculus is complete if it does not omit any “certificate” (proof). In our case it means that to be complete our method should have the following property: If S is UNSAT then there is a sequence (certificate) $S = S_0, \dots, S_k \ni \square$ ($S_i \rightsquigarrow S_{i+1}$ only adding resolvents from clauses in S_i).

Our refutation, resolution-based, method we’ll only be **REFUTATIONALLY** complete, we’ll explain what it means later.

5.1.1 Refutational Completeness of the Resolution Principle (proof of J.A. Robinson)

Theorem 5.1.5. A (possibly infinite) set of clauses S is unsatisfiable iff $\square \in \mathcal{R}^*(S)$ where $\mathcal{R}^*(S)$ is defined as follows: $\leftarrow$ (that iff is by correctness and completeness)

$$\begin{aligned} \mathcal{R}(S) &:= S \cup \{D : D = \mathbb{R}(C_1, C_2; L, \bar{L}) \text{ for some } C_1, C_2 \in S, L \in C_1, \bar{L} \in C_2\} \\ \mathcal{R}^2(S) &:= \mathcal{R}(\mathcal{R}(S)) \\ &\vdots \\ \mathcal{R}^{t+1}(S) &:= \mathcal{R}(\mathcal{R}^t(S)) \\ &\vdots \\ \mathcal{R}^*(S) &:= \bigcup_{\substack{i \in \omega \\ i \in \mathbb{N}}} \mathcal{R}^i(S) \end{aligned}$$

For simplicity:

$$\begin{aligned} \mathcal{R}^1(S) &:= \mathcal{R}(S) \\ \mathcal{R}^0(S) &:= S \end{aligned}$$

Remark 5.1.6. Assume we proved the last theorem. If S is a finite set of clauses, then $\mathcal{R}^*$ provides us with a decision procedure for establishing whether S is SAT or UNSAT. That is, in both cases, the procedure building $\mathcal{R}^*(S)$ ends in a finite number of steps (finite amount of time) providing always the correct answer.

Proof. If S is finite then in S there occur only finitely many distinct propositional letters. Let us write $Var(S) \subseteq \{p_1, p_2, \dots, p_n\}$ for some $n \in \mathbb{N}$.

The literals writable using only $p_1, p_2, \dots, p_n$ are exactly $2n$. The clauses writable using only literals on $p_1, p_2, \dots, p_n$ are exactly 2^{2n} .

Key observation: the application of the resolution rule (the **formation of new resolvents**) **never introduces any new literals**. So, the sequence:

$$(\dagger) \quad S = \mathcal{R}^0(S) \subseteq \mathcal{R}^1(S) \subseteq \dots \subseteq \mathcal{R}^k(S) \subseteq \dots$$

is such that every $\mathcal{R}^i(S)$ is a subset of the 2^{2n} clauses writable on $p_1, p_2, \dots, p_n$. Let us write $\mathcal{C}^{(n)}$ for the set of all clauses on $p_1, \dots, p_n$

$$\mathcal{R}^i(S) \subseteq \mathcal{C}^{(n)} \quad \text{for each } i \in \omega$$

Then there must be $t \in \omega$ such that $\mathcal{R}^t(S) = \mathcal{R}^{t+1}(S)$ (since no $\mathcal{R}^i(S)$ can have more than 2^{2n} clauses, $t \leq 2^{2n}$).

Then:

$$\mathcal{R}^t(S) = \mathcal{R}^{t+1}(S) = \mathcal{R}(\mathcal{R}^t(S)) = \mathcal{R}(\mathcal{R}^{t+1}(S)) = \mathcal{R}^{t+2}(S) = \dots$$

So we have that

$$\mathcal{R}^t(S) = \mathcal{R}^{t+1}(S) = \dots = \mathcal{R}^*(S)$$

Then: At each generic step $i \rightsquigarrow i + 1$ of the construction:

- If we found $\square \in \mathcal{R}^{i+1}(S)$ then stop $\implies$ output S UNSAT
- If $\square \notin \mathcal{R}^{i+1}(S)$ then
 - if $\mathcal{R}^{i+1}(S) = \mathcal{R}^i(S)$ then stop and output S SAT
 - else go to the next step ($i + 1 \rightsquigarrow i + 2$)

This algorithm **always terminates when S is finite**.

□

Note: If S is **infinite**, but it happens that $\mathcal{R}^t(S) = \mathcal{R}^{t+1}(S)$ (and we can check this), the algorithm can stop and say that S is SAT. The procedure is complete, but it's a semi-decision procedure, since it might not end.

Theorem 5.1.7.

$$\begin{aligned} S \text{ UNSAT} &\Leftrightarrow \square \in \mathcal{R}^*(S) \\ &\Leftarrow \text{correctness (soundness)} \\ &\Rightarrow \text{completeness} \end{aligned}$$

Proof.

Correctness

Assuming that $\square \in \mathcal{R}^*(S)$, then, by definition of $\mathcal{R}^*(S) = \bigcup_{i \in \omega} \mathcal{R}^i(S)$, there is $i \in \omega$ such that $\square \in \mathcal{R}^i(S)$. So $\mathcal{R}^i(S)$ is UNSAT, by definition of $\square$. But $\mathcal{R}^i(S) \equiv \mathcal{R}^{i-1}(S) \equiv \dots \equiv \mathcal{R}^0(S) \equiv S$ (by Correctness Lemma), so also S is UNSAT.

□

(Refutational) Completeness

We have to prove that S UNSAT implies $\Box \in \mathcal{R}^*(S)$.

Proof. Since $S \in \text{UNSAT}$ (by assumption), by compactness theorem, there must exist a $S_{fin} \subseteq_\omega S$ such that S_{fin} is UNSAT. In S_{fin} there occur finitely many (distinct) propositional letters, say $\text{Var}(S_{fin}) \subseteq \{p_1, \dots, p_n\} \subseteq \mathcal{L}$, for some $n \in \omega$.

Remark 5.1.8. $S_{fin} \subseteq \mathcal{C}^{(n)}$ (we call $\mathcal{C}^{(n)}$ the set of all clauses over the letters $p_1, \dots, p_n$).

Remark 5.1.9. $\mathcal{C}^{(0)} = \{\Box\}$ ($|\mathcal{C}^{(0)}| = 2^{2^0} = 1$).

A fortiori (“from stronger reasons”) we can say that:

$$\left. \begin{array}{l} S_{fin} \subseteq \mathcal{C}^{(n)} \\ S_{fin} \subseteq S \end{array} \right\} \implies S_{fin} \subseteq \mathcal{C}^{(n)} \cap S \stackrel{(*1)}{\subseteq} \mathcal{C}^{(n)} \cap \mathcal{R}^*(S) \quad (*1) \text{ (since } S \subseteq \mathcal{R}^*(S))$$

It follows:

$$(\dagger\dagger) \quad \mathcal{C}^{(n)} \cap \mathcal{R}^*(S) \text{ is UNSAT}$$

Since it contains S_{fin} , which is UNSAT by definition. We shall prove that for each $k = n, n-1, \dots, 0$ that $\mathcal{C}^{(k)} \cap \mathcal{R}^*(S)$ is UNSAT ($\dagger$). If we can prove ($\dagger$) for all k , then ($\dagger$) holds for $k = 0$ too. That is $\mathcal{C}^{(0)} \cap \mathcal{R}^*(S)$ is UNSAT, and thus $\{\Box\} \cap \mathcal{R}^*(S)$ is UNSAT.

If we prove $\{\Box\} \cap \mathcal{R}^*(S)$ is UNSAT ($\dagger\dagger\dagger$) then

$$\{\Box\} \cap \mathcal{R}^*(S) = \begin{cases} \emptyset & \rightarrow \{\Box\} \cap \mathcal{R}^*(S) \text{ SAT contradicting } (\dagger\dagger\dagger) \\ \text{or} \\ \{\Box\} & \rightarrow \{\Box\} \cap \mathcal{R}^*(S) \text{ UNSAT} \end{cases}$$

but

$$\{\Box\} \cap \mathcal{R}^*(S) = \{\Box\} \text{ iff } \Box \in \mathcal{R}^*(S)$$

And this is the end of completeness.

We shall prove by **descending induction** on the index $k = n, n-1, \dots, 0$, that ($\dagger$): $\mathcal{C}^{(k)} \cap \mathcal{R}^*(S)$ is UNSAT.

Base: $k = n$, then $\mathcal{C}^{(k)} \cap \mathcal{R}$ UNSAT, which is already proven, by ($\dagger\dagger$).

Step: Assume the statement ($\dagger$) is true for $k = n, n-1, \dots, k+1$ and we want to prove it for k . So $\mathcal{C}^{(n)} \cap \mathcal{R}^*(S)$ UNSAT, $\mathcal{C}^{(n-1)} \cap \mathcal{R}^*(S)$ UNSAT, $\dots$ $\mathcal{C}^{(k+1)} \cap \mathcal{R}^*(S)$ UNSAT, and we want to **prove** $\mathcal{C}^{(k)} \cap \mathcal{R}^*(S)$ UNSAT, too.

By **contradiction**: we assume $\mathcal{C}^{(k)} \cap \mathcal{R}^*(S)$ is SAT. Then there is $v : \mathcal{L} \rightarrow \{0, 1\}$ such that for all $C \in \mathcal{C}^{(k)} \cap \mathcal{R}^*(S)$, $v \models C$ ($\tilde{v}(C) = 1$).

Let us define the two following variants $v^+, v^- : \mathcal{L} \rightarrow \{0, 1\}$ of v :

- $v^+(p_{k+1}) := 1$
- $v^-(p_{k+1}) := 0$
- $v^+(p_i) := v(p_i) =: v^-(p_i)$ for all $i \neq k+1$

By I.H. $\mathcal{C}^{(k+1)} \cap \mathcal{R}^*(S)$ is UNSAT, then $v, v^+, v^- \not\models \mathcal{C}^{(k+1)} \cap \mathcal{R}^*(S)$. Then there is at least a clause $C_1 \in \mathcal{C}^{(k+1)} \cap \mathcal{R}^*(S)$ such that $\tilde{v}^+(C_1) = 0$ ($v^+ \not\models C_1$) ($\star$), analogously there must be a $C_2 \in \mathcal{C}^{(k+1)} \cap \mathcal{R}^*(S)$ such that $\tilde{v}^-(C_2) = 0$ ($v^- \not\models C_2$) ($\star\star$).

Key Observation: p_{k+1} , as a propositional letter, occurs in C_1 and, precisely, only in the literal $\neg p_{k+1}$ (same thing but not negated in C_2)

$$\neg p_{k+1} \in C_1, \quad p_{k+1} \notin C_1$$

Proof.

- p_{k+1} cannot occur in C_1 as a literal, for otherwise $v^+(p_{k+1}) = 1$ implies $\tilde{v}^+(C_1) = 1$, contradicting $(\star)$
- p_{k+1} must occur in C_1 in the literal $\neg p_{k+1}$, for otherwise $p_{k+1} \notin C_1$, $\neg p_{k+1} \notin C_1$, so $C_1 \in \mathcal{C}^{(k)} \cap \mathcal{R}^*(S)$, and so $\tilde{v}(C_1) = 1$, and since p_{k+1} does not occur as a variable in C_1 , $\tilde{v}^+(C_1) = \tilde{v}(C_1) = 1$ that is $\tilde{v}^+(C_1) = 1$, contradicting $(\star)$.

So $\neg p_{k+1} \in C_1$, $p_{k+1} \notin C_1$.

□

Analogously: p_{k+1} must be in C_2 in the form p_{k+1} . Then $\neg p_{k+1} \in C_1$, $p_{k+1} \notin C_1$, $p_{k+1} \in C_2$, $\neg p_{k+1} \notin C_2$.

But then there **exists the resolvent**:

$$\begin{aligned} D &= \mathbb{R}(C_1, C_2; \neg p_{k+1}, p_{k+1}) \\ &= (C_1 \setminus \{\neg p_{k+1}\}) \cup (C_2 \setminus \{p_{k+1}\}) \\ &= (C_1 \cup C_2) \setminus \{p_{k+1}, \neg p_{k+1}\} \end{aligned}$$

Notice that D is such that $\neg p_{k+1} \notin D$ and $p_{k+1} \notin D$, so

$$D \in \mathcal{C}^{(k)} \cap \mathcal{R}^*(S) \implies \tilde{v}(D) = 1$$

Then v must satisfy at least one literal in $M \in D$, ($\tilde{v}(M) = 1$).

So $M \in D = (C_1 \cup C_2) \setminus \{p_{k+1}, \neg p_{k+1}\}$, and $\tilde{v}(M) = 1$, so we have two cases:

1. $M \in C_1 \setminus \{\neg p_{k+1}\}$, then $\tilde{v}(C_1) = 1$, and then since p_{k+1} does not occur in $C_1 \setminus \{\neg p_{k+1}\}$ (as a letter), $\tilde{v}^+(C_1) = \tilde{v}(C_1) = 1$, contradicting $(\star)$.
2. $M \in C_2 \setminus \{p_{k+1}\}$, then $\tilde{v}(C_2) = 1$, and then since p_{k+1} does not occur in $C_2 \setminus \{p_{k+1}\}$ (as a letter), $\tilde{v}^-(C_2) = \tilde{v}(C_2) = 1$, contradicting $(\star\star)$.

As there are no more cases to consider we have concluded our proof by contradiction, showing that it is not the case that $\mathcal{C}^{(k)} \cap \mathcal{R}^*(S)$ is SAT, implying, in any case, that $\mathcal{C}^{(k)} \cap \mathcal{R}^*(S)$ is UNSAT. This concludes induction step and proof by induction. We have proved that $\mathcal{C}^{(k)} \cap \mathcal{R}^*(S)$ is UNSAT for all $k = n, \dots, 0$.

□

Remark 5.1.10. Say S is an **infinite theory**. In the first step of the proof we reduced S to $S_{fin} \subseteq_{\omega} S$ by compactness theorem.

This application of compactness theorem doesn't provide us with a decision procedure for the input S , when S is infinite, since, in general, we don't know anything useful on the concrete nature of S_{fin} (we have no clues on how to choose such $S_{fin} \subseteq_{\omega} S$, S_{fin} UNSAT). The best thing we can do is to **build a possibly infinite sequence of finite subsets** of S :

$$S_1 \subseteq_{\omega} S_2 \subseteq_{\omega} \dots \subseteq_{\omega} S \text{ s.th. } \bigcup_{i \in \omega} S_i = S$$

Then, for each S_i , which are all finite, compute $\mathcal{R}^*(S_i)$.

Remark 5.1.11. If S is infinite, at step i :

- If $\square \in \mathcal{R}^m(S_i)$ for some $m \in \omega$:

$$\begin{aligned} &\implies \square \in \mathcal{R}^*(S_i) \\ &\implies \square \in \mathcal{R}^*(S) \\ &\implies S \text{ UNSAT} \end{aligned}$$

- If we find $t \in \omega$, $\square \notin \mathcal{R}^t(S_i)$ and $\mathcal{R}^t(S_i) = \mathcal{R}^{t+1}(S_i)$, then S_i is SAT, go to the next step $i + 1$

If S_i UNSAT (which is decidable) then S UNSAT and stop, but if S_i is SAT (also decidable) then go on. This is a semi-decidability procedure for “ S SAT”.

Definition 5.1.12. A **deduction by resolution** of a clause C from a set of clauses S ($S \vdash_R C$) is a finite sequence $C_1, C_2, \dots, C_u$ of clauses such that

1. $C_u = C$
2. For all indexes $i \in \{1, \dots, u\}$ either
 - $C_i \in S$
 - $C_i = \mathbb{R}(C_j, C_k; L, \bar{L})$ for some $j, k < i$ (and some literals $L \in C_j, \bar{L} \in C_k$)

$S \vdash_R C$ by correctness, implies that $S \models C$.

A **deduction by resolution** of $\square$ from S ($S \vdash_R \square$) is called a **refutation** of S (and then S is UNSAT, i.e. $S \models \square$).

The **Refutational Completeness** of $\vdash_R$ shows:

$$\begin{array}{ll} S \text{ UNSAT} & \text{iff } S \vdash_R \square \\ \text{equivalently } S \models \square & \text{iff } S \vdash_R \square \end{array}$$

A calculus C is **refutationally complete** iff $\Gamma \models \perp$ implies $\Gamma \vdash_C \perp$ (whenever Γ is contradictory, i.e., is UNSAT, then the calculus C proves such contradiction). A calculus C is “**tout-court**”/fully complete (just complete) iff $\Gamma \models A$ implies $\Gamma \vdash_C A$ for any $A \in \mathcal{F}_{\mathcal{L}}$.

Example 5.1.13. Examples of deductions (not necessarily refutations) in $\vdash_R$

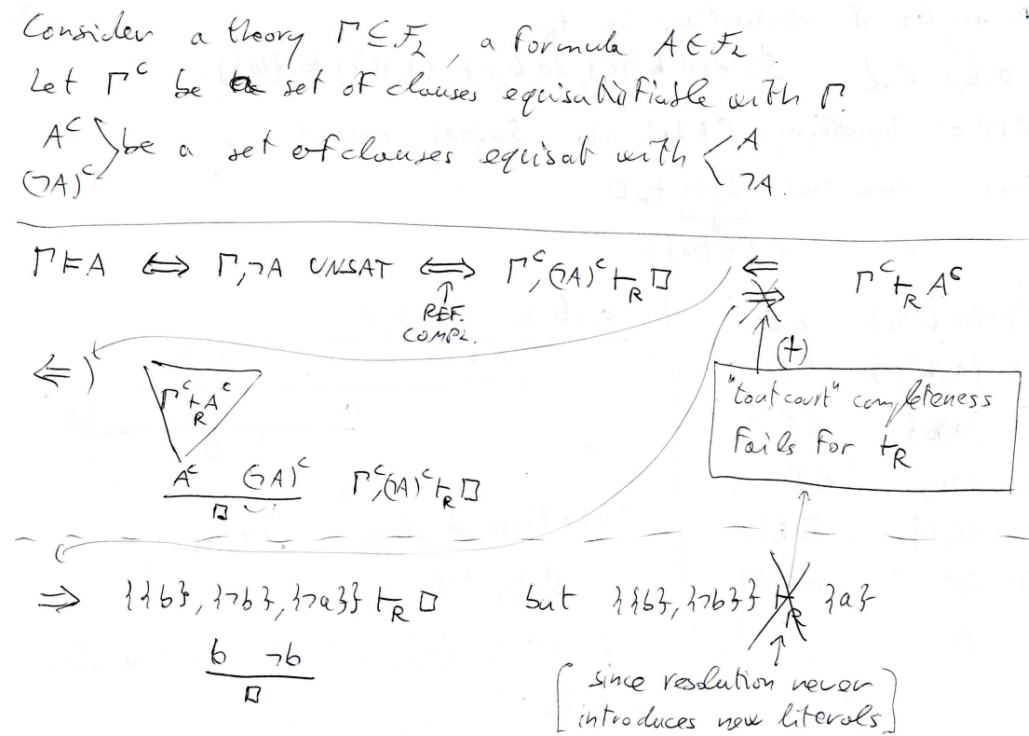
$$a, b, c \in \mathcal{L} \quad S = \{\{a, b, \neg c\}, \{a, b, c\}, \{a, \neg b\}\} \models^? \{\{a\}\}$$

Steps:

0. Transform $S \models^? \{a\}$ into $S \cup \{\neg a\}$ UNSAT? That is, show that $S, \neg a \vdash_R \square$
1. Choose a formula from S , $\{a, b, \neg c\} \in S$

1) $\{a, b, \neg c\} \in S$
 2) $\{a, b, c\} \in S$
 3) $\{a, \neg b\} \in S$
 4) $\{\neg a\} \in S$
 5) $\{a, b\} \quad R(1,2)$
 6) $\{a\} \quad R(3,5)$
 7) $\square \quad R(4,6)$

The First 6 steps contradict
 $S \vdash_R \{a\}$
 A deduction by resolution of $\{a\}$ from $S, \neg a$



So $(\dagger)$ shows that $\vdash_R$ is only **refutationally complete**, that is

$$\Gamma \models \perp \Leftrightarrow \Gamma \vdash_C \perp$$

$$(S \models \square \Leftrightarrow S \vdash_R \square)$$

but it's **not** complete

$$\Gamma \models A \Leftrightarrow \Gamma \vdash_C A$$

$$(S \models C \not\Leftrightarrow S \vdash_R C)$$

This is not a problem from the computational point of view, because, via a suitable and efficient "pre-processing", we reduce $\Gamma \models^? A$ to a refutational problem $\Gamma^C, (\neg A)^C \vdash_R^? \square$.

Actually, from the computational point of view, calculi that are just refutationally complete, may be preferable for efficiency reasons with respect to fully complete calculi.

← (there's an exercise in the last slide)

5.2 Axiomatic Systems (Hilbert's style calculus)

The Hilbert's style calculus $\mathcal{H}$ is another formal logical system that can be used for mathematical logic. It's **based on a set of axioms** and inference rules that are designed to formalize mathematical reasoning. It's relatively short and **not well suited to automated deduction**, due to its nature.

The **Axioms** are:

- $A_1 : A \rightarrow (B \rightarrow A)$
- $A_2 : (A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$
- $A_3 : (\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B)$

Inference rule(s):

- **Modus Ponens:**

$$\frac{A \quad A \rightarrow B}{B}$$

← (Axioms are tautologies, and recall that $\{\neg, \rightarrow\}$ is functionally complete)

A **Proof** of $A \in \mathcal{F}_{\mathcal{L}}$, from some theory $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$ in the **Hilbert Style calculus** $\Gamma \vdash_{\mathcal{H}} A$ is a **finite sequence** $A_1, \dots, A_u$ **of formulas such that**:

1. $A_u = A$
2. For each $i \in \{1, 2, \dots, u\}$ it holds that either:
 - $A_i \in \Gamma$, or
 - A_i is an instance of one of the axiom schemes A_1, A_2, A_3
 - there are $j, k < i$ such that $A_j = A_k \rightarrow A_i$ so that

$$\frac{A_k \quad A_k \rightarrow A_i}{A_i}$$

We have proof that $\mathcal{H}$ is **sound** and **complete** (fully, not just refutationally)

$$\Gamma \models A \text{ iff } \Gamma \vdash_{\mathcal{H}} A$$

Example 5.2.1. As an **example** Using both refutational and hilbert calculi, lets verify that

$$\neg\neg A \models^? A$$

Refutational $\mathcal{R}$

First of all we substitute the formula A for a propositional letter, obtaining

$$\neg\neg p \stackrel{?}{\models} p$$

We then change the problem in an unsatisfiability one

$$\{\neg\neg p, \neg p\} \text{ UNSAT}$$

Applying one step of resolution using $D = \mathbb{R}(\{p\}, \{\neg p\}; p, \neg p)$ we have that:

$$\{\{p\}, \{\neg p\}\} \vdash_R \square$$

Hilbert $\mathcal{H}$

In $\mathcal{H}$ we do not have any preprocessing to do. Steps:

$$\begin{array}{c} Ax_1 \frac{}{\neg\neg A \rightarrow (\neg\neg\neg\neg A \rightarrow \neg\neg A)} \quad \neg\neg A \\ MP \frac{}{\neg\neg\neg\neg A \rightarrow \neg\neg A} \\ MP \frac{}{\neg A \rightarrow \neg\neg\neg A} \quad Ax_3 \frac{}{(\neg\neg\neg\neg A \rightarrow \neg\neg A) \rightarrow (\neg A \rightarrow \neg\neg\neg A)} \\ MP \frac{}{\neg A \rightarrow \neg\neg\neg A} \quad Ax_3 \frac{}{(\neg A \rightarrow \neg\neg\neg A) \rightarrow (\neg\neg A \rightarrow A)} \\ MP \frac{}{\neg\neg A \rightarrow A} \quad A \\ MP \frac{}{\neg\neg A} \end{array}$$

$\leftarrow$ (EXERCISE: try doing the same thing trying to verify that $\models A \rightarrow A$)

5.3 Refutational procedure by David Putnam (DPP)

Problem: Maintaining (refutational) completeness by selecting among all possible resolvents.

Definition 5.3.1. A clause C_1 **subsumes** C_2 (or C_2 is subsumed by C_1) **iff** $C_1 \subset C_2$ (Notice: $\models C_1 \rightarrow C_2$, equivalently $\{C_1, C_2\} \equiv \{C_1\}$).

Subsumption rule: Let S be a set of clauses and let S' be obtained from S , by removing all subsumed clauses. Then $S \equiv S'$, preserving logical equivalence (Notice: S' cannot be further reduced by subsumption).

Definition 5.3.2. A clause C is **trivial** iff it contains L and $\bar{L}$ for some literal L . Notice $\models C$, if $C \in S$ then $S \equiv S \setminus \{C\}$.

Elimination of trivial clauses rule: Let S be a set of clauses and let S' be obtained from S by removing all trivial clauses. Then $S \equiv S'$.

Notice: If $D = \mathbb{R}(C_1, C_2; L, \bar{L})$ and C_1, C_2 are not trivial

$$(C_1 \setminus \{L\}) \cup (C_2 \setminus \{\bar{L}\}) = (C_1 \cup C_2) \setminus \{L, \bar{L}\}$$

While if it was trivial

$$\begin{aligned} \mathbb{R}(\{a, \neg a\}, \{a, \neg a\}, a, \neg a) &= (\{a, \neg a\} \setminus \{a\}) \cup (\{a, \neg a\} \setminus \{\neg a\}) \\ &= \{\neg a\} \cup \{a\} = \{a, \neg a\} \\ &\neq (\{a, \neg a\} \cup \{a, \neg a\}) \setminus \{a, \neg a\} = \square \end{aligned}$$

So, **trivial clauses do not carry meaningful information** and removing them allows us to be more efficient

5.3.1 Steps of DPP

Input: A finite set of clauses S , S is already “cleaned” from **subsumed and trivial clauses** We call the process of “cleaning”

$$S = \text{clean}(S)$$

Then DPP consists of the following steps:

1. **Choose** a propositional letter $p \in \mathcal{L}$ among those occurring in clauses of S , ($p \in \text{Var}(S)$). p is called the “**pivot**” the step. How do we choose? No fixed way, heuristically.

2. **Form**

$$S' = \{C \in S : p \notin C \text{ and } \neg p \notin C\}$$

S' is the set of **p -free clauses** of S .

3. **Form**

$$S'' = \{D = \mathbb{R}(C_1, C_2; p, \neg p) : C_1, C_2 \in S \setminus S'\}$$

S'' is the set of **p -resolvents** of S .

4. **Form**

$$S''' = \text{clean}(S' \cup S'')$$

The cleaned version of S'' .

5. **Output** S''' and go to the **next step**, i.e., repeat the process with another literal.

If at the end we get $\square$ then S UNSAT, otherwise we get $\emptyset$ and S is SAT.

$\leftarrow$ (Mind that whenever you find *square* you can stop and output UNSAT, not necessarily the last step)

Examples $S = \{\{a, b, c\}, \{a, b, d\}, \{a, b, c, d\}, \{a, d\}, \{a, c, d\}, \{c\}\}$

$S_0 = \text{clean}(S) = S$

1) Pivot: c

2) c-free: $\{\{a, b, d\}, \{a, d\}\}$

3) c-resolvents: $\{\{a, b\}, \{a, b, d\}, \{a, d\}\}$

4) clean: $\{\{a, b, d\}, \{a, d\}, \{a, b\}, \{a, b, d\}, \{a, d\}\}$

1) Pivot: a

2) a-free: $\emptyset$

3) a-resolvents: $\{\{b, d\}, \{b, d\}, \{b, d\}, \{b, d\}, \{b, d\}, \{b, d\}\}$

4) clean: $\{\{b, d\}, \{b, d\}, \{b, d\}, \{b, d\}\}$

1) Pivot: b

2) b-free: $\emptyset$

3) b-resolvents: $\{\{d\}, \{d\}, \{d\}, \{d\}\}$

4) clean: $\{\{d\}, \{d\}\}$

1) Pivot: d

2) d-free: $\emptyset$

3) d-resolvents: $\{\Box\}$

4) clean: $\{\Box\}$

We have reached $\{\Box\}$
DPP stops: S UNSAT.

Example: $S = \{\{a, b, c\}, \{b, c\}, \{a, c, d\}, \{b, c, d\}, \{a, b, d\}, \{a, d, b\}, \{b, c, d\}, \{c, c\}\}$

$S_0 = \text{clean}(S)$

1) Pivot: b

2) b-free: $\{\{a, c, d\}\}$

3) b-resolvents: $\{\{a, c\}, \{a, c, d\}, \{a, c, d\}, \{c, d\}, \{a, c, d\}, \{a, c, d\}\}$

4) clean: $\{\{a, c\}, \{a, c, d\}, \{c, d\}\} = S_1$

1) Pivot: c

2) c-free: $\emptyset$

3) c-resolvents: $\{\{a, d\}, \{a, d\}\}$

4) clean: $\{\{a, d\}\} = S_2$

1) Pivot: a

2) a-free: $\emptyset$

3) a-resolvents: $\emptyset$

4) clean: $\emptyset$ $S_3 = \emptyset$

This implies, we shall
show why, that S
is SAT.

We have reached $S_3 = \emptyset$, DPP stops

5.3.2 Model Building

Referring to the last example. We shall use the DPP proof to get a **satisfying assignment** for S . If a **letter** has **never been used as a pivot** in any step let us call it “**outsider**”. To the outsiders we assign **whatever truth value**. So, in our earlier example $v(d) = 0$.

Let us **substitute the values already assigned to each step**, proceeding **backwards from the last step** $\neq \emptyset$. So we start with:

$$S_2 = \{\{a, \neg d\}\}$$

With $v(d)$ known, define $v(a)$ in order to satisfy S_2 : $v(a) = 0$. Going backwards:

$$S_1 = \{\{a, c, d\}, \{\neg c, \neg d\}, \{a, c, \neg d\}\}$$

Pivot of $S_1 \rightsquigarrow S_2$ is c . We now find a suitable $v(c)$ and looking at the first clause we can say that $v(c) = 1$. Last step:

$$S_0 = \{\{a, \neg b, c\}, \{b, \neg c\}, \{a, c, \neg d\}, \{\neg b, \neg d\}, \{a, b, d\}, \{\neg a, b, d\}\}$$

The pivot of $S_0 \rightsquigarrow S_1$ is b . We need to chose $v(b)$, looking at the already satisfied ones we see that the choice must be $v(b) = 1$.

5.3.3 Termination

Proof. Let us display the DPP over input S as the sequence of the outputs of the steps $DPP(S) = S_0, S_1, \dots, S_k$ with $S_0 = \text{clean}(S)$, we have that:

- S_i is obtained from S_{i-1} by one DPP step on pivot $q_i \in \{p_1, p_2, \dots, p_n\}$.
- Observe that S_i does not contain any occurrence of q_i (as a propositional letter).
- After at most $k \leq n$ steps it holds that $\text{Var}(S_k) = \emptyset$ (there are no more letters).
- So, either $S_k = \{\square\}$ or $S_k = \emptyset$.

□ ← (Exercise)

5.3.4 Correctness and refutational completeness of DPP

We call k the last step of DPP on input S , obviously $k \leq n$ where $\text{Var}(s) \subseteq \{p_1, \dots, p_n\}$.

To prove Correctness we need to prove one of two forms:

- If we reach $\square$, $S_k = \{\square\}$, then S_0 is UNSAT
- if S_0 is SAT then $S_k = \emptyset$

To prove Refutational completeness we need to prove one of two forms:

- If S_0 is UNSAT then $S_k = \{\square\}$
- if $S_k = \emptyset$ then S_0 is SAT (we'll prove this form)

Correctness

If $S_k = \{\square\}$ then S_0 is UNSAT.

Proof. It's just a corollary of the correctness theorem 5.1.2, since $S_i \models S_{i+1}$ then $S_0 \models \{\square\}$, that is S_0 is UNSAT. □

← (its not so direct since in this case we are in DPP but just think about it and its pretty easy also to prove that $S_i \models S_{i+1}$ also in DPP)

Refutational Completeness

We have to prove that $S_k = \emptyset$ implies S_0 is SAT.

Proof. By descending induction on k . We shall prove S_i SAT for $i = k, k-1, \dots, 1, 0$.

Base: $i = k, S_i = S_k = \emptyset$. By definition of $\emptyset$, S_i this is SAT.

Step: Assume S_{i+1} is SAT, with the aim to prove S_i SAT. Since S_{i+1} is SAT, we know that there is $v : \mathcal{L} \rightarrow \{0, 1\}$ such that $v \models S_{i+1}$. We define two variants of v , $v^-, v^+ : \mathcal{L} \rightarrow \{0, 1\}$:

$$\begin{cases} v^+(q_{i+1}) = 1 \\ v^-(q_{i+1}) = 0 \end{cases} \quad v^+(p) = v = v^-(p) \text{ for all other } p \in \mathcal{L}, p \neq q_{i+1}.$$

q_{i+1} is the pivot in step $S_i \rightsquigarrow S_{i+1}$. We shall show that either $v^+ \models S_i$ or $v^- \models S_i$, that is, S_i is SAT ($\dagger$).

By contradiction we assume that $v^+ \not\models S_i$ and $v^- \not\models S_i$ ($\dagger\dagger$). Then there are $C_1, C_2 \in S_i$ such that $v^+ \not\models C_1$ and $v^- \not\models C_2$.

Key observation: $\neg q_{i+1} \in C_1$ and $q_{i+1} \notin C_1$ (note that C_1 is not trivial).

Proof. For otherwise, if $q_{i+1} \in C_1$ (as a literal) then $v^+(C_1) = 1$ contradicting ($\dagger\dagger$). Then $q_{i+1} \notin C_1$ (as a literal).

If $\neg q_{i+1} \notin C_1$, too, then C_1 does not contain q_{i+1} , so $C_1 \in S_{i+1}$, then $v \models C_1$, but then $v^+ \models C_1$ (since $q_{i+1} \notin \text{Var}(C_1)$), reaching another contradiction of ($\dagger\dagger$). $\square$

Analogously $q_{i+1} \in C_2$ and $\neg q_{i+1} \notin C_2$ (noting that C_2 is not trivial).

Then we can form the resolvent

$$D = \mathbb{R}(C_1, C_2; \neg q_{i+1}, q_{i+1})$$

and D is not trivial since C_1 and C_2 are not. So $D \in q_{i+1}$ -resolvents $\implies$ either $D \in S_{i+1}$ or there is $D' \in S_{i+1}$ and D' subsumes D . In both cases $v \models D$.

$v \models D$ implies $v^+ \models D$ and $(v^- \models D)$, since $q_{i+1} \notin \text{Var}(D)$, $\implies v^+(C_1) = 1$ ($v^-(C_2) = 1$), contradicting ($\dagger\dagger$).

We have reached in any case contradiction. We proved by contradiction that either $v^+ \models S_i$ or $v^- \models S_i$, that is S_i SAT. Then S_0 is SAT. $\square$

Remark 5.3.3. Let S be a finite set of clauses. Then $\text{Var}(S) \subseteq \{p_1, \dots, p_n\}$ for some $n \leq ||S||$. Then

$$DPP(S) = S_0, S_1, \dots, S_k = \begin{cases} \{\square\} & \implies S \text{ UNSAT} \\ \emptyset & \implies S \text{ SAT} \end{cases}$$

with $k \leq n \leq ||S||$.

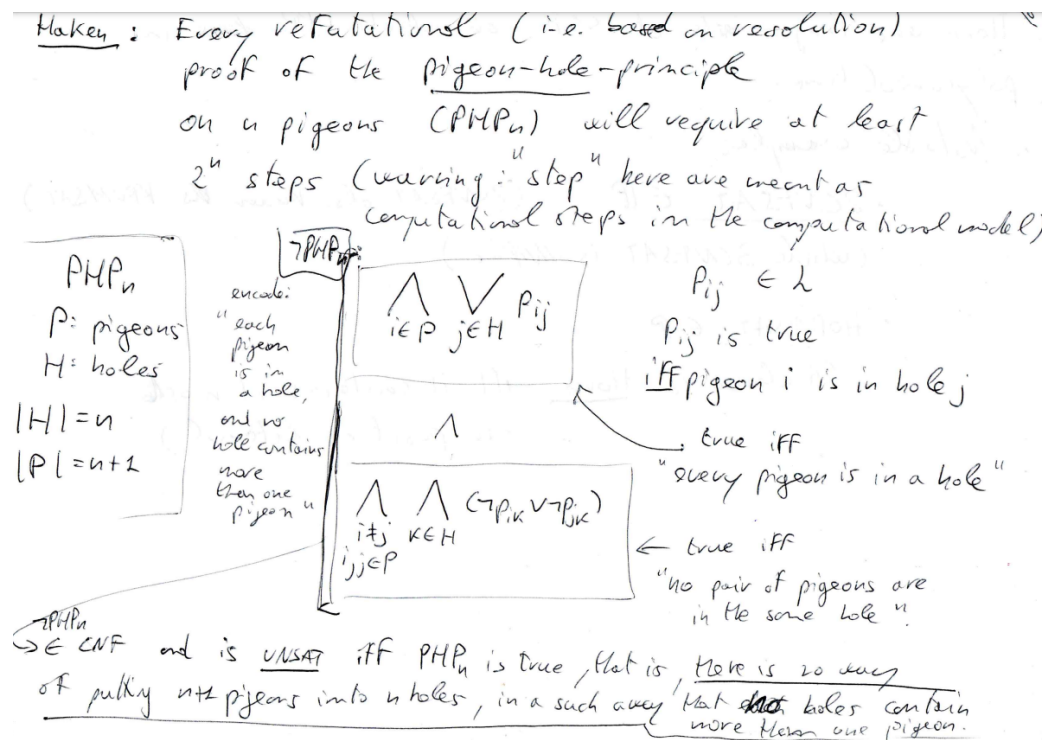
If we can guarantee that $||S_i||$ is always **polynomially bounded** by the size of the input $||S||$ then the overall length of the DPP proof, $||S_0, \dots, S_k||$ would be **polynomially short with respect to $||S||$** .

If the **choice of the pivot** at each step is **polynomially short** with respect to $||S||$ the **overall time** used by $DPP(S)$ would be **polynomial with respect to $||S||$** , too.

$\leftarrow$ (but in general DPP in NOT polynomial the length of S_i could explode exponentially)

There are fragments of CNF on which DPP runs on polynomial time. Notable examples:

- $2CNFSAT \in \mathcal{P}$ (also known as KROMSAT)
- $HORNSAT \in \mathcal{P}$ (A clause is Horn iff it contains at most one positive literal)



Improvements on DPP: DPLL (Davis-Putnam-Loveland-Logemann): On input S (finite set of clauses) tries to build a satisfying assignment by **extending a partial assignment**. Split rule: at a certain point assign at the pivot either 0 or 1, **fork the possibilities for eventual backtracking**. Saves space with respect to DPP, by using backtracking.

Unit propagation:

- unit resolution: resolvents only among singletons and Clauses $\mathbb{R}(\{L\}, C, L, \bar{L}) = C \setminus \{\bar{L}\}$
- unit subsumption: if $L \in C \implies$ delete C (if you're assuming $v(L) = 1$)

Improvement on DPLL: CDCL: Conflict Driven Clause Learning: DPLL with **non-chronological backtracking** (backjumping). Building an implication graph while building the DPLL which is able to find conflict clauses (not satisfied by the partial assignment of the DPLL). Learn the clause: add a clause recording the conflict to the input and backtrack to the first choice that originated the conflict

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✂ First-Order Logic

Introductory example:

- Every man is mortal (p)
- Socrates is a man (q)

-
- Therefore Socrates is a mortal (r)

They are all atomic, we **can't relate them using propositional logic**. To handle reasoning such as the syllogism about Socrates we must adopt richer (more expressive) language.

Syntax: We'll enrich syntax by adding **new** kinds of **symbols**:

- Quantifiers
- Predicate symbols
- Function symbols (including constants)
- Variables

Semantics: We attach **new meanings** to new concepts, to assign a formal way to interpret the new “ingredients” of the language, with the aim of being able to express more refined aspects of the possible “worlds” that we want to model.

Try to develop **calculi fit to be automated**, to (semi) decide problems like the earlier syllogism.

6.1 Syntax of First Order Logic

At propositional level, we considered a unique language (the set of propositional letters $\mathcal{L}$). From this we derived the set $\mathcal{F}_{\mathcal{L}}$ of propositional formulas.

In First order logic we'll consider several distinct languages called **elementary languages**

Definition 6.1.1. An **elementary language** $\mathcal{L} = (\mathcal{P}, \mathcal{F}, \alpha, \beta)$, where:

- $\mathcal{P}$ is the set of all “**Predicates**” ($\mathcal{P} \neq \emptyset$)
- $\mathcal{F}$ is the set of symbols, called “**Function symbols**” ($\mathcal{P} \cap \mathcal{F} = \emptyset$)
- Arity functions:

$$\begin{cases} \alpha : \mathcal{P} \rightarrow \mathbb{N} & \text{assigns the \textbf{arity} of \textbf{each} } p \in \mathcal{P} \\ \beta : \mathcal{F} \rightarrow \mathbb{N} & \text{assigns the \textbf{arity} of \textbf{each} } f \in \mathcal{F} \end{cases}$$

“**Ingredients**” common to every elementary language:

- $\mathcal{V}$: an infinite set of symbols ($\mathcal{V} \cap \mathcal{P} \neq \emptyset$, $\mathcal{V} \cap \mathcal{F} \neq \emptyset$), called “**individual variables**”
- **Connectives:** $\wedge, \vee, \neg, \rightarrow$ (and all derived connectives $\perp, \top, \leftrightarrow, \dots$)
- **Quantifiers:** $\forall, \exists$

The formulas of the formal language $\mathcal{F}_{\mathcal{L}}$ (formulas over the elementary language $\mathcal{L}$) will be defined from $\mathcal{L} : (\mathcal{P}, \mathcal{F}, \alpha, \beta)$ together with the common “ingredients”.

← (ci sono degli esempi sulle slide se vuoi sono un po' inutili)

Key stipulation: If $= \in \mathcal{P}$, then $\alpha(=) = 2$ (its **arity is always 2**), and the language with $= \in \mathcal{P}$ is called a **language with identity**.

6.1.1 Formulas in $\mathcal{F}_{\mathcal{L}}$

The definition of formulas of $\mathcal{F}_{\mathcal{L}}$ ($F \in \mathcal{F}_{\mathcal{L}}$) is given **by layers** and **inductively** on each one of the two layers.

First layer: $\mathcal{L}$ -terms.

Definition 6.1.2. Let $\mathcal{L} : (\mathcal{P}, \mathcal{F}, \alpha, \beta)$. Then the **set $\mathcal{T}_{\mathcal{L}}$ of $\mathcal{L}$ -terms** is defined as follows:

- **Base:**
 - Every $x \in \mathcal{V}$ (x is an **individual variable**, is an $\mathcal{L}$ -term, $x \in \mathcal{T}_{\mathcal{L}}$)
 - Every $C \in \mathcal{F}$, such that $\beta(c) = 0$ is an $\mathcal{L}$ -term ($c \in \mathcal{T}_{\mathcal{L}}$, it's a **constant symbol**)
- **Step:** If $f \in \mathcal{F}$ is such that $\beta(f) = n > 0$ and $t_1, t_2, \dots, t_n \in \mathcal{T}_{\mathcal{L}}$ then $f(t_1, t_2, \dots, t_n)$ is an $\mathcal{L}$ -term ($\in \mathcal{T}_{\mathcal{L}}$)

Nothing else is an $\mathcal{L}$ -term.

Second Layer: $\mathcal{L}$ -formulas.

Definition 6.1.3. Let $\mathcal{L} : (\mathcal{P}, \mathcal{F}, \alpha, \beta)$. The **set $\mathcal{F}_{\mathcal{L}}$ of $\mathcal{L}$ -formulas** is defined as follows:

- **Base:** if $P \in \mathcal{P}$ and $\alpha(P) = n$, and $t_1, t_2, \dots, t_n \in \mathcal{T}_{\mathcal{L}}$ then $P(t_1, t_2, \dots, t_n)$ is an $\mathcal{L}$ -formula called Atomic (cannot split an atomic formula into another formula, only into terms)
- **Step:**
 - If $A, B \in \mathcal{F}_{\mathcal{L}}$ then $(A \wedge B), (A \vee B), (\neg A), (A \rightarrow B)$ are $\mathcal{L}$ -formulas.
 - If $A \in \mathcal{F}_{\mathcal{L}}$ and $x \in \mathcal{V}$ then $(\forall x A)$ and $(\exists x A)$ are $\mathcal{L}$ -formulas ($\in \mathcal{F}_{\mathcal{L}}$)

Nothing else is an $\mathcal{L}$ -formula.

Remark 6.1.4. For simplicity's sake, we shall omit parentheses when there is **no risk of ambiguity**.

Terminology 6.1.5. An $\mathcal{L}$ -term is called **“ground”** if it is built without using variables.

Terminology 6.1.6. An occurrence of a variable $x \in \mathcal{V}$ in a formula $A \in \mathcal{F}_{\mathcal{L}}$ is **“bound”** (“constrained”) if it occurs in a subformula $B \preceq A$ of the form $(\forall x C)$ or $(\exists x C)$ (x is in the scope of some quantifier, i.e., it “can be seen” from a $\forall$ or $\exists$).

Terminology 6.1.7. If an occurrence of $x \in \mathcal{V}$ in $A \in \mathcal{F}_{\mathcal{L}}$ is not bound then x **occurs free** in A . A **variable** $x \in \mathcal{V}$ **occurs free** in $A \in \mathcal{F}_{\mathcal{L}}$ if some of its occurrences are free (otherwise the variable is bound). We denote with $FV(A)$ the **set of variables occurring free** in A .

Definition 6.1.8. An $\mathcal{L}$ -sentence or **closed $\mathcal{L}$ -formula** is an $\mathcal{L}$ -formula **without free variables** ($FV(A) = \emptyset$).

Definition 6.1.9. With the notation: $A[t/x]$ where $A \in \mathcal{F}_{\mathcal{L}}$, $x \in \mathcal{V}$, $t \in \mathcal{T}_{\mathcal{L}}$ we mean the $\mathcal{L}$ -formula obtained by **replacing** simultaneously all and only the **free occurrences** of x with t .

Exercise: Provide a notion of “certificate” for $\mathcal{L}$ -terms and $\mathcal{L}$ -formulas analogous to the notion of $\mathcal{L}$ -constructions for propositional logic.

$$\leftarrow (FV((\forall x(R(x, y)) \vee P(X)) = \{x, y\}, FV(\forall x(R(x, y))) = \{y\})$$

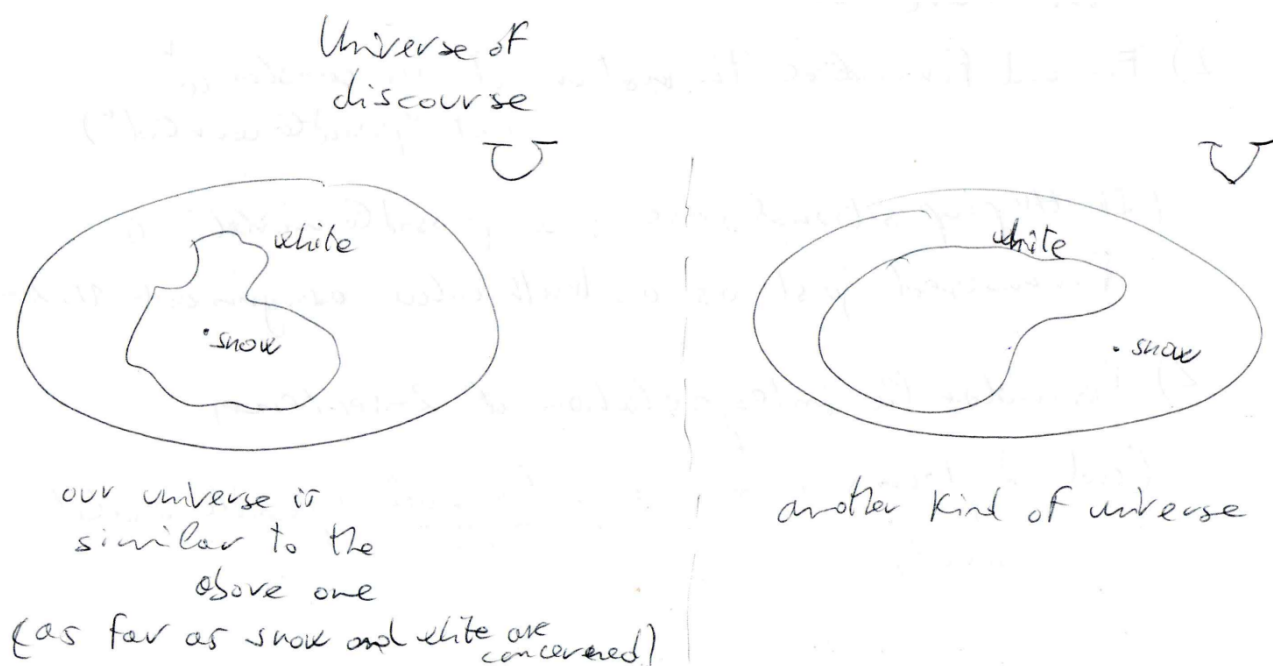
6.2 Semantics of First order logic

To provide an **interpretation of an $\mathcal{L}$ -sentence** (whether it's true or false) in every given “circumstance” (or “possible world”) we have to:

1. Fix and formalize the **notion of “circumstance”** (or “possible world”, in the propositional case, a “possible world” is formalized just as a truth value assignment)
2. Formalize the **interpretation of $\mathcal{L}$ -sentences** (and (ground) $\mathcal{L}$ -terms) in every formal “possible world”.

Idea (Tarskian semantics): The **truth** is the **correspondence with the state of things**, e.g., the sentence:

“The snow is white” is true (in a given world) iff the actual snow is actually white.



6.2.1 $\mathcal{L}$ -Structures

Let $\mathcal{L} = (\mathcal{P}, \mathcal{F}, \alpha, \beta)$. An $\mathcal{L}$ -structure is a pair $\mathcal{A} = (A, I)$, where A is just a set, called “universe of discourse” (or “domain”). It is **required** $A \neq \emptyset$. And I is a function called “Interpretation”, defined as follows:

- For all $P \in \mathcal{P}$, $\alpha(P) = n$

$$I(P) \subseteq A^n$$

$I(P)$ is an n -ary relation, that is, it's a set of n -tuples of elements of A .

- For all $f \in \mathcal{F}$, $\beta(f) = n$

$$I(f) : A^n \rightarrow A$$

$I(f)$ is a function from the set of all n -tuples of elements of A to A .

Remark 6.2.1. If $c \in \mathcal{F}$, $\beta(c) = 0$, if c is a constant, then

$$I(c) : A^0 \rightarrow A$$

but $A^0 = \{f : \emptyset \rightarrow A\} = \{\emptyset\}$. It takes the set containing only the $\emptyset$ and fixes an element of A to it. The “left side” ($\emptyset$ part) carries no information, so we identify $f : A^0 \rightarrow A$ with the element $f(\emptyset) \in A$, $I(c) \in A$.

Remark 6.2.2. If $P \in \mathcal{P}$, $\alpha(P) = 0$

$$I(P) \subseteq A^0 \implies I(P) \subseteq \{\emptyset\}$$

So $I(P)$ can be either $\emptyset$ or $\{\emptyset\}$ (we can call them 0 and 1 respectively).

Remark 6.2.3. To each relation we can associate its characteristic function:

$$Q \in \mathcal{P} \quad \alpha(Q) = n$$

$$\chi_{I(Q)} : A^n \rightarrow \{0, 1\} : \quad \chi_{I(Q)}(a_1, a_2, \dots, a_n) = 1 \quad \text{iff} \quad (a_1, a_2, \dots, a_n) \in I(Q)$$

Remark 6.2.4. Let $\mathcal{L} = (\mathcal{P}, \mathcal{F}, \alpha, \beta)$ be a language with identity, $= \in \mathcal{P}$, $\alpha(=) = 2$. The semantics of “=” is fixed. Let $\mathcal{A} = (A, I)$ be an $\mathcal{L}$ -structure, then:

$$I(=) \subseteq A^2 \quad I(=) := \{(a, a) : a \in A\}$$

Each element is in relation with only itself, this is the meaning of “=”.

We shall see later that $(I(a), I(b)) \in I(=)$ iff $I(a) = I(b)$ (assuming $a, b \in \mathcal{T}_{\mathcal{L}}$).

Define the **semantics** of $\mathcal{L}$ -sentences (and **ground $\mathcal{L}$ -terms**) in **every $\mathcal{L}$ -structure**. We proceed by layers

- layer for (ground) $\mathcal{L}$ -terms
- layer of $\mathcal{L}$ -sentences ($\mathcal{L}$ -formulas)

and inductively on each of these two layers. We have to capture the Tarskian “idea”. We’ll need to say when an $\mathcal{L}$ -structure such as $\mathcal{A} = (A, I)$ “**models**” a certain property, e.g.,

$$\mathcal{A} \models W(s)$$

6.2.2 Interpretation

We are going to establish the **interpretation of quantifiers**. We want, for instance, to establish whether in a given $\mathcal{L}$ -structure $\mathcal{A} = (A, I)$:

$$\mathcal{A} \models \exists x F$$

is true **iff** it exists at least one element $a \in A$ such that $\mathcal{A} \models F[a/x]$. While:

$$\mathcal{A} \models \forall x F$$

is true **iff** for all $a \in A$ it holds that $\mathcal{A} \models F[a/x]$.

We’re **substituting** a variable (x) with an element of the universe (a). These expressions make no sense: we cannot substitute variables with objects of the universe. We can overcome this difficulty if every object in the universe has a name (that is, there is a term whose interpretation is that object).

← (The structure “models”/“renders true” the sentence on the right. We need to define this notation)

In general, there is no need to have a name for each object of the universe we want to talk about. In many practical cases, it's usually not the case. For instance, if we want to deal in first order logic with the real numbers we want to do it with an elementary language $\mathcal{L}$ containing just finitely many symbols and able to name just enumerably many reals.

So, in order to give the semantics to $\mathcal{L}$ -sentences over universes “too large” to have names for all objects in the universe, we “cheat” expanding the language $\mathcal{L}$ to give names to all objects. This expansion of the language will only be “instrumental” to give the semantics.

Side effect: variables will be mere “placeholders”, they will be just syntactic devices, without any direct semantics.

Expansion $\mathcal{L}$ to $\mathcal{L}_{\mathcal{A}}$:

1. Let $\mathcal{L} = (\mathcal{P}, \mathcal{F}, \alpha, \beta)$.
 2. Let $\mathcal{A} = (A, I)$ be an $\mathcal{L}$ -structure.
-
3. Define the expansion $\mathcal{L}_{\mathcal{A}}$ of $\mathcal{L}$ with respect to $\mathcal{A}$. $\mathcal{L}_{\mathcal{A}}$ is **defined** as follows:
 - (a) For all $a \in A$ add to the set of constants .. a new symbol $\bar{a}$ (the “name” of a)
 - (b) we shall interpret each $\bar{a}$ into a

Now we need create a new $\mathcal{L}$ -structure on $\mathcal{L}_{\mathcal{A}}$

1. Given:

$$\begin{aligned}\mathcal{L}_{\mathcal{A}} &:= (\mathcal{P}, \mathcal{F}^{\mathcal{A}}, \alpha, \beta^{\mathcal{A}}) \\ \mathcal{F}^{\mathcal{A}} &:= \mathcal{F} \cup \{\bar{a} : a \in A\} \quad (\bar{a} \notin \mathcal{F}) \\ \beta^{\mathcal{A}} &= \beta \text{ for all } f \in \mathcal{F} \\ \beta^{\mathcal{A}}(\bar{a}) &= 0 \text{ for all } a \in A\end{aligned}$$

2. $\bar{\mathcal{A}} = (A, \bar{I})$, $\bar{\mathcal{A}}$ is an $\mathcal{L}_{\mathcal{A}}$ -structure. $\bar{I}$ coincides with I on $\mathcal{L}$, while $\bar{I}(\bar{a}) := a$, for all $a \in A$

So to sum up the whole procedure: First there is $\mathcal{L} = (\mathcal{P}, \mathcal{F}, \alpha, \beta)$, then there could be infinitely many $\mathcal{L}$ -structures:

$$\left\{ \begin{array}{l} \vdots \\ \mathcal{A} = (A, I) \rightsquigarrow \mathcal{L}_{\mathcal{A}} \rightsquigarrow \bar{\mathcal{A}} = (A, \bar{I}) \quad \mathcal{L}_{\mathcal{A}}\text{-structure} \\ \mathcal{B} = (B, I) \rightsquigarrow \mathcal{L}_{\mathcal{B}} \rightsquigarrow \bar{\mathcal{B}} = (B, \bar{I}) \quad \mathcal{L}_{\mathcal{B}}\text{-structure} \\ \vdots \end{array} \right.$$

Interpretation of $\mathcal{L}_{\mathcal{A}}$ -ground terms

Inductively:

- **Base:**

- old constants $\rightarrow c \in \mathcal{F}, \beta(c) = 0 \rightarrow \bar{I}(c) := I(c)$
- new constants $\rightarrow \bar{a} \in \mathcal{F}^{\mathcal{A}} \setminus \mathcal{F} \rightarrow \bar{I}(\bar{a}) := a \ (a \in A)$

- **Step:** Let $F \in \mathcal{F}$, $\beta(F) = n > 0$ and let $t_1, \dots, t_n$ be $\mathcal{L}_{\mathcal{A}}$ -terms, that are **ground**.
Then

$$\bar{I}(F(t_1, \dots, t_n)) := (I(F))(\bar{I}(t_1), \dots, \bar{I}(t_n))$$

$\leftarrow$ (remember that $I(F) : A^n \rightarrow A$)

By induction, for each **ground** $\mathcal{L}_{\mathcal{A}}$ -term, the interpretation of t is guaranteed to be an element of A , $\bar{I}(t) \in A$.

Interpretation of $\mathcal{L}_{\mathcal{A}}$ -sentences

Inductively we define $\overline{\mathcal{A}} \models F$ ($\overline{\mathcal{A}}$ “modes”/“renders true” F) for each $F \in \mathcal{L}_{\mathcal{A}}$ -sentence.

- **Base**; ground atomic $\mathcal{L}_{\mathcal{A}}$ formulas (an atomic formula is a sentence iff it is ground)

$$F = P(t_1, \dots, t_n) \text{ for } P \in \mathcal{P}, \alpha(P) = n \quad t_1, \dots, t_n \text{ are } \mathcal{L}_{\mathcal{A}}\text{-ground terms}$$

. Then we can define (remeber the iff is by definition)

$$\overline{\mathcal{A}} \models P(t_1, \dots, t_n) \text{ iff } (\bar{I}(t_1), \dots, \bar{I}(t_n)) \in I(P)$$

- Step:

$$\begin{aligned} \overline{\mathcal{A}} \models F_1 \wedge F_2 & \text{ iff } \overline{\mathcal{A}} \models F_1 \text{ and } \overline{\mathcal{A}} \models F_2 \\ \overline{\mathcal{A}} \models F_1 \vee F_2 & \text{ iff } \overline{\mathcal{A}} \models F_1 \text{ or } \overline{\mathcal{A}} \models F_2 \\ \overline{\mathcal{A}} \models \neg F_1 & \text{ iff } \overline{\mathcal{A}} \not\models F_1 \\ \overline{\mathcal{A}} \models F_1 \rightarrow F_2 & \text{ iff } \overline{\mathcal{A}} \not\models F_1 \text{ or } \overline{\mathcal{A}} \models F_2 \end{aligned}$$

- Step (again):

$$\begin{aligned} \overline{\mathcal{A}} \models \forall x F & \text{ iff } \text{for all } a \in A, \overline{\mathcal{A}} \models F[\bar{a}/x] \\ \overline{\mathcal{A}} \models \exists x F & \text{ iff } \text{there is } a \in A \text{ such that } \overline{\mathcal{A}} \models F[\bar{a}/x] \end{aligned}$$

← (note we are using $\bar{a}$ (the name) not a)

By induction, for every $\mathcal{L}_{\mathcal{A}}$ -sentence F the **truth value of F** in the structure $\overline{\mathcal{A}}$ is **completely determined**, either $\overline{\mathcal{A}} \models F$ xor $\overline{\mathcal{A}} \not\models F$.

We have completed the inductive definition of the **relation** $\overline{\mathcal{A}} \models F$ for each $\mathcal{L}$ -structure $\mathcal{A}$ and $\mathcal{L}_{\mathcal{A}}$ -sentence F . For all $F \in \mathcal{L}$ -sentence, we have that $\mathcal{A} \models F$ (as an $\mathcal{L}$ -structure) iff $\overline{\mathcal{A}} \models F$ (as an $\mathcal{L}_{\mathcal{A}}$ -structure).

← (important)

Remark 6.2.5. $F \in \mathcal{F}_{\mathcal{L}}$ but F is not in $\mathcal{L}$ -sentence. Assume

$$FV(F) \neq \emptyset \quad FV(F) \subseteq \{x_1, x_2, \dots, x_n\}$$

that is, F has some free variables. How can we define the notion

$$\mathcal{A} \models^? F$$

If F is a sentence we already specified, but if there are free variables we simply do:

$$\forall[F] := \forall x_1 \forall x_2 \dots \forall x_n F$$

we call this “universal closure of F ”, $\implies \forall[F]$ is an $\mathcal{L}$ -sentence (so $FV(\forall[F]) = \emptyset$). By definition we say: $\mathcal{A} \models F$ iff $\mathcal{A} \models \forall[F]$.

Remark 6.2.6. If $\mathcal{L}$ is a **language with identity** $\mathcal{L} = (\mathcal{P}, \mathcal{F}, \alpha, \beta)$, $= \in \mathcal{P}$, $\alpha(=) = 2$, and in every $\mathcal{L}$ -structure (with identity) $\mathcal{A} = (A, I)$ we have the same interpretation of $=$,

$$I(=) = \{(a, a) : a \in A\} \subseteq A^2$$

Then t_1, t_2 ground $\mathcal{L}$ -terms ($\mathcal{L}_{\mathcal{A}}$ -terms) then we have that

$$\mathcal{A} \models t_1 = t_2 \quad (\overline{\mathcal{A}} \models t_1 = t_2)$$

$$\text{iff} \quad \text{iff}$$

$$I(t_1) = I(t_2) \quad (\bar{I}(t_1) = \bar{I}(t_2))$$

Alternative (but equivalent) definition of $\mathcal{A} \models F$: We'll say $\mathcal{A} \models' F$ to distinguish them, but they're functionally the same. Here, the variables will play a major role (instead of being just placeholders). The two definitions $\models, \models'$ are completely equivalent $\models = \models'$ ($\mathcal{A} \models F$ iff $\mathcal{A} \models' F$).

Definition 6.2.7. An (interpretational) **environment** in an $\mathcal{L}$ -structure $\mathcal{A} = (A, I)$ is just a function $v : \mathcal{V} \rightarrow A$ (map every variable to an element of the universe).

Definition 6.2.8. Variant of an environment, having $v : \mathcal{V} \rightarrow A$, for a variable $x \in \mathcal{V}$ and an element of the universe $a \in A$

$$v[a/x](y) = \begin{cases} v(y) & \text{if } y \neq x \\ a & \text{if } y = x \end{cases}$$

$v[a/x]$ is the variant of v with respect to x, a . Clearly it's an environment.

Interpretation of $\mathcal{L}$ -terms (also non-ground terms):

- For all $x \in \mathcal{V}$ and each environment $v : \mathcal{V} \rightarrow A$

$$I_v(x) := v(x)$$

- While $I_v(F(t_1, \dots, t_n)) := (I(F))(I_v(t_1), \dots, I_v(t_n))$

Definition 6.2.9. $\mathcal{A} \models_v F$, it's the **interpretation with regard to some environment v** , “ $\mathcal{A}$ models F in the environment v ”.

- $\mathcal{A} \models_v P(t_1, \dots, t_n)$ iff $(I_v(t_1), \dots, I_v(t_n)) \in I(P)$

$$\begin{cases} \mathcal{A} \models_v F_1 \wedge F_2 \text{ iff } \mathcal{A} \models_v F_1 \text{ and } \mathcal{A} \models_v F_2 \\ \vdots \end{cases}$$

- $\mathcal{A} \models_v \forall x \in F$ iff for all $a \in A$ $\mathcal{A} \models_{v[\frac{a}{x}]} F$.

- $\mathcal{A} \models_v \exists x \in F$ iff for all $a \in A$ $\mathcal{A} \models_{v[\frac{a}{x}]} F$

But we want to define the notion without dependence on the environment so we say that;

$$\mathcal{A} \models' F \text{ iff for all } v : \mathcal{V} \rightarrow A: \mathcal{A} \models_v F$$

← (complete as exercise $\mathcal{A} \models' F$ iff $\mathcal{A} \models F$)

First definition:

$\mathcal{L} \longrightarrow \left\{ \begin{array}{l} \text{infinitely many} \\ \mathcal{L}\text{-structures} \end{array} \right. \left\{ \begin{array}{l} \mathcal{A} = (A, I) \rightsquigarrow \mathcal{L}_{\mathcal{A}} \rightsquigarrow \overline{\mathcal{A}} \models (\mathcal{L}_{\mathcal{A}}\text{-structure}) \\ \mathcal{B} = (B, J) \rightsquigarrow \mathcal{L}_{\mathcal{B}} \rightsquigarrow \overline{\mathcal{B}} \models (\mathcal{L}_{\mathcal{B}}\text{-structure}) \\ \vdots \end{array} \right.$

Second definition:

$\mathcal{L} \longrightarrow \left\{ \begin{array}{l} \text{infinitely many} \\ \mathcal{L}\text{-structures} \end{array} \right. \left\{ \begin{array}{l} \mathcal{A} = (A, I) \rightsquigarrow \left\{ \begin{array}{l} \mathcal{A} \models_v \\ \mathcal{A} \models_w \end{array} \right\} \rightarrow \mathcal{A} \models \\ \mathcal{B} = (B, J) \rightsquigarrow \left\{ \begin{array}{l} \mathcal{B} \models_{-} \\ \mathcal{B} \models_{-} \end{array} \right\} \rightarrow \mathcal{B} \models \end{array} \right.$

6.3 Notions, Terminology

Definition 6.3.1. An $\mathcal{L}$ -sentence F is called a **logical truth**, or is a valid sentence, **iff** for all $\mathcal{L}$ -structures $\mathcal{A}$ we have that $\mathcal{A} \models F$, in this case we write $\models F$ (same notation for tautologies at propositional level, it's kind of like a tautology, but not exactly).

There are some differences with propositional tautologies, for example $c = c$ are just propositional letters, but using the fixed semantics of $=$ we can say that this is a valid sentence, same with $\forall x(x = x)$.

Other examples:

- $(\forall xP(x)) \rightarrow (\forall xP(x))$ is valid and is a tautology, since it's in the form $A \rightarrow A$
- $(\forall xP(x)) \rightarrow (\exists xP(x))$, it's valid and a logical truth, but not a tautology at the propositional level, to say that it's valid we have to reason at first order

Definition 6.3.2. Logical equivalence: Given the $\mathcal{L}$ -sentences A, B we say that A is logically equivalent with B , written as $A \equiv B$, **iff** A and B have (exactly) the same models, i.e., for all $\mathcal{L}$ -structures $\mathcal{A}$, $\mathcal{A} \models A$ **iff** $\mathcal{A} \models B$. Equivalently, $A \equiv B$ **iff** $\models (A \rightarrow B) \wedge (B \rightarrow A)$ (same as $\models (A \leftrightarrow B)$).

Definition 6.3.3. Equisatisfiability: Given the $\mathcal{L}$ -sentences A, B . We say that A equisat B **iff** (A has a model **iff** B has a model). Similar to propositional logic.

Definition 6.3.4. An $\mathcal{L}$ -theory Γ is just a **set of $\mathcal{L}$ -sentences**. If Γ is an $\mathcal{L}$ -theory and $\mathcal{A} = (A, I)$ is an $\mathcal{L}$ -structure we say that $\mathcal{A}$ is a model of the theory Γ (models/renders true/satisfies Γ), written as $\mathcal{A} \models \Gamma$ **iff** $\mathcal{A} \models \gamma$ for all $\gamma \in \Gamma$.

Definition 6.3.5. An $\mathcal{L}$ -theory Γ is **satisfiable** **iff** Γ has at least one model, i.e., there is at least one $\mathcal{A}$ such that $\mathcal{A} \models \Gamma$. Otherwise Γ is **unsatisfiable**.

Definition 6.3.6. Let Γ be an $\mathcal{L}$ -theory and F be an $\mathcal{L}$ -sentence. Then F is **logical consequence** of Γ ($\Gamma \models F$) **iff** F is true in every model of Γ , i.e., if $\mathcal{A} \models \Gamma$ then $\mathcal{A} \models F$.

We shall be concerned in ascertaining whether $\Gamma \models F$ or not, for a given $\Gamma \subseteq \mathcal{F}_{\mathcal{L}}$, $F \in \mathcal{F}_{\mathcal{L}}$.

Examples of $\mathcal{L}$ -theories Order, Partial Order, R of arity 2 in a language with identity

Example 6.3.7. Partial order R of arity 2, in language with identity

$$\Gamma_{po} := \begin{cases} \forall x R(x, x) & \text{Reflexivity} \\ \forall x \forall y (R(x, y) \wedge R(y, x)) \rightarrow x = y & \text{Antisymmetry} \\ \forall x \forall y \forall z (R(x, y) \wedge R(y, z)) \rightarrow R(x, z) & \text{Transitivity} \end{cases}$$

We can also do a total order adding this:

$$\Gamma_{to} := \Gamma_{po} \cup \{\forall x \forall y (R(x, y) \vee R(y, x))\}$$

Obviously we have:

$$\begin{aligned} (\mathbb{N}, \leq) &\models \Gamma_{po}, \models \Gamma_{to} \\ (\mathbb{N}, \geq) &\models \Gamma_{po}, \models \Gamma_{to} \\ (\mathbb{N}, =) &\models \Gamma_{po}, \not\models \Gamma_{to} \\ (\mathbb{N}, |(divisibility)|) &\models \Gamma_{po}, \not\models \Gamma_{to} \end{aligned}$$

← (Exercises remember to see them!!)

$$\leftarrow (\mathbb{N}, \leq) = (\mathcal{A} := (A, I(R)))$$

ADDITIONAL EXERCISES

Example 6.3.8. Congruence relation, given $\mathcal{L} = (\mathcal{P}, \mathcal{F}, \alpha, \beta)$, $R \in \mathcal{P}$, $\alpha(R) = 2$

$$\Gamma_{eq} := \begin{cases} \forall x R(x, x) & \text{Reflexivity} \\ \forall x \forall y R(x, y) \rightarrow R(y, x) & \text{Symmetry} \\ \forall x \forall y \forall z (R(x, y) \wedge R(y, z)) \rightarrow R(x, z) & \text{Transitivity} \end{cases}$$

1. $\forall x_1 \dots \forall x_n \forall y_1 \dots \forall y_n$, with $F \in \mathcal{F}$, $\beta(F) = n$:

$$[R(x_1, y_1) \wedge R(x_2, y_2) \wedge \dots \wedge R(x_n, y_n)] \rightarrow R(F(x_1, x_2, \dots, x_n), F(y_1, y_2, \dots, y_n))$$

2. $\forall x_1 \dots \forall x_n \forall y_1 \dots \forall y_n$, with $P \in \mathcal{P} \setminus \{R\}$, $\alpha(P) = n$:

$$[R(x_1, y_1) \wedge R(x_2, y_2) \wedge \dots \wedge R(x_n, y_n)] \rightarrow (P(x_1, x_2, \dots, x_n) \rightarrow P(y_1, y_2, \dots, y_n))$$

Every model of $\Gamma_{eq} \cup \{1., 2.\}$ is a congruence with respect to any $F \in \mathcal{F}$ and any $P \in \mathcal{P} \setminus \{R\}$.

Example 6.3.9. Group Theory, A group is an algebraic structure with a binary operation which is associative, and such that every element is invertible.

First formulation, $\star \in \mathcal{F}$, $e \in \mathcal{F}$, $()^{-1} \in \mathcal{F}$, $\beta(star) = 2$, $\beta(e) = 0$, $\beta(()^{-1}) = 1$ also the language has **identity** :

$$\Gamma_{G1} := \begin{cases} \forall x \forall y \forall z ((x \star y) \star z) = (x \star (y \star z)) \\ \forall x (x \star e = x) \wedge (e \star x = x) \\ \forall x (x \star x^{-1} = e \wedge x^{-1} \star x = e) \end{cases}$$

Obviously we have:

$$(G, \cdot, ()^{-1}, e) \models \Gamma_{G1} \text{ iff } G \text{ is a group}$$

$$\left. \begin{array}{l} (\mathbb{Z}, +, -, 0) \models \Gamma_{G1} \\ (\mathbb{Q}, +, -, 0) \models \Gamma_{G1} \\ (\mathbb{Q} \setminus \{0\}, \cdot, \frac{1}{\cdot}, 1) \models \Gamma_{G1} \end{array} \right\} \text{All these examples satisfy also } \forall x \forall y x \star y = y \star x : \text{comm} \quad \leftarrow (\Gamma_{G1} \cup \{comm\} \text{ makes it the theory of abelian groups.})$$

Second formulation:

$$\Gamma_{G2} := \begin{cases} \forall x \forall y \forall z ((x \star y) \star z = (x \star (y \star z)) \\ \forall x (x \star e = x) \wedge (e \star x = x) \\ \forall x \exists y (x \star y = e \wedge y \star x = e) \end{cases}$$

Obviously we have:

$$(\mathbb{Z}, +, 0) \models \Gamma_{G2} \quad (\mathbb{Q}, +, 0) \models \Gamma_{G2} \quad (\mathbb{Q} \setminus \{0\}, \cdot, 1) \models \Gamma_{G2}$$

$$(\mathbb{Z}, +, -, 0) \models \Gamma_{G2}$$

$$(\mathbb{Z}, +, - + 1, 0) \models \Gamma_{G2}$$

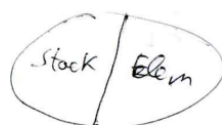
$$(\mathbb{Z}, +, - + 1, 0) \not\models \Gamma_{G1}$$

Data Structures: Stack

$$\mathcal{L}: (\mathcal{D}, \mathcal{F}, \alpha, \beta) \quad \mathcal{D} = \{\text{Stack}, \text{Elem}, =\} \\ \mathcal{F} = \{\text{push}, \text{pop}, \text{top}, \text{nil}\}$$

$$\begin{cases} \alpha(\text{Stack}) = 1 \\ \alpha(\text{Elem}) = 1 \\ \alpha(=) = 2 \\ \beta(\text{push}) = 2 \\ \beta(\text{pop}) = 1 \\ \beta(\text{top}) = 1 \\ \beta(\text{nil}) = 0 \end{cases} \quad \forall$$

$$\Gamma_{\text{STACK}} := \begin{cases} \forall x (\text{Stack}(x) \vee \text{Elem}(x)) \\ \forall x (\neg \text{Stack}(x) \vee \neg \text{Elem}(x)) \end{cases}$$



$$\cup \begin{cases} \forall x \forall y (\text{Stack}(x) \wedge \text{Elem}(y) \rightarrow \text{top}(\text{push}(x, y)) = y) \\ \forall x \forall y (\text{Stack}(x) \wedge \text{Elem}(y) \rightarrow \text{pop}(\text{push}(x, y)) = x) \\ \forall x (\text{Stack}(x) \rightarrow (x = \text{nil} \vee \text{push}(\text{pop}(x), \text{top}(x)) = x)) \end{cases}$$

Arithmetic: Peano (PA)

$$\mathcal{L}_{\text{PA}}: (\mathcal{D}, \mathcal{F}, \alpha, \beta) \quad \mathcal{D} = \{=, 0, S\} \quad \alpha(=) = 2 \\ \mathcal{F} = \{+, \times, S, 0\} \quad \begin{cases} \beta(+) = \beta(\times) = 2 \\ \beta(S) = 1, \beta(0) = 0 \end{cases}$$

$$\text{PA1: } \forall x \neg (0 = S(x)) \quad \equiv \leftarrow (\neg \exists x (0 = S(x))) \quad \text{"0 is not a successor of any natural."}$$

$$\text{PA2: } \forall x \forall y (S(x) = S(y) \rightarrow x = y) \quad \text{"S is injective"}$$

$$\text{PA3: } \forall x \quad x + 0 = x$$

$$\text{PA4: } \forall x \forall y \quad x + S(y) = S(x + y)$$



$$\text{PA5: } \forall x \quad x \times 0 = 0$$

$$\text{PA6: } \forall x \forall y \quad x \times S(y) = x + (x \times y)$$

contains infinitely many axioms

$$\text{PA7(F): } (F[0] \wedge \forall x (F \rightarrow F[Sx])) \rightarrow \forall x F$$

induction schema

where **F** is any formula with $\text{FV}(F) \subseteq \{x\}$

simplify notation

$$(F(0) \wedge \forall x (F(x) \rightarrow F(Sx))) \rightarrow \forall x F(x)$$

for instance

$\forall x \neg (x = Sx)$ is true only if we add PA7 to our theory

6.3.1 Normal forms

Our Goal Similarly to propositional logic, is trying to decide whether $\Gamma \models^? A$.

If Γ is infinite we can use first-order compactness theorem to study finite approximations of Γ . If $\Gamma = \{\gamma_1, \dots, \gamma_n\}$ is a finite theory. Then $\Gamma \models A$ iff $\gamma_1 \wedge \gamma_2 \wedge \dots \wedge \gamma_n \wedge \neg A$ is UNSAT (in the first-order meaning, i.e., if it doesn't have a model). We need to develop a method to (semi-)decide whether a first-order sentence is UNSAT.

First Step: put the sentence in a suitable normal form.

Prenex Normal Form PNF

Definition 6.3.10. An $\mathcal{L}$ -sentence is in PNF iff it is of the form $Q_1x_1 \dots Q_nx_nM$, where each Q_i is one of the quantifiers, $Q_i \in \{\forall, \exists\}$ and M is an $\mathcal{L}$ -formula free of quantifiers (called Matrix). Notice that $FV(M) \subseteq \{x_1, \dots, x_n\}$.

Fact 6.3.1. Every $\mathcal{L}$ -sentence A is logically equivalent (has the same models, true under the same $\mathcal{L}$ -structures) to an $\mathcal{L}$ -sentence $\in PNF$

$$A \equiv A^P, A^P \in PNF$$

The sketch of proof A^P is built (in polynomial time with respect to $\|A\|$) using iteratively the following notable logical equivalences:

Fact 6.3.2. Let P be any $\mathcal{L}$ -formula and let z be a variable **not** occurring in P (z is “new”).

1. Renaming:

$$\begin{cases} \exists xP & \equiv \exists zP[z/x] \\ \forall xP & \equiv \forall zP[z/x] \end{cases}$$

2. De Morgan:

$$\begin{cases} \neg \forall xP & \equiv \exists x \neg P \\ \neg \exists xP & \equiv \forall x \neg P \\ \forall xP & \equiv \neg \exists x \neg P \\ \exists xP & \equiv \neg \forall x \neg P \end{cases}$$

← (can see $\forall$ as a form of generalized conjunction and $\exists$ as a generalized disjunction)

Proof. Lets prove one case of the De Morgan. Starting from an $\mathcal{L}$ -structure we want to prove $\mathcal{A} \models \neg \forall xP$ iff $\mathcal{A} \models \exists x \neg P$:

iff $\mathcal{A} \models \neg \forall xP$

iff $\mathcal{A} \not\models \forall xP$

by the semantics of negation

iff it's not the case that for all $a \in A, \mathcal{A} \models P[\bar{a}/x]$

by the semantics of $\forall$

iff for some $a \in A, \mathcal{A} \not\models P[\bar{a}/x]$

by the semantics of $\forall$

iff for some $a \in A, \mathcal{A} \models \neg P[\bar{a}/x]$

by the semantics of negation

iff $\mathcal{A} \models \exists x \neg P$

by the semantics of $\exists$

□

← (Notice!, in general $\forall x \exists y P \not\equiv \exists y \forall x P$)

3. Redundant quantification and exchange

$$\begin{cases} \forall x \forall y P \equiv \forall y \forall x P \\ \exists x \exists y P \equiv \exists y \exists x P \\ \forall x P \equiv P \text{ if } x \notin FV(P) \\ \exists x P \equiv P \text{ if } x \notin FV(P) \end{cases}$$

4. Quantifiers shifting (w.r.t $\wedge, \vee$)

$$\begin{cases} \forall x(P_1 \wedge P_2) \equiv \forall x P_1 \wedge \forall x P_2 \\ \exists x(P_1 \vee P_2) \equiv \exists x P_1 \vee \exists x P_2 \\ \forall x(P_1 \vee P_2) \equiv \forall x P_1 \vee P_2 \text{ if } x \notin FV(P_2) \\ \exists x(P_1 \wedge P_2) \equiv \exists x P_1 \wedge P_2 \text{ if } x \notin FV(P_2) \end{cases}$$

$$\left. \begin{array}{l} \forall x(P_1 \vee P_2) \not\equiv \forall x P_1 \vee \forall x P_2 \\ \forall x(P_1 \wedge P_2) \not\equiv \forall x P_1 \wedge \forall x P_2 \end{array} \right\} \text{In general}$$

Example 6.3.11. Given $\mathcal{N} = \{\mathbb{N}, E, O\}$ where E is interpreted as the set of Even number and O of Odd numbers

$$\begin{cases} \mathcal{N} \models \forall x(E(x) \vee O(x)) \\ \mathcal{N} \not\models \forall x E(x) \vee \forall x O(x) \\ \mathcal{N} \not\models \exists x(E(x) \vee O(x)) \\ \mathcal{N} \models \exists x E(x) \vee \exists x O(x) \end{cases}$$

5. When in doubt, substitute, $Q_1, Q_2 \in \{\forall, \exists\}$

$$\begin{cases} Q_1 x P_1 \vee Q_2 x P_2 \equiv Q_1 x Q_2 z (P_1 \vee P_2[z/x]) \\ Q_1 x P_1 \vee Q_2 x P_2 \equiv Q_1 x Q_2 z (P_1 \vee P_2[z/x]) \end{cases}$$

6. Indirect implication, all these are true only if $x \notin FV(Q)$

$$\begin{cases} \forall x P \rightarrow Q \equiv \exists x (P \rightarrow Q) \\ \exists x P \rightarrow Q \equiv \forall x (P \rightarrow Q) \\ Q \rightarrow \forall x P \equiv \forall x (P \rightarrow Q) \\ Q \rightarrow \exists x P \equiv \exists x (P \rightarrow Q) \end{cases} \leftarrow \text{(si son veri pensaci)}$$

Fact 6.3.3. Given an $\mathcal{L}$ -sentence A , there is a polynomially short (in $\|A\|$) sequence

$$A = A_0 \rightsquigarrow A_1 \rightsquigarrow \dots \rightsquigarrow A_u, \quad A_u \in PNF \leftarrow \text{(NOT PROVEN)}$$

6.3.2 Skolemization

We're up to the point in which: $\mathcal{L}$ -sentence A , $A \rightsquigarrow A^P \in PNF$.

Now we want to get rid of all occurrences of $\exists$ from A^P . This process is called **Skolemization** (From Thoralf Skolem).

Definition 6.3.12. An $\mathcal{L}$ -sentence $A \in PNF$ is in **Skolem Normal Form SNF** iff it does not contain any occurrences of $\exists$; they are universal sentences, this is also called universal normal form.

$\leftarrow$ (Exercise in Slides)

Skolemisation

Given $A: \forall x_1 \forall x_2 \dots \forall x_n \exists y B$ where $\exists y$ is the outermost occurrence of $\exists$ and y is not free in B (without loss of generality).

Then $A^S := \forall x_1 \forall x_2 \dots \forall x_n B[\frac{\bar{F}(x_1 \dots x_n)}{y}]$

where $\bar{F}$ is a new function symbol of arity n .

$\mathcal{L} \rightsquigarrow \mathcal{L}^S : (\mathcal{D}, \mathcal{F} \cup \{\bar{F}\}, \alpha, \beta^S)$ where β^S extends β by putting $\beta^S(\bar{F}) = \bar{u}$.

where β^S extends β by putting $\beta^S(\bar{F}) = \bar{u}$.

0) $A \in \text{PNF}$ ✓
 1) for sake of simplicity in A there are no occurrences of redundant quantifiers (that is in B there are no occurrences of $\forall x_i \exists x_i, \forall y \exists y$)
 2) In B there may occur other quantifiers

Let $\mathcal{A}$ be an $\mathcal{L}$ -structure $\mathcal{A} = (U, I)$

Then we write $(\mathcal{A}, h)$ for the $\mathcal{L}^S$ -structure $(\mathcal{A}, h) = (U, I_h)$

such that I_h extends I ($I(g) = I_h(g)$ for all $g \in \mathcal{F}$)

by setting $I_h(\bar{F}) = h$, for $h: U^n \rightarrow U$ such otherwise h is the interpretation of $\bar{F}$.

Notice: Every $\mathcal{L}^S$ -structure arises as a structure of the form $(\mathcal{A}, h)$ for some $\mathcal{L}$ -structure $\mathcal{A} = (U, I)$ and some function $h: U^n \rightarrow U$.

← (i mean kinda you get the point)

Example 6.3.13. $A \neq A^S$, $P \in \mathcal{P}$, with $\alpha(P) = 1$, $A := \exists x P(x)$ and $A^S := P[\bar{c}/x]$ we replace the quantifier with a function of arity equal to the number of quantifiers before that, in this case it's a new constant. Given this structure $\mathcal{A} = \{U, I\}$, $U = \{e_1, e_2\}$, $I(P) = \{e_2\}$

$$(\mathcal{A}, e_2) \models A$$

$$(\mathcal{A}, e_2) \models A$$

$$(\mathcal{A}, e_2) \not\models A^S \quad ((\mathcal{A}, e_2) \models P(\bar{c}) \text{ iff } I(\bar{c}) \in I(P) \text{ iff } e_1 \in \{e_2\} \text{ NO!})$$

$$(\mathcal{A}, e_2) \models A^S \quad ((\mathcal{A}, e_2) \models P(\bar{c}) \text{ iff } I(\bar{c}) \in I(P) \text{ iff } e_2 \in \{e_2\} \text{ SI!})$$

The by (†) $(\mathcal{A}, e_2) \models A$ but $(\mathcal{A}, e_2) \not\models A^S$,
 that is $A \neq A^S$ ($\not\models A \rightarrow A^S$).

$\models A^S \rightarrow A$ $A: \forall x_1 \dots \forall x_n \exists y B$ $A^S: \forall x_1 \dots \forall x_n B[\frac{\bar{F}(x_1 \dots x_n)}{y}]$

$(\mathcal{A}, h) \models A^S$ iff for all $a_1, \dots, a_n \in U^n$ $(\mathcal{A}, h) \models B[\frac{\bar{F}(a_1 \dots a_n)}{y}][\frac{a_1}{x_1}] \dots [\frac{a_n}{x_n}]$
 iff for all $a_1, \dots, a_n \in U^n$ $(\mathcal{A}, h) \models B[\frac{a_1}{x_1}] \dots [\frac{a_n}{x_n}][\frac{\bar{F}(a_1 \dots a_n)}{y}]$

Put $b := h(a_1 \dots a_n)$ ($I_h(\bar{F}) = h$)

iff for all $a_1, \dots, a_n \in U^n$ $(\mathcal{A}, h) \models B[\frac{a_1}{x_1}] \dots [\frac{a_n}{x_n}][\frac{b}{y}]$
 ($I_h(\bar{F}(a_1 \dots a_n)) = b$)

iff for all $a_1, \dots, a_n \in U^n$ ~~$(\mathcal{A}, h) \models B[\frac{a_1}{x_1}] \dots [\frac{a_n}{x_n}][\frac{b}{y}]$~~
 there is $b \in U$ s.t. $(\mathcal{A}, h) \models B[\frac{a_1}{x_1}] \dots [\frac{a_n}{x_n}][\frac{b}{y}]$

iff $(\mathcal{A}, h) \models \forall x_1 \dots \forall x_n \exists y B$

iff $(\mathcal{A}, h) \models A$

So: $(\mathcal{A}, h) \models A$ implies $(\mathcal{A}, h) \models A^S \Rightarrow \models A^S \rightarrow A$.

We now have to show that A and A^S are equisatisfiable, we do this by showing that if $(\mathcal{A}, h) \models A$ then there is $g : U^n \rightarrow U$ (in general $g \neq h$) such that $(\mathcal{A}, g) \models A^S$.

Notice: for each function $\ell : U^n \rightarrow U$, $(\mathcal{A}, \ell) \models A$ implies $\mathcal{A} \models A$ since A does not contain occurrences of $\bar{f}$. $(\mathcal{A}, h) \models A$ then $\mathcal{A} \models A$ (remember that $A : \forall x_1 \dots \forall x_n \exists y B$).

$\mathcal{A} \models A$ iff for all $(a_1, \dots, a_n) \in U^n$ there is $b \in U$ such that $\mathcal{A} \models B[\bar{a}_1/x_1] \dots [\bar{a}_n/x_n][\bar{b}/y]$. Put $g(a_1, \dots, a_n) := b$ and call it $(\dagger)$. The collection of all $(\dagger)$ s, for all $(a_1, \dots, a_n) \in U^n$ defines a function $g : U^n \rightarrow U$ (maybe $g \neq h$)

Let us check that indeed $(\mathcal{A}, g) \models A^S$ ($A^S : \forall x_1 \dots \forall x_n B[\frac{\bar{f}x_1 x_n}{y}]$) 13

$(\mathcal{A}, g) \models A^S$ iff For all $(a_1, \dots, a_n) \in U^n$ $(\mathcal{A}, g) \models B[\frac{\bar{f}a_1 \dots a_n}{y}][\frac{\bar{a}_1}{x_1}] \dots [\frac{\bar{a}_n}{x_n}]$
 iff " " " " $(\mathcal{A}, g) \models B[\frac{\bar{a}_1}{x_1}] \dots [\frac{\bar{a}_n}{x_n}][\frac{\bar{f}a_1 \dots a_n}{y}]$

Recall that $I_g(\bar{f}a_1, \dots, a_n) = g(a_1, \dots, a_n) = b = I_g(\bar{b}) = I_g(\bar{f})$
 ($\dagger$)

iff For all $(a_1, \dots, a_n) \in U^n$ $(\mathcal{A}, g) \models B[\frac{\bar{a}_1}{x_1}] \dots [\frac{\bar{a}_n}{x_n}][\frac{\bar{f}}{y}]$

iff For all $(a_1, \dots, a_n) \in U^n$ there is $b \in U$ s.t.
 $(\mathcal{A}, g) \models B[\frac{\bar{a}_1}{x_1}] \dots [\frac{\bar{a}_n}{x_n}][\frac{\bar{f}}{y}]$

iff $(\mathcal{A}, g) \models \forall x_1 \dots \forall x_n \exists y B$

iff $(\mathcal{A}, g) \models A$

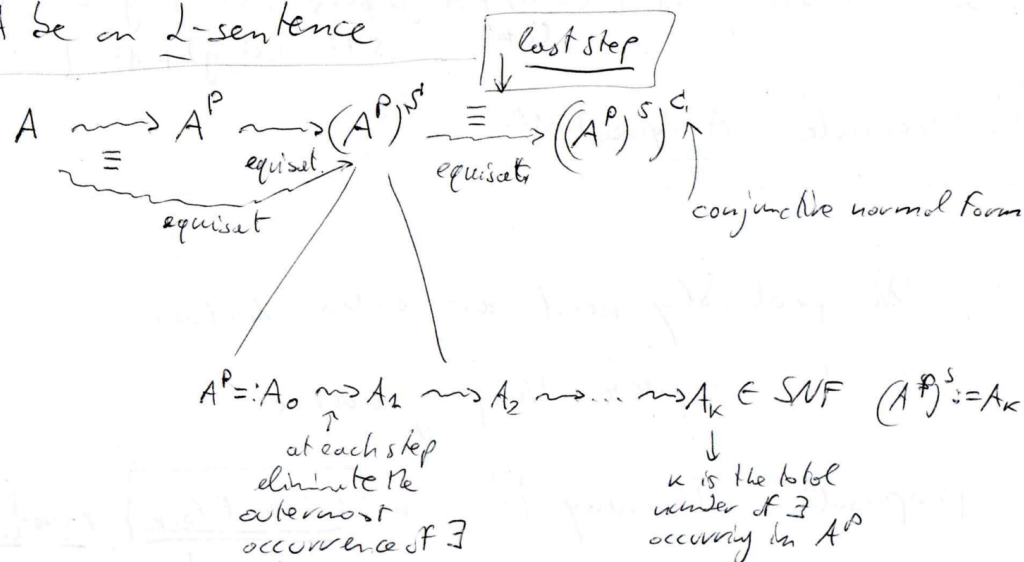
iff $\mathcal{A} \models A$ (since $\bar{f}$ does not occur in A)

But since $\mathcal{A} \models A$, then $(\mathcal{A}, g) \models A^S$.
 (by our assumption)

From $\models A^S \rightarrow A$ and $[(\mathcal{A}, h) \models A \text{ implies there is } g : U^n \rightarrow U \text{ such that } (\mathcal{A}, g) \models A^S]$ we conclude A equisat A^S

The last step consists in putting the obtained sentence in CNF.

Let A be an L -sentence



Let us consider any atomic formula $P(t_1, \dots, t_k)$ as a propositional letter (even if the formula contains variable names). Mind that distinct atomic formulas (distinct as strings of symbols) are distinct propositional letters. Then in $\forall x_1 \dots \forall x_n M \in SNF$, the matrix M is considered as a propositional formula over its atomic formulas as propositional letters. One can use any propositional algorithm to produce $M^C \in CNF$ such that M^C equisat M (or even $M^C \equiv M$ if an algorithm that preserves logical equivalence is used). Then $\forall x_1 \dots \forall x_m M^C$ is in first-order conjunctive normal form. We would like to semidecide the following statement

$$\Gamma \stackrel{?}{\models} A$$

assuming Γ is finite $\Gamma = \gamma_1, \dots, \gamma_u$ (otherwise we have to use compactness to focus on growing finite subsets of Γ). As we already know, this amounts to asking:

$$\Gamma, \neg A \text{ UNSAT} \rightsquigarrow \gamma_1, \wedge \dots \wedge \gamma_u \wedge \neg A \text{ UNSAT} \rightsquigarrow (\gamma_1, \wedge \dots \wedge \gamma_u \wedge \neg A)^{PS^C} \text{ UNSAT}$$

If $F \in PNF \cap SNF \cap CNF$ then F is of the form

$$F: \forall x_1 \dots \forall x_m (C_1 \wedge C_2 \wedge \dots \wedge C_k)$$

$$(\text{C}_i \text{ is a 1st order clause}) \quad C_i = l_{i1} \vee l_{i2} \vee \dots \vee l_{i\alpha_i}$$

$$(\text{l}_{ij} \text{ is a 1st order literal}) \quad \text{each } l_{ij} = \begin{cases} P(t_1, \dots, t_k) \\ \text{or} \\ \neg P(t_1, \dots, t_k) \end{cases} \quad \begin{array}{l} \text{for } P \in \mathcal{P} \quad \alpha(P)=k \\ t_1, \dots, t_k \in \mathcal{T}_L \\ (\text{not necessarily ground}) \end{array}$$

We now simplify the notation:

1. We forget all quantifiers: writing all quantifiers $\forall x_1 \dots \forall x_m$ does not add any information, as we already assume every variable is universally quantified. Alternatively we can say that $\mathcal{A} \models G$ iff $\mathcal{A} \models \forall[G]$.
2. We use set theoretic notations for CNFs as already introduced in a previous chapter, thus:

$$F \in PNF \cap SNF \cap CNF \rightsquigarrow S_F = C_1, \dots, C_v$$

where S_F is a set of clauses and

$$C_i = l_{i1}, \dots, l_{iu}$$

This way we get the following advantages:

- We can put any formula in our original theory in our normal form independently one from another (explained better in further exercises)
 - if γ is an infinite theory, then S_F is going to be just an infinite set of clauses
3. Finally we will use sequent form for clauses: take a clause

$$C_i = l_{i1}, \dots, l_{iu}$$

and consider the two subsets of the negative and positive atomic formulas in C_i

$$N_i = \{l_{ij} : l_{ij} = \neg P(t_1, \dots, t_k) \in C_i\}$$

$$P_i = \{l_{ij} : l_{ij} = P(t_1, \dots, t_k) \in C_i\}$$

observe that this is a partition, and as such:

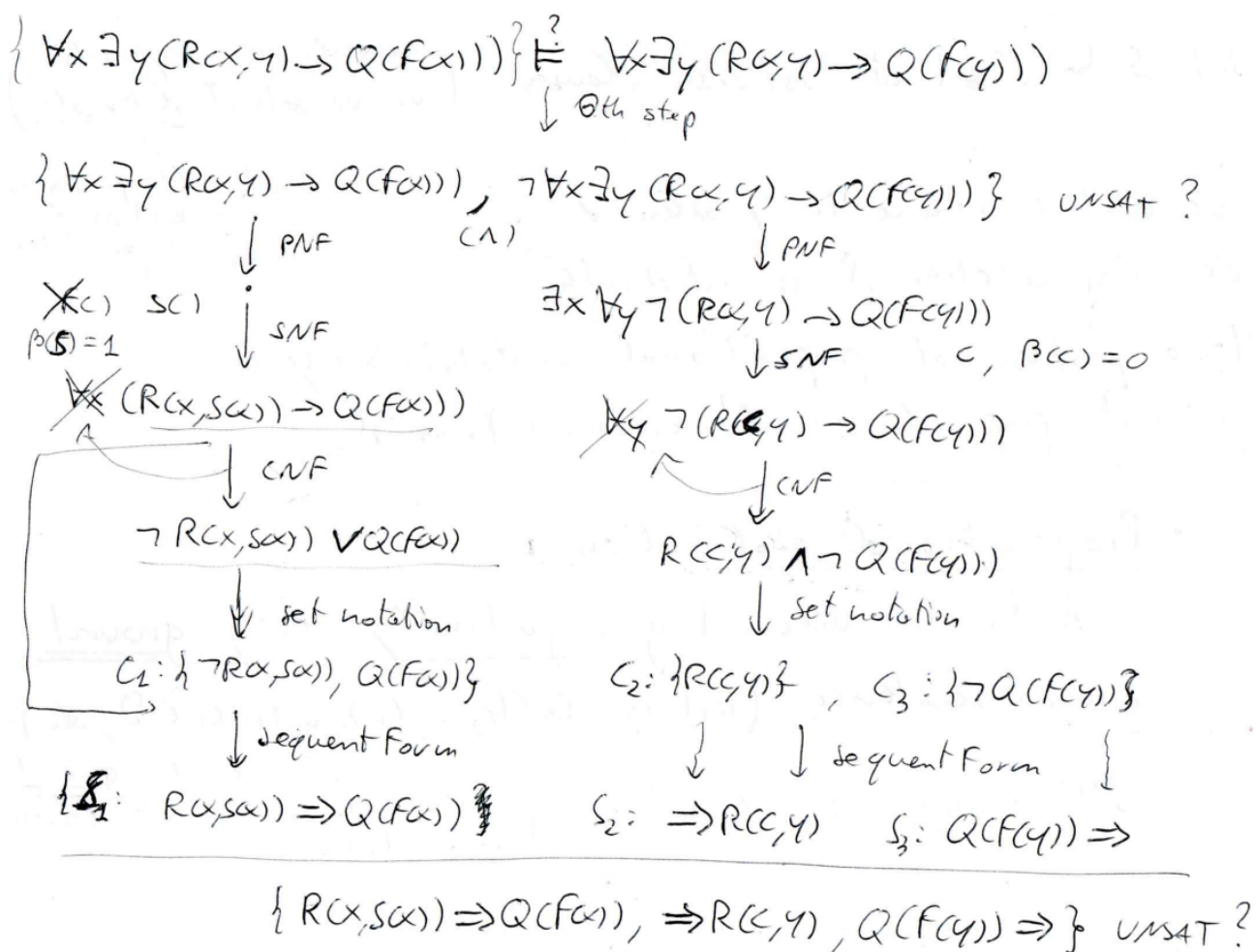
$$P_i \cup N_i = C_i \quad P_i \cap N_i = \emptyset$$

the sequent form of C_i is then:

$$N_i \Rightarrow P_i$$

← (think about it its literally just a implication, also the empty clause $\square$ is just $\{\} \Rightarrow \{\}$)

Example 6.3.14.



6.3.3 Herbrand Theory

Let S be a set of first-order clauses (written as sequents or as sets of literals). We want to reduce the problem of deciding whether S is satisfiable to the problem of propositional satisfiability on a set of propositional clauses S' obtained in some way from S . This process will follow the following step:

- **Propositional Abstraction:** we use a function associating injectively every ground atomic sentence (that is $Q(t_1, \dots, t_n)$ with $Q \in \mathcal{P}$, $\alpha(Q) = n$ and $t_1, \dots, t_n$ ground terms) with a propositional letter $p_{Q(t_1, \dots, t_n)}$.

Given a clause $C = \{A_1, \dots, A_n \Rightarrow B_1, \dots, B_p\}$, where all A_i, B_i are atomic and ground, then applying propositional abstraction we get:

$$C^* = \{p_{A_1}, p_{A_2}, \dots, p_{A_n} \Rightarrow p_{B_1}, p_{B_2}, \dots, p_{B_p}\}$$

with p_{A_1}, p_{B_1} propositional letters.

Theorem 6.3.15 (Herbrand). A set of first order clauses $\mathcal{S}$ is unsatisfiable **iff** there exists a finite set of ground instances of clauses of $\mathcal{S}$ such that their propositional abstraction is propositionally unsatisfiable.

Definition 6.3.16. Let $\mathcal{S}$ be a set of 1st order $\mathcal{L}$ -clauses and let G be a set of ground $\mathcal{L}$ -terms. Then, by $\mathcal{S}/G$ we mean the **ground instantiation** of $\mathcal{S}$ over G , that is the set of all $\mathcal{L}$ -clauses (which are necessarily ground by construction) obtained from each $C \in \mathcal{S}$, by substituting in C every occurrence of any variable x , with some $t \in G$. This happens simultaneously, by which we mean that every instance of some variable x is substituted by the same t , and in each possible way.

Example 6.3.17.

Examples : $\mathcal{S} = \{ \Rightarrow P(x), D(x), P(x), D(x) \Rightarrow \}$ $\leftarrow \begin{matrix} F \\ E \\ F \\ P \end{matrix}$

$G = \{a, b, c\}$ $(a, b, c \text{ are ground } \mathcal{L}\text{-terms})$

$$\mathcal{S}/G = \{ \Rightarrow P(a), D(a), \Rightarrow P(b), D(b), \Rightarrow P(c), D(c), \\ P(a), D(a) \Rightarrow, P(b), D(b) \Rightarrow, P(c), D(c) \Rightarrow \}$$

Notice: $\Rightarrow P(a), D(b)$ is not in $\mathcal{S}/G$

$$G' = \{a, f(b, c)\}$$

$$\mathcal{S}/G' = \{ \Rightarrow P(a), D(a), \Rightarrow P(f(b, c)), D(f(b, c)), \\ P(a), D(a) \Rightarrow, P(f(b, c)), D(f(b, c)) \Rightarrow \}$$

Definition 6.3.18 (Herbrand Universe). $H_{\mathcal{S}}$ of a set $\mathcal{S}$ of 1st order $\mathcal{L}$ -clauses.

- Let $F_{\mathcal{S}}^0$ be the set of all function symbols (including constants) occurring in $\mathcal{S}$;
- We divide in two cases:
 - If $F_{\mathcal{S}}^0$ does not contain any constant, then set $F_{\mathcal{S}} := F_{\mathcal{S}}^0 \cup \{\uparrow\}$ where $\uparrow$ is a new constant
 - Otherwise $F_{\mathcal{S}} := F_{\mathcal{S}}^0$

Then $H_{\mathcal{S}}$ is the set of all (necessarily ground by construction) terms built only using the symbols in $F_{\mathcal{S}}$

Example 6.3.19.

Examples $\mathcal{S} = \{U(x) \Rightarrow M(x), \Rightarrow U(x), M(x) \Rightarrow\}$ $\checkmark$

$F_S^0 = \{s\}$ $F_S = F_S^0 = \{s\}$, $H_S = \{s\}$

$\mathcal{S} = \{\Rightarrow D(x), P(x), P(x), D(x) \Rightarrow\}$

$F_S^0 = \emptyset$ $F_S = F_S^0 \cup \{x\} = \{x\}$ $H_S = \{x\}$

$\mathcal{S} = \{\Rightarrow D(x), P(F(x))\}$

$F_S^0 = \{f\}$, $F_S = F_S^0 \cup \{x\} = \{f, x\}$, $H_S = \{x, f(x), f(f(x)), \dots, f(\dots f(x))\dots\}$

Notice:

- If F_S^0 contains at least one function symbol of arity > 0 , then H_S is infinite.
- $H_S \neq \emptyset$.

Example 6.3.20.

Examples: $\mathcal{S} = \{U(x) \Rightarrow M(x), \Rightarrow U(x), M(x) \Rightarrow\}$ UNSAT? $\nearrow$ Yes.

$H_S = \{s\}$

$\mathcal{S}/_{H_S} = \{U(s) \Rightarrow M(s), \Rightarrow U(s), M(s) \Rightarrow\}$

$\downarrow \text{PROP ABST}$
 $\downarrow P_{U(s)}$
 $\downarrow P_{M(s)}$
 $\downarrow P_{U(s)}$
 $\downarrow P_{M(s)}$

$(\mathcal{S}/_{H_S})^* = \{p \Rightarrow q, \Rightarrow p, q \Rightarrow\}$ PROP. UNSAT.

$\frac{\{ \neg p, q \} \quad \{ p \}}{\{ q \} \quad \{ \neg q \}} \quad \square$

$\frac{p \Rightarrow q \quad \Rightarrow p}{\Rightarrow q} \quad q \Rightarrow$

$\Rightarrow$

(Yes)

Definition 6.3.21. Let $\mathcal{S}$ be a set of first order clauses, then $\mathcal{S}$ is unsatisfiable iff $(\mathcal{S}/_{H_S})^*$ is propositionally unsatisfiable.

Remark 6.3.22. If H_S is infinite then $(S/H_S)^\times$ is infinite.

Remark 6.3.23. If $G_1 \subseteq G_2$ then $S/G_1 \subseteq S/G_2$

The two theorems of Herbrand's theory that we stated both follow as corollaries of the main theorem of Herbrand's semantics. Herbrand's semantics are a refinement of Tarskian semantics.

Definition 6.3.24. Given:

$$\Gamma \stackrel{?}{\models} A$$

we shall consider, instead of all $\mathcal{L}$ -structures, just some of them, which are "canonical" (da spiegare), where the universe is H_S and the interpretation of terms is fixed and reflects that the universe that we consider is made of terms. Herbrand's semantics flattens semantics to syntax.

• An Herbrand's $\mathcal{L}$ -structure is a special kind of
(Tarski's) $\mathcal{L}$ -structure
as follows:

Tarski	Herbrand
$\mathcal{L}$ -structure $\mathcal{A} = (U, I)$	$\mathcal{L}$ -structure $\mathcal{H} = (H_S, H)$
• $U \neq \emptyset$ (U is arbitrarily chosen)	• $H_S \neq \emptyset$ (fixed, determined by S)
$\beta(c) = 0$ • $I(c) \in U$ (arbitrarily fixed element)	$\beta(c) = 0$ • $H(c) \in H_S$, $H(c) := c$
$\beta(F) = n$ • $I(F) : U^n \rightarrow U$ (arbitrarily fixed function) $n > 0$	$\beta(F) = n$ • $H(F) : H_S^n \rightarrow H_S$ $(H(F))(H(t_1), \dots, H(t_n)) :=$ $(HI(F))(t_1, \dots, t_n) := f(t_1, \dots, t_n)$
$\alpha(P) = n$ • $I(P) \subseteq U^n$ (arbitrarily fixed relation)	$HI(F(t_1, \dots, t_n)) = f(t_1, \dots, t_n) \in H_S^n$ $\alpha(P) = n$ • $HI(P) \subseteq H_S^n$ (arbitrarily fixed relation)!